shAnalysis Workflow & Theory Guide

GRACE / GRACE-FO spherical-harmonic processing in base MATLAB — theory, best practices, and step-by-step recipes

Toolbox version	shAnalysis v3.5.0
Edition	5 (adds: ICGEM time-series download, the standard chain and custom filter design as single entry points, and what real provider files actually look like — FORTRAN D-exponents, the ICGEM 2.0 column order, ragged record groups)
Scope	ICGEM I/O, spectral diagnostics, synthesis & analysis, destriping, Gaussian / DDK / tvANS filtering, VCE, basin averages & deconvolution, climatology & AR(1) significance, GIA, degree-1/C20 chain, Slepian localization, SINEX covariances, mascon comparison, load deformation, gradient tensor, multi-center combination, sea-level fingerprints, Love-number I/O incl. frame conversion, ICGEM series download, pole-tide conventions, pointwise synthesis at satellite altitude
Dependencies	base MATLAB only (no toolboxes); Java optionally for gz streaming
Validation	every numerical method independently validated in Python (numpy/scipy) before MATLAB implementation; 169 unit tests
License	MIT (see LICENSE; bundled DDK3 Wbd file MIT from its upstream repository; ITSG/GFZ fixtures remain their providers' property)
Provenance	Developed by Matthias Weigelt with the help of Claude (Fable 5), 2026-08-07

How to read this guide. Part I develops the theory exactly as implemented (equation forms match the code, including storage conventions and numerically validated tolerances). Part II condenses the best practices the implementation enforces or enables. Part IV is the complete API reference — every public function and method with typed, dimensioned inputs/outputs, defaults and a real-data example, generated from the source arguments blocks at build time. Part III gives copy-paste step-by-step recipes for the standard products. The HTML reference (`doc shAnalysis`) documents every function signature; this guide explains why and in which order.

Part I — Theory as implemented

1. Spherical harmonics, normalization, storage

The disturbing potential is expanded in fully (4π-)normalized real spherical harmonics:

$$V(r,\vartheta,\lambda) = \frac{GM}{R}\sum_{n=0}^{N}\left(\frac{R}{r}\right)^{n+1}\sum_{m=0}^{n}\left[\bar{C}_{nm}\cos m\lambda + \bar{S}_{nm}\sin m\lambda\right]\bar{P}_{nm}(\cos\vartheta)$$

with the normalization $\int|Y_{nm}|^2\,d\Omega = 4\pi$, the ICGEM convention. Throughout the toolbox coefficients live in lower-triangular matrices with 1-based indexing `C(n+1, m+1)`; vectorized orderings are always mediated by `shLowLevel.shIndex` (fields n, m, cs, P; MinDegree defaults to 2 for GRACE work, 0 for analysis). GM and R ride with the data (read from the gfc header); the defaults (3.986004415e14, 6378136.3) are overridable, never baked-in physics. Load Love numbers k_n for water-storage kernels must be supplied explicitly — the toolbox refuses to guess them.

2. Legendre functions: scaled Holmes–Featherstone recursion

Fully normalized associated Legendre functions are computed by the standard order-diagonal seed plus degree recursion. The naive diagonal seed underflows near the poles at high degree; `shLowLevel.legendreALF` therefore implements the globally scaled Holmes–Featherstone variant (scale 10^{-280}, rescaling on the fly), verified stable to degree 2190: the sum rule and parity

$$\sum_{m=0}^{n}\bar{P}_{nm}^2(\cos\vartheta) = 2n+1 \quad (\sim 10^{-11}\ \text{at}\ n=2190), \qquad \bar{P}_{nm}(-\varphi) = (-1)^{n+m}\,\bar{P}_{nm}(\varphi)$$

hold across all latitudes; the parity relation halves the recursion cost on hemispherically symmetric grids (the parity trick).

3. Synthesis

Spatial synthesis evaluates, per latitude, the order sums

$$A_m(\varphi) = \sum_{n=m}^{N} f_n\,\bar{C}_{nm}\,\bar{P}_{nm}(\varphi), \qquad B_m(\varphi)\ \text{analogously with}\ \bar{S}_{nm},$$

then the longitude sum

$$g(\varphi,\lambda) = \sum_{m=0}^{N}\left[A_m(\varphi)\cos m\lambda + B_m(\varphi)\sin m\lambda\right].$$

For uniform full-circle longitude grids this last sum is an inverse FFT after folding aliased orders into mod(m, n_{lon}) bins — exact, validated against the direct product to 3e-13 (`Method="auto"|"fft"|"direct"`).

Memory model (v2.2). The Legendre stack $(n_{max}+1)^2 \times n_{lat} \times 8$ bytes is the memory driver: 7 GB at n_{max} 2190 over 181 latitudes. Above the 'MaxMemGB' budget (default 4), synthesis streams latitude bands: compute the Legendre block, fold into A_m/B_m, discard. Chunked equals monolithic exactly (Python-validated, max deviation 0.0); the class-level Legendre cache is bypassed automatically when the stack would not fit.

4. Quantity kernels

A single degree factor f_n maps Stokes coefficients to the target quantity (`shLowLevel.kernelFactors`):

quantity	f_n	unit
geoid	R	m
potential	GM/R	m²/s²
gravity disturbance	$\frac{GM}{R^2}(n+1)$	m/s²
gravity anomaly	$\frac{GM}{R^2}(n-1)$	m/s²
EWH (water)	$R\frac{\rho_{ave}}{3\rho_w}\frac{2n+1}{1+k_n}$	m
surface density *	$R\frac{\rho_{ave}}{3}\frac{2n+1}{1+k'_n}$	kg/m²

quantity	f_n	unit
bottom pressure *	$g_0 R \frac{\rho_{ave}}{3} \frac{2n+1}{1+k'_n}, \quad g_0 = \frac{GM}{R^2}$	Pa
vertical deformation *	$R \frac{h'_n}{1+k'_n}$	m
gravity gradient T_{rr} *	$\frac{GM}{R^3}(n+1)(n+2)$	1/s²

(* new in v2.3.) The 'Height' option continues potential-type kernels upward by $(R/r)^p$ (p = n+1 potential, n+2 anomaly/disturbance, n+3 T_{rr}; validated against -d/dr and d²/dr² of the continued potential) — evaluate directly at satellite altitude. Surface quantities reject Height by contract.

The EWH kernel divides by $(1+k_n)$: the observed potential contains the solid-Earth loading response, which must be removed to recover surface mass. k_n is always user-supplied (e.g. Wahr et al. 1998 or your loading-theory set of choice).

5. Elastic load deformation (v2.3)

A surface load deforms the elastic Earth; the displacement at the surface, expressed through the load-induced (residual) Stokes coefficients, is (Wahr et al. 1998):

$$u_{\mathrm{up}} = R\sum_{n,m} \frac{h'_n}{1+k'_n} \Delta\bar{C}_{nm}\, \bar{Y}_{nm}, \qquad u_{\mathrm{north}} = R\sum_{n,m} \frac{l'_n}{1+k'_n} \Delta\bar{C}_{nm} \frac{\partial \bar{Y}_{nm}}{\partial\varphi}, \qquad u_{\mathrm{east}} = R\sum_{n,m} \frac{l'_n}{1+k'_n} \Delta\bar{C}_{nm} \frac{1}{\cos\varphi} \frac{\partial \bar{Y}_{nm}}{\partial\lambda}$$

The vertical is a pure degree factor ('deformation_up'); the horizontals need the exact spherical-harmonic gradient. `shLowLevel.legendreALFDeriv` provides dPbar/dϕ through a frozen identity — the (n,m) coefficient pattern (with √2 corrections at m=0 and m=1 from the normalization of $Pbar_{n0}$/$Pbar_{n1}$) was calibrated by least squares against numerical differentiation and then frozen; residuals sit at the finite-difference floor (~1e-9). `shLowLevel.shSynthesisDeformation` / `g.deformation` evaluate all three components on grids or GNSS station lists (Mode="points", LatType="geodetic"); north/east validated point-wise against brute-force gradients to ~5e-10. Degree 0 is excluded; degree 1 is opt-in and only meaningful with consistent geocenter (CF/CE frame) handling; all three Love-number sets must come from ONE loading model. East is NaN at the poles (undefined direction).

6. Analysis (the inverse problem)

`shLowLevel.shAnalysisGrid` / `shCoefficients.analysis` estimate coefficients from data. Two regimes:

- **Ring grids** (uniform full-circle longitudes, arbitrary latitude rings): an FFT along longitude decouples the normal equations per order; each order solves a small least-squares over latitude rings. Exact recovery of band-limited input to ~8.7e-15 when the sampling theorem is met ($n_{lat} \geq n_{max}+1$, $n_{lon} \geq 2n_{max}+1$).
- **Scattered points**: chunked accumulation of full normal equations. Under-determined or ill-conditioned samplings (polar gaps, regional patches) need Kaula regularization: a diagonal prior

$$\sigma^2_{\mathrm{prior}}(n) = \left(\frac{K}{n^2}\right)^2$$

stabilizes the inversion at the cost of damping unconstrained coefficients toward zero. For regional data the mathematically cleaner route is Slepian localization (Section 12) — estimate only the ~N concentrated linear combinations the data can actually see.

7. Destriping and Gaussian smoothing

GRACE monthly solutions carry correlated north–south stripes: within one order and parity, coefficients of successive degrees correlate. The classic Swenson–Wahr destriper (`shLowLevel.shDestripe`) fits and removes a low-order polynomial across same-parity degree series per order (defaults: orders ≥ 8, polynomial order 2, window 5, all configurable). Gaussian smoothing (`shLowLevel.shGaussianFilter`) multiplies degree-wise by Jekeli weights from the recursion

$$b = \frac{\ln 2}{1-\cos(r/R)}, \qquad W_0 = 1, \qquad W_1 = \frac{1+e^{-2b}}{1-e^{-2b}} - \frac{1}{b}, \qquad W_{n+1} = -\frac{2n+1}{b} W_n + W_{n-1}$$

with the half-response radius r as the single tuning knob. Both are provided primarily as the baseline against which tvANS and DDK are judged.

8. DDK anisotropic decorrelation (v2.2)

The DDK filters (Kusche 2007; Kusche et al. 2009: DDK1–DDK8, ordered strong→weak) replace the scalar Gaussian weight by a full matrix per (order, C/S) block, derived from a GRACE error covariance and a signal prior — anisotropic, order-dependent smoothing that suppresses stripes with less signal attenuation than an equivalent Gaussian. `shLowLevel.readDDK` parses the released binary Wbd files natively (Rietbroek/Kusche BIN format; endian autodetect; verified to 4.4e-16 against the repository-documented DDK3 reference values; DDK1..8 = a_1d14p_4, 1d13, 1d12, 5d11, 1d11, 5d10, 1d10, 5d9; Nmax= truncates an Lmax-120 filter to n96 series), plus .mat containers and an ASCII exchange layout; `applyDDK` applies block-wise c → M·c, passing uncovered degrees through unchanged. Formal errors are invalidated by the filter (coefficients become correlated); propagate uncertainties with `shLowLevel.mcPropagate`.

Note. The real DDK3 file (Wbd_2-120.a_1d12p_4, MIT-licensed, from github.com/strawpants/GRACE-filter) ships in tests/test_data; the unit tests pin the parser to the repository-documented pack values. Degrees below the filter's Lmin (2) pass through unfiltered, as in the reference implementation.

9. The tvANS filter (time-variable anisotropic noise/signal Wiener filter)

The toolbox's flagship decorrelation filter. Model per month t: x = signal + noise with signal covariance S (estimated from the data themselves) and noise covariance s_tN (a shape N, e.g. from a SINEX covariance or the built-in order/parity model, scaled monthly by VCE factors s_t). The Wiener estimate of the residual field is

$$\hat{x}_t = W_t\, x_t, \qquad W_t = S\,(S + s_t N)^{-1}$$

Implementation: one generalized symmetric eigendecomposition gives, for every month simultaneously,

$$SU = NU\Lambda \;\Rightarrow\; W_t = V\,\mathrm{diag}\!\left(\tfrac{\lambda_i}{\lambda_i + s_t}\right)U^{\mathsf{T}}, \qquad V = U^{-\mathsf{T}}$$

— O(P²) per month instead of a fresh P³ solve. The deterministic model (bias/trend/annual/semi-annual) is removed first and restored after filtering; VCE factors are estimated from high-degree residuals via a robust median χ^2 estimator (`shLowLevel.vceRescale`). With a block-diagonal N (order/parity blocks), the whole machinery runs per block (Blocks="auto"), tractable to Lmax ≈ 120. External noise models join the block path when they are block-diagonal in the (order, C/S, parity) partition (verified, not assumed — v2.2.2); dense covariances fall back to the full path (auto) or error loudly (on).

Per-order-band VCE (v2.2). One s_t per month is coarse: striping strength varies with order. VCEBands=[0 16 33 61] estimates independent monthly factors per order band (block path only — band-wise scaling breaks the single global eigendecomposition, which the code asserts). The per-block factors are honored consistently in the filter application, the posterior sigmas, and the basin deconvolution covariance (identity of the banded block filter with the directly constructed W validated to 5.6e-16).

Posterior uncertainty. The filtered-residual variance

$$\mathrm{diag}\big((I - W_t)\,S\big) = (V \circ V)\,\tfrac{s_t\lambda}{\lambda + s_t}$$

(validated 3.5e-16) is exposed as per-coefficient/month sigma stacks (`tsF.sigmaCs/Ss`). With hard constraints (v2.5) the EXACT diagonal of (W–I)S(W–I)' + s·WNW' is used (O(P²q) in the eig basis, validated 2e-15 against the brute-force covariance; the constrained-direction identity Ac'·Cov·Ac = s·Ac'NAc holds exactly). Honest correction: the earlier formula UNDERESTIMATED along constrained directions (ratios down to ~0.72) — the pre-v2.5 'upper bound' note had the direction wrong.

10. Basin averages and deconvolution

A basin kernel b (Section 11) gives the filtered basin series $a_t = b^{\mathsf{T}} x_t / (b^{\mathsf{T}} b)$. Because filtering attenuates and leaks, `shLowLevel.basinDeconvolve` solves the small linear system that undoes the filter's action on a set of basin kernels B:

$$c_t = A^{-1} B^{\mathsf{T}} \hat{x}_t, \quad A = B^{\mathsf{T}} W_t B \;\; (K \times K); \qquad \mathrm{cov}(c_t) = A^{-1}\big[G_v^{\mathsf{T}}\,\mathrm{diag}(s_{\mathrm{vec}}\, g^2)\, G_v\big] A^{-\mathsf{T}}$$

where g are the spectral gains and s_{vec} the (possibly band-wise) monthly noise factors — the v2.2 form is exactly the v2.1 formula when bands are off. Ill-conditioned kernel sets accept a relative ridge (λ = Ridge·||A||), with the covariance consistently using the regularized inverse. Since v2.5 the basin sigma also carries the OLS parameter uncertainty of the restored deterministic part (design leverage h_t × per-coefficient residual variance, carried on the operator; Monte-Carlo: emp/pred 1.18 without the term, 1.00 with it, 1.26 → 1.08 at seasonal leverage peaks), and with constraints the deconvolution noise covariance uses the exact constrained operator. The deterministic-part errors remain correlated across epochs through the shared fit (pointwise sigma; documented).

11. Basin kernels: buffering and tapering (v2.2)

`shLowLevel.basinKernel` builds b = Y^T(w·mask) by exact Gauss–Legendre quadrature from a polygon, mask, or f(lat,lon) handle. Two leakage controls:

- **BufferKm**: grow/shrink the region by exact great-circle distance before quadrature (trade leakage-in against leakage-out; computed in latitude chunks, practical to Lmax ~ 96).
- **TaperKm**: multiply the kernel spectrally by Jekeli Gaussian weights — soft edges suppress the ringing of the truncated indicator. Rule of thumb: taper radius ≈ the coefficient filter's half-response radius.

The quadrature area fraction is returned; b(1) equals it exactly (Y_{00} = 1), which the tests pin down.

12. Slepian localization (v2.2)

For a region R, the concentration problem maximizes the energy ratio over band-limited g = Yx:

$$\lambda = \frac{\int_R g^2\, d\Omega}{\int_\Omega g^2\, d\Omega} \;\rightarrow\; \max, \qquad K = Y^{\mathsf{T}} \mathrm{diag}(w \cdot \mathrm{mask})\, Y, \qquad N_{\mathrm{Shannon}} = \sum_i \lambda_i = \frac{A_R}{4\pi} P$$

Eigenvectors (Slepian tapers) are orthonormal; eigenvalues are concentrations in [0,1]; the Shannon number (exact by the addition theorem — the trace identity the tests verify) counts the tapers the region supports. Regional analysis then estimates ~N Slepian coefficients a = G^Tx instead of P Stokes coefficients — well-posed by construction, no Kaula prior needed. Validated on the 30° polar cap: λ bounds to 4e-16, orthonormality 3e-15.

13. Climatology, alias periods, AR(1) significance (v2.2)

`ts.climatology()` fits bias, trend, annual and semi-annual terms (plus optional extra periods) per coefficient. GRACE tidal aliasing puts S2 at 161 d, K2 at 3.66 yr, K1 at 7.48 yr: Periods=[161/365.25, 3.66, 7.48] keeps these out of trend and semi-annual estimates. Per-coefficient 1-sigma significance comes from the OLS covariance (weighted and robust variants available) and rides on every component accessor, enabling significance-masked maps.

AR(1) correction. GRACE residuals are temporally correlated; white-noise sigmas are optimistic. ARCorrect=true estimates the lag-1 residual autocorrelation r_1 per coefficient and inflates sigmas by

$$\sigma_{\mathrm{AR(1)}} = \sigma \sqrt{\frac{1+r_1}{1-r_1}}$$

— the standard first-order effective-sample-size correction, since v2.5 applied to the Kendall bias-corrected $r_1' = r_1 + (1+3r_1)/T$. Monte-Carlo validation: uncorrected sigmas underestimate the empirical trend scatter by 2.00× (ϕ = 0.6, T = 120); with the raw-r_1 inflation the ratio is 1.065, with the bias-corrected r_1 1.028 (1.152 → 1.079 at T = 60). Still first-order and slightly conservative — the residual few percent stems from the regression absorbing low-frequency noise.

14. GIA correction (v2.2)

Secular Stokes trends mix present-day mass change with glacial isostatic adjustment. `ts.removeGIA(gia)` subtracts rate·(t - T_0) per epoch; `clim.removeGIA(gia)` corrects a fitted trend directly — both routes agree identically (tested to 1e-13). GIA models (ICE-6G_D, Caron et al., A/W13, ...) are user-supplied gfc rate files, never embedded. The model is treated as exact: its uncertainty, often the dominant regional trend error (Laurentia, Fennoscandia, Antarctica), is not propagated — difference several models as a spread estimate.

15. Geocentric vs. geodetic latitude (v2.2)

All SH mathematics here runs in geocentric latitude; maps, GIS products and mascon grids are geodetic:

$$\tan\varphi' = (1-f)^2 \tan\varphi$$

with the WGS84 flattening as overridable default. The difference peaks at 0.192° (~21 km) near 45° — large enough to visibly bias basin averages and point comparisons. Synthesis, analysis and shAnalysisGrid accept LatType="geodetic" and convert internally.

16. The degree-1 / C20 completion chain

GRACE does not observe geocenter motion (degree 1), and its C20 is weak: the standard product chain is GSM + TN-14 (C20/C30 from SLR) + TN-13 (degree-1). `shLowLevel.readTN14` and `shLowLevel.readTN13` parse the official technical notes (TN-13 verified against the real GFZ RL06 file: 256 paired months 2002-04 to 2026-04, including the 2017.4–2018.4 gap); `ts.addDegree1` and the C20/C30 replacement close the chain with epoch

matching and provenance notes in the history.

17. Normal field and (GM, R) conventions (v2.4)

Geoid maps need the DISTURBING field: subtract the ellipsoidal normal potential. `shLowLevel.normalFieldCS` computes the even zonals from the WGS84/GRS80 defining constants (GM, a, f, ω) via the Heiskanen–Moritz closed form (q_0 series → J_2 → J_{2n}) — no coefficient tables; validated against NIMA TR8350.2 to all published digits. Crucially, WGS84 GM (3.986004418e14) and a (6378137.0) DIFFER from the ICGEM conventions (3.986004415e14, 6378136.3): `g.subtractNormalField` rescales the ellipsoid field to the model's (GM, R) first via

$$\bar{C}'_{nm} = \bar{C}_{nm}\,\frac{GM_1}{GM_2}\left(\frac{R_1}{R_2}\right)^n$$

(`shLowLevel.rescaleGMR` / `g.toReference`; the physical field is invariant — verified by synthesizing the potential at a common radius from both representations). Skipping the rescaling costs millimetres. The arithmetic operators refuse GM/R-mismatched operands (shCoefficients:constantsMismatch) instead of silently mixing conventions. Permanent tide: the ellipsoid is tide-free by construction; the model's C20 tide system (ICGEM header) is NOT converted silently — zero-tide vs tide-free differ by ~4.2e-9 in $\bar{C}_{20}$.

18. Multi-center combination (v2.4)

Several centers observe the same monthly field with different noise. One variance factor per (center, month) — the COST-G granularity — estimated by Foerstner iteration with PARTIAL redundancies:

$$\hat{x} = \left(\sum_c w_c N_c^{-1}\right)^{-1} \sum_c w_c N_c^{-1} x_c, \qquad s_c^2 = \frac{v_c^{\mathrm{T}} N_c^{-1} v_c}{r_c}, \qquad r_c = P - \mathrm{tr}\left(H\, w_c N_c^{-1}\right)$$

Without r_c the factors bias low (v_c is correlated with x■). Noise shapes N_c come from each center's own climatology residuals as (order, C/S, parity) blocks; the whole iteration runs block-wise. Robust=true replaces the global ratio by the median of per-block unbiased estimates q_b/r_b. The empirical shapes are trace-normalized, so the factors carry each center's absolute noise power and are directly comparable (weights invariant; v2.4.1). Two limits are structural: common-mode errors (shared K-band data, AOD1B, tides) are invisible to inter-center VCE — posterior sigmas are a LOWER bound — and inter-center correlations violate independence; info.interCenterCorr quantifies the optimism. Combine GSM before TN-14/TN-13, then apply the replacements once.

19. Elastic sea-level fingerprints (v2.4)

A melting ice mass does not raise the ocean uniformly: the ocean surface follows the perturbed geoid minus the deformed crust, under global mass conservation (Farrell & Clark 1976, elastic limit):

$$S = \mathcal{O}\,(N - U + \kappa), \qquad \int_{\mathrm{ocean}} \rho_w\, S\, dA = -\int_{\mathrm{land}} \sigma\, dA$$

`shLowLevel.seaLevelFingerprint` iterates this on the toolbox's exact Gauss–Legendre quadrature pair. Validation: mass conserved to machine precision, ~11 iterations, and the classic pattern — sea level FALLS near the melting mass (lost self-attraction plus rebound; validation run: near field -0.3 mm vs far field +0.8 mm about a +0.6 mm eustatic mean per unit load). Elastic only, no rotational feedback, fixed coastlines.

20. Gravity gradient tensor (v2.4)

`shLowLevel.shSynthesisGradientTensor` synthesizes all six NEU components at altitude; second Legendre derivatives come from the frozen first-derivative stencil applied twice (exact identity). Every call self-checks the Laplace invariant trace(G) = 0, validated at 7e-16 of max|G|; G_{uu} is bit-consistent with the 'gravity_gradient_rr' kernel route. 1 Eotvos = 1e-9 s^{-2}.

Part II — Best practices

Processing chain order

The order below is not arbitrary; each step assumes the previous ones:

- **1.** Read GSM monthlies (shSeries.read, gz transparent, ICGEM 1.0/2.0 incl. gfct).
- **2.** Replace C20 (and C30 in the accelerometer-degraded years) from TN-14; add degree-1 from TN-13 (addDegree1). Do this BEFORE filtering — filters see the stripes, not the low-degree completions.
- **3.** Bridge the GRACE↔GRACE-FO gap explicitly: dropNaN / select; never fit a climatology across NaNs.
- **4.** Filter: tvANS (or DDK for benchmarking; Gaussian as baseline).
- **5.** Remove GIA (trend work only).
- **6.** Fit climatology with tidal alias periods and ARCorrect=true; or form basin series with deconvolution.
- **7.** Synthesize maps / averages; propagate sigmas (analytic where derived, mcPropagate everywhere else).

Choosing a filter

situation	recommendation
global EWH maps, standard use	tvANS (Blocks="auto"); Gaussian 300 km only as sanity baseline
cross-paper comparability	add DDK3-DDK6 variants via readDDK (binary Wbd, Nmax=96) + applyDDK
strong order-dependent striping	tvANS with VCEBands=[0 16 33 61] (v2.2)
small basins (< ~200,000 km²)	expect leakage; taper kernels, consider basinDeconvolve with several neighboring kernels
regional studies	Slepian basis instead of filtering a global field (Part III, G4)

Uncertainty accounting — what is honest and what is approximate

- Formal per-coefficient sigmas (readers) → exact as published.
- tvANS posterior sigmas → exact for the unconstrained AND (v2.5) the constrained filter.
- Basin sigmas → noise part consistent with (banded) VCE and the exact (constrained) operator; deterministic-fit parameter uncertainty included since v2.5 (pointwise; epoch-correlated through the shared fit).
- Climatology sigmas → OLS/weighted/robust; AR(1) correction first-order with Kendall-corrected r_1 (v2.5; residual ~3% optimism from regression absorption).
- GIA → model treated exact; use model spread.
- Multi-center combination → posterior sigmas are a LOWER bound (common mode invisible; see info.interCenterCorr).
- DDK-filtered fields → no analytic sigmas; use mcPropagate.
- Anything else → mcPropagate: independent-sigma or full SINEX covariance sampling validates every analytic formula empirically (the toolbox's own tests do exactly this).

Validation habits the toolbox is built around

- Every numerical method was validated in Python (numpy/scipy) before the MATLAB implementation; the validation numbers are quoted in the help texts. Keep the habit for your own extensions.
- Ring-grid analysis↔synthesis roundtrips are exact to ~1e-14: use them as a self-test on your grids (Part III, G1 step 5).
- Real provider files beat synthetic fixtures: the test suite auto-discovers any *.gfc / *.snx(.gz) you drop into tests/test_data and sanity-checks them (large SINEX are streamed, v2.2).
- runAllTests is the gate: run it after every toolbox update; perf_log.csv accumulates machine-local timing history.

Common silent errors this toolbox guards against

error	guard
geodetic latitudes fed to SH math (~21 km bias)	LatType="geodetic" + conversion helpers (v2.2)
hardcoded Love numbers	kn is a required explicit input for EWH
climatology across the 2017-2018 gap NaNs	assertClean errors loudly; dropNaN/select

error	guard
trend maps without GIA	removeGIA one-liner (v2.2)
optimistic trend sigmas	ARCorrect=true (v2.2)
tidal alias leakage into trends	Periods=[161/365.25, 3.66, 7.48]
zero-variance rows collapsing the noise floor	positive-diagonal floor + shLowLevel:buildNoiseCov:allZeroVariance
8-column ICGEM 1.0 acos/asin ambiguity	EIGEN-style period read; documented in README
sine sectorals S_{nn} blanked in difference triangles	fixed v2.4.2; rendered-CData regression test

Part III — Step-by-step guides

G1. Monthly EWH anomaly maps from ITSG files

```
% 1. read the monthly GSM series (gz ok, ICGEM 1.0/2.0)
ts = shSeries.read("data/ITSG-Grace2018_n96_*.gfc");

% 2. low-degree completion BEFORE filtering
tn14 = shLowLevel.readTN14("TN-14_C30_C20_SLR_GSFC.txt");
ts   = ts.applyTN14(tn14);                  % C20 (+C30 where flagged)
tn13 = shLowLevel.readTN13("TN-13_GEOC_GFZ_RL06.txt");
ts   = ts.addDegree1(tn13);

% 3. bridge the mission gap explicitly
ts = ts.dropNaN();

% 4. anomalies w.r.t. a mean field, then filter
ts  = ts - ts.mean();                       % anomalies
tsF = ts.filter("tvANS", Blocks="auto");    % or .gaussian(300)

% 5. EWH synthesis (Love numbers user-supplied!)
kn  = readmatrix("loadLoveNumbers.txt");    % your set, degrees 0..nmax
lat = -89.5:1:89.5; lon = 0.5:1:359.5;
ewh = tsF.at(24).synthesis(lat, lon, quantity="ewh", kn=kn);  % [m]

% self-test: analysis roundtrip on a ring grid should be ~1e-14
gChk = shCoefficients.analysis(ewh, lat, lon, tsF.nmax, ...
          quantity='ewh', kn=kn);
```

Interpretation: values are equivalent water height anomalies relative to the removed mean, in meters. The FFT synthesis path engages automatically on this uniform grid; check `tsF.at(24).history` for the full provenance chain.

G2. Basin time series with uncertainties

```
idx = shLowLevel.shIndex(ts.nmax, MinDegree=2);
% kernel from a polygon [lat lon], buffered + tapered (v2.2)
amazon = readmatrix("amazon_polygon.txt");
[b, info] = shLowLevel.basinKernel(idx, amazon, BufferKm=150, TaperKm=350);

% filtered series + filter operator
[tsF, op] = ts.filter("tvANS", Blocks="auto", VCEBands=[0 16 33 61]);

% deconvolved basin average with 1-sigma (banded-VCE-consistent, v2.2)
[avg, out] = shLowLevel.basinDeconvolve(b, op);
shLowLevel.plotBasinSeries(op.tYears, avg(1,:)', out.sigma(1,:)', Units="cm");
```

The deconvolution undoes the filter's attenuation of the kernel; the sigma comes from the posterior covariance with per-band noise factors. For several basins pass a kernel matrix B — the joint solve also redistributes mutual leakage. (Edition-2 erratum fixed: the signature is `basinDeconvolve(B, op)`, epochs live in `op.tYears`, averages in the first output.)

G3. Significance-masked trend maps (GIA- and AR(1)-honest)

```
gia = shCoefficients.read("ICE-6G_D_rates.gfc");       % rate [1/yr]
tsG = tsF.removeGIA(gia);
clim = tsG.climatology(Periods=[161/365.25, 3.66, 7.48], ...
                       ARCorrect=true);
tr   = clim.trend();                        % shCoefficients WITH sigmas
kn   = readmatrix("loadLoveNumbers.txt");
rate = tr.synthesis(lat, lon, quantity="ewh", kn=kn);         % [m/yr]
mc   = shLowLevel.mcPropagate(@(g) g.synthesis(lat, lon, ...
          quantity="ewh", kn=kn), tr, N=300, Seed=1);
sigma = mc.sigma;
mask = abs(rate) > 2 * sigma;               % 2-sigma significance
rate(~mask) = NaN; imagesc(lon, lat, rate);
```

Compare against a second GIA model (Caron) and show the difference as the model-spread contribution to the error budget — usually dominant over formal errors in formerly glaciated regions.

G4. Regional analysis with Slepian functions

```
idx = shLowLevel.shIndex(60, MinDegree=0);
region = @(la,lo) double(la>-90 & la<-60);      % Antarctica-ish belt
[G, lam, infoS] = shLowLevel.slepianBasis(idx, region);
fprintf("Shannon %.1f -> keeping %d tapers\n", infoS.shannon, numel(lam));

x  = shLowLevel.vecFromCS(g.C, g.S, idx);              % one month, idx order
a  = G' * x;                                    % Slepian coefficients
xR = G * a;                                     % regional part of x
[Cr, Sr] = shLowLevel.csFromVec(xR, idx);
gR = shCoefficients(Cr, Sr, GM=g.GM, R=g.R);
```

Estimating a (dimension ≈ Shannon number) instead of all P Stokes coefficients is well-posed on regional data by construction; concentrations λ near 1 tell you which tapers the region genuinely constrains.

G5. Working with released covariances (SINEX)

```
idx = shLowLevel.shIndex(96, MinDegree=2);
% full read (monthly solutions to ~30 MB): covariance in idx order
snx = shLowLevel.readSINEX("ITSG-..._n96_2008-04.snx.gz", ...
                    Output="covariance", Index=idx);
N   = snx.M;                                    % noise shape for tvANS
[tsF, op] = ts.filter("tvANS", NoiseCov=N, Blocks="off");

% multi-100-MB NEQ SINEX: stream just the estimates in seconds (v2.2)
est = shLowLevel.readSINEX("ITSG-..._operational_n96_2020-01.snx.gz", ...
                    Only="estimate");

% propagate the full covariance through ANY functional (v2.2)
out = shLowLevel.mcPropagate(@(gs) b' * shLowLevel.vecFromCS(gs.C,gs.S,idx), ...
                      g0, Cov=snx.M, Idx=idx, N=2000);
```

G6. Comparing against mascons

```
mas = shLowLevel.readMascon("GRCTellus.JPL...MSCNv03CRI.nc");
% mascon lats are GEODETIC -> one switch avoids the 21-km bias (v2.2)
kn = readmatrix("loadLoveNumbers.txt");
k  = find(abs(mas.epoch - 2010.29) < 0.05, 1);     % match an epoch
ewhSH = tsF.at(k).synthesis(mas.lat', mas.lon', ...
          quantity="ewh", kn=kn, LatType="geodetic");  % [m]
d = 100*ewhSH - mas.ewh(:,:,k);                    % mascon units: cm
fprintf("RMS diff %.2f cm\n", sqrt(mean(d(:).^2, 'omitnan')));
```

Expect coastal and small-basin differences: mascons carry their own regularization. Compare basin-integrated series rather than pointwise fields for a fair statement; use the same GIA convention on both sides (JPL mascons ship GIA-corrected — remove GIA from the SH side too).

G7. GNSS loading comparison (v2.3)

```
% residual monthly series (mean removed), filtered
ts  = ts - ts.mean();                       % anomalies
tsF = ts.filter("tvANS", Blocks="auto");

% load Love numbers k', h', l' - ONE consistent model, user-supplied;
% v2.5 reader: layouts, sparse pchip interpolation, CF-frame conversion
LN = shLowLevel.readLoveNumbers("loadLoveNumbers_PREM.txt", ...
    Columns="n k h l", MaxDegree=tsF.nmax, Interp="pchip", ...
    InFrame="CE", OutFrame="CF");               % match TN-13 (CF)
kn = LN.kn; hn = LN.hn; ln = LN.ln;

% GNSS stations (geodetic coordinates!)
sta = readmatrix("stations.txt");               % [lat lon]
T = tsF.nEpochs;
up = zeros(size(sta,1), T); no = up; ea = up;
for k = 1:T
    [up(:,k), no(:,k), ea(:,k)] = tsF.at(k).deformation( ...
        sta(:,1)', sta(:,2)', kn=kn, hn=hn, ln=ln, ...
        Mode="points", LatType="geodetic");
end
% compare up(i,:) against the GNSS vertical of station i (same epochs);
% typical GRACE-elastic amplitudes: mm to ~1 cm vertical, ~1/3 horizontal
```

Degree 1 is excluded by default; include it only when both the GRACE side (TN-13 in the CF frame) and the Love numbers use the same frame convention. The elastic prediction excludes GIA and poroelastic effects — detrend both series or remove GIA explicitly before comparing trends.

G7b. Data management (v2.4.1, temporal catalogue v2.4.2)

```
shLowLevel.dataFolder("D:/geodata/shLowLevel");      % once; persists (getpref)
shLowLevel.fetchDDK(1:8);                    % all released DDK filters
W5 = shLowLevel.readDDK("DDK5");             % by name, from the data folder
T = shLowLevel.listICGEM();                  % 180+ static models as a table
f = shLowLevel.fetchICGEM("EGM2008");        % -> dataFolder/icgem/EGM2008.gfc
g = shCoefficients.read(f);
shLowLevel.fetchITSG(2010:2016);             % -> dataFolder/itsg_series
shLowLevel.fetchITSG("2008-04", Product="daily"); % Kalman daily n40 (v2.4.1)

% FULL temporal catalogue: all ~70+ series, all centers
Tt = shLowLevel.listICGEM(Type="temporal");
% fetch a WHOLE monthly series (v3.1.1) - one row of that catalogue
fs = shLowLevel.fetchICGEM(17, Type="temporal");
ts = shSeries.fromFolder(fileparts(fs(1)));   % straight into a series
```

The ICGEM static catalogue is parsed from icgem.gfz.de (fixture-tested offline). The temporal catalogue lists every series of every center — groups 01_GRACE, 02_COST-G, 03_other, 04_SLR — with columns group, center, series, path, url and zip. Since v3.1.1 a temporal row is not just browsable but fetchable: `fetchICGEM(idx, Type="temporal")` downloads the whole series into `<dataFolder>/icgem/series/<group>_<center>_<series>/`, which `shSeries.fromFolder` and `shLowLevel.standardChain` consume directly. For ITSG monthlies `shLowLevel.fetchITSG` remains the convenient route.

Mode: one request, not three hundred

The default `Mode="auto"` takes the server's whole-series ZIP in a SINGLE request and unpacks it. This is not an optimization but a courtesy requirement: icgem.gfz.de rate-limits, and fetching a 283-file series one file at a time earns HTTP 429 and a tarpit (field-observed as "too many connections"). If the archive is unavailable the fetcher falls back to per-file mode automatically — resumable, throttled with `Pause=` and `Retries=` exponential backoff, each file verified by parse before it replaces anything. `Mode="archive"|"files"` force either path; `info.mode` reports which one actually ran, so a script can tell a fresh download from a no-op. `Files=` filters the series, and `FileList=` injects a catalogue table for offline mirrors and subsets.

```
% forced per-file, filtered to a window, gentle on the server
fs = shLowLevel.fetchICGEM(17, Type = "temporal", Mode = "files", ...
        Files = "*2008*.gfc", Pause = 3, Retries = 3);
[ts, rep] = shLowLevel.standardChain(fileparts(fs(1)), Filter = "DDK3");
```

G8. Multi-center combination (v2.4)

```
shLowLevel.fetchITSG(2010:2016);             % download once (websave)
tsI = shSeries.fromFolder("ITSG/");    % GSM level, same nmax
tsC = shSeries.fromFolder("CSR/");
tsG = shSeries.fromFolder("GFZ/");
[tsComb, info] = shLowLevel.combineCenters({tsI, tsC, tsG}, Robust=true);

figure; plot(info.epochs, info.weights');     % weight time series
legend(info.centers); ylabel("relative weight");
figure; imagesc(info.interCenterCorr);        % honesty diagnostic
% THEN degree-1 / C20-C30 replacements, once:
tsComb = tsComb.applyTN14("TN-14.txt");
tsComb = tsComb.addDegree1("TN-13.txt");
```

Watch the weight series for regime changes (single-accelerometer era, GRACE-FO transition) and treat posterior sigmas as lower bounds — the common mode is invisible.

G9. Sea-level fingerprint (v2.4)

```
idx = shLowLevel.shIndex(96, MinDegree=0);
LN = readmatrix("loadLoveNumbers.txt"); kn = LN(:,2); hn = LN(:,3);
greenland = @(la,lo) -280 * inGreenland(la, lo);   % kg/m^2/yr, say
oceanF = @(la,lo) oceanMaskFun(la, lo);            % 1 over ocean
[S, grid, info] = shLowLevel.seaLevelFingerprint(greenland, oceanF, idx, ...
    kn=kn, hn=hn);
shLowLevel.plotSHMap(info.S2D / info.eustatic, grid.latDeg, grid.lonDeg, ...
    Units="S / eustatic", Title="Greenland fingerprint");
```

Plot S normalized by the eustatic mean: near-field values drop below zero, far-field plateaus around 1.1–1.3 — if not, check the ocean mask and that idx uses MinDegree = 0.

G10. The standard chain in one call

```
% the whole post-processing chain, in the ONE correct order
gia = shCoefficients.read("ICE-6G_D_trend.gfc");   % a TREND field [1/yr]
[ts, rep] = shLowLevel.standardChain("D:/grace/monthly", ...
    TN14File = "TN-14_C30_C20_SLR_GSFC.txt", ...   % TN14 itself is a FLAG
    Degree1 = "CSR", ...                           % provider, not a path
    GIA = gia, Filter = "DDK3");
disp(rep.steps')          % what was applied, in order, with provenance
```

Order is not a matter of taste here. The SLR C20/C30 replacement and the degree-1 restoration must happen on the UNFILTERED coefficients — a filter mixes degrees, so replacing a coefficient afterwards inserts an unfiltered value into a filtered field and the low degrees no longer mean anything consistent. GIA is a trend in the same (corrected) coefficients, so it comes next; filtering is last. `standardChain` encodes exactly that order and returns REP, a provenance record naming every step, the files it used and the toolbox version that ran — the thing you paste into a paper's data section. Every stage is optional: leave `GIA=` out and the report says so, rather than silently skipping a step you assumed had happened.

Two contracts worth reading before the first call: `TN14` is a logical FLAG and the path goes in `TN14File`, while `Degree1` names the TN-13 provider ("GFZ" | "CSR" | "JPL" | "none") with the path in `Degree1File`; and `GIA` takes an `shCoefficients` TREND field in [1/yr], not a filename — it is applied as (t – GIAEpoch) × GIA. Left at their defaults the files come from the persistent data folder that `setup_shAnalysis` populates.

G11. Designing your own anisotropic filter

```
% a DDK-class filter built from YOUR error covariance
g  = shCoefficients.read("ITSG-Grace2018_n60_2008-04.gfc");
W  = shLowLevel.designFilter(g.sigmaC, g.sigmaS, Kaula = 1e-5);
[Cf, Sf] = shLowLevel.applyDDK(g.C, g.S, W);   % same block format as readDDK
```

The released DDK filters are built from a specific center's normal equations. If you have your own sigmas or a released covariance, `designFilter` builds the same kind of operator, $W = (N + a S^{-1})^{-1} N$, in the block format `readDDK` produces — so it drops into `applyDDK`, `g.applyDDK` and `standardChain(Filter=W)` unchanged. Note that `Noise=` expects a COVARIANCE, not sigmas: passing standard deviations where a variance is expected is the classic way to get a filter that looks plausible and smooths by the square root of what you intended.

G12. Correcting leakage

```
% forward modelling: only the FILTER has to be known
ewh = g.synthesis(-89:89, 0:359, quantity = "ewh", kn = kn);
[m, info] = shLowLevel.leakageCorrect(ewh, -89:89, 0:359, ...
    Filter = "gauss300", Mask = basinMask, Gain = 2);

% or scaling factors, if you trust a model's spatial PATTERN
k = shLowLevel.gridScaling(modelEWH, -89:89, 0:359, Filter = "gauss300");
corrected = k .* ewh;                % NaN where the model is blind
```

A filter removes real signal and spreads the rest across basin boundaries; a 6° disc under a 500 km Gaussian keeps about half its peak. The two corrections differ in what you have to BELIEVE, which is how to choose between them. Forward modelling (`leakageCorrect`) needs only the filter and, usefully, a mask saying where mass can exist — it then has to explain the observation with mass inside that region, so leakage outside is removed rather than redistributed. Scaling factors (`gridScaling`) need a model series, and inherit the correctness of its spatial pattern: the factors are invariant to the model's amplitude but not to its shape.

Two behaviours worth knowing before you read the output. Convergence is judged on the change of the SOLUTION, not on the residual: with a mask the problem is inconsistent, so the residual settles at a floor while the

solution is converged. And `gridScaling` returns NaN where the model carries no signal instead of a ratio of two numerical zeros — a model that does not reach your basin shows up as missing coverage rather than as noise multiplying your data.

G13. What real provider files look like

Format specifications describe what files ought to contain. The reader is written against what they actually contain, and three separate field failures are worth knowing about, because they are silent — you get numbers, just not the right ones.

- **FORTRAN D-exponents.** Providers switch from `1.23E-10` to `1.23D-10` partway through large files (EIGEN-6C4 does it above degree ~370). MATLAB's `str2double` returns NaN for those, so a line-by-line parser quietly corrupts the high degrees of exactly the models where the high degrees are the point. The reader normalizes D/d to E/e before conversion.
- **ICGEM 2.0 column order.** The real-world layout of an `acos`/`asin` line is `... sigC sigS t0 t1 period` — period LAST, verified against CNES/GRGS files. A period-first assumption mis-parses every such file and produces a time-variable model whose seasonal terms are wrong without ever raising an error.
- **Ragged record groups.** EIGEN-5S and 5C carry a single `gfc` line with a trailing epoch among thousands of uniform ones. Bulk `sscanf(txt, '%f', [nc Inf])` does not complain about that — it silently re-flows the whole group by one column and asks for a coefficient matrix indexed by a date. The reader checks rows-out == lines-in per group, subgroups by numeric width when ragged, and keeps an n/m sanity net.

The practical rule: when a new provider or a new release enters your pipeline, read one file and check `g.C(3,1)` and the highest degrees against the provider's own published values before you trust a thousand of them. Bulk parsing is fast — EIGEN-6C4 at 177.7 MB and degree 2190 in about 5 s, a 73.6 MB GRGS mean field with 674k variable terms in about 7 s — but fast and correct are different properties.

Validation against GravIS

This chapter is a worked, reproducible comparison of the toolbox against a PUBLISHED product: the GravIS Greenland ice-mass basin averages (https://gravis.gfz.de/gis). It is written out in full because the numbers only mean something together with the choices behind them — and because two of those choices cost a factor that would otherwise have been mistaken for a bug in the toolbox.

V1. The reference

GravIS publishes seven Greenland drainage basins as ASCII time series in Gt. Summing them and fitting bias, trend, annual and semi-annual gives the reference trend. It must be recomputed over the SPAN YOU ACTUALLY USE: over the full record COST-G RL02 gives –220.8 Gt/yr, but over the shorter span for which the auxiliary correction tables were available it is –231.1 Gt/yr. Comparing against the wrong span is a 5% error before any processing starts.

V2. Matching the GravIS processing

GravIS Level-3 basin averages come from Level-2B coefficients, which are Level-2 plus four corrections (https://gravis.gfz.de/corrections). Applying anything less makes the comparison meaningless. All four are read with `shLowLevel.readSHM` or as plain tables:

```
% (1) their low-degree replacement: C20, C30, C21, S21
% (2) their geocenter (Swenson approximation, NOT TN-13)
% (3) their mean field, 2002/04-2020/03
% (4) GIA: ICE-6G_D (VM5a), a GRDOTA rate model in 1/yr
Mn  = shLowLevel.readSHM("GRAVIS-2B_..._MEAN_2002095-2020091_NFIL.gz", Nmax = 90);
Gi  = shLowLevel.readSHM("GRAVIS-2B_..._GIA_ICE-6G_D_VM5a.gz", Nmax = 90);

ts = shSeries.fromFolder(costgFolder);
ts = ts.select(ts.epochs <= maxTableEpoch + 0.05);   % tables trail the data
% ... insert their C20/C30/C21/S21 and C10/C11/S11 per epoch ...
for k = 1:ts.nEpochs                    % mean field and GIA, per epoch
    Cs(:,:,k) = Cs(:,:,k) - Mn.C - (ts.epochs(k) - 2011) * Gi.C;
    Ss(:,:,k) = Ss(:,:,k) - Mn.S - (ts.epochs(k) - 2011) * Gi.S;
end
ts = ts.removeAlias();                  % (5) the S2 161-day alias
```

V3. The basin average

```
idx  = shLowLevel.shIndex(ts.nmax, MinDegree = 0);
[b, info] = shLowLevel.basinKernel(idx, gisPolygon);   % NaN-separated rings
area = info.areaFraction * 4*pi*6371^2;                % NOT b(1)
kf   = shLowLevel.kernelFactors("ewh", ts.nmax, GM, R, kn = kn);
avg  = (b .* kf(idx.n+1))' * X / (b'*b);               % EWH [m] per epoch
mass = avg(:) * area * 1e6 * 1000 / 1e12;              % -> Gt
```

V4. Results, and what each step is worth

Method	Gt/yr	vs published
naive kernel average	–196.4	–15%
naive grid integral	–170.3	–26%
forward modelling, **Greenland-only** mask	–259.8	+12%
forward modelling, **union** mask	–240.2	+4%
union mask + S2 alias removed	–240.0	+4%
GravIS COST-G RL02, matching span	**–231.1**	—

V4b. Describe the filter the FORWARD MODEL sees

The table above hides the single largest error I made. GravIS basin averages start from UNFILTERED Level-2B coefficients, so I passed `Filter="none"` to `leakageCorrect`. But Dahle et al. (2025), Sect. 2.2.1, list the basin-average steps: spectral masking, then **low-pass filtering with a Wiener optimal filter, approximately a Gaussian of 4° latitude spatial half-width**, then the conversion to surface mass, then the least-squares adjustment. The forward model is filtered IDENTICALLY to the observations — that is the whole point of the method

— so the filter to declare is the Wiener one, not "none".

```
r = 4 * 111.195;                          % 4 deg latitude ~ 445 km
wG = shLowLevel.shGaussianWeights(nmax, r);
obs = shCoefficients(Ct .* wG(:), St .* wG(:)).synthesis( ...
        lat, lon, quantity = "ewh", kn = kn, nmin = 0);
m = shLowLevel.leakageCorrect(obs, lat, lon, Filter = "gauss445", ...
        Mask = unionMask, Gain = 2, MaxIter = 300);
```

Declared filter	Gt/yr	vs published
`"none"` (wrong)	–234.9	+1.6%
`"gauss445"` (Wiener equivalent)	–**227.4**	–**1.6%**
GravIS COST-G RL02, matching span	–**231.1**	—

Declaring the filter correctly turns a 1.6% overshoot into a 1.6% undershoot, and the iteration also drifts less with continued iteration (2.0 Gt/yr from 300 to 900 steps, against 2.5 unfiltered) because the filter regularises the inverse problem. At that point the agreement is as close as this reproduction can meaningfully be read: the remaining difference is smaller than the spread between GravIS product versions and smaller than the effect of the stopping point.

The paper also specifies the mask GravIS uses, which is NOT the bare drainage-basin outline: the spectral mask is 1 until 200 km outside the grounding line and tapers smoothly to 0 by 600 km. That is `basinKernel(..., BufferKm = 200, TaperKm = 400)`, not a hard boundary.

V5. The lessons

The mask must cover every region that can hold mass, not only the one you are measuring. A Greenland-only mask tells the inversion that mass exists nowhere else, so Canadian Arctic, Iceland and Svalbard signal is forced into Greenland: +12% instead of +4%. In the union solution the neighbours absorb –112 Gt/yr — mass the first mask had nowhere to put. The union mask also converges faster. This is a usage rule, not a bug: `Mask=` always accepted a union.

The filter you declare must be the filter the FORWARD MODEL sees, not the filter the input file carries. The inputs here are genuinely unfiltered, and declaring that was still wrong, because the method filters both sides. This cost more than the mask, the polygon resolution and the alias put together.

The S2 alias removal is worth 0.2 Gt/yr on a twenty-year trend — that is, nothing. A 161-day harmonic is very nearly orthogonal to a linear trend over two decades, so removing it barely moves the trend even though it is a real and necessary correction for the MONTHLY series. Reported here because it was expected to explain part of the residual and did not: a hypothesis removed is worth as much as one confirmed.

V6. What the remaining 4% is, honestly

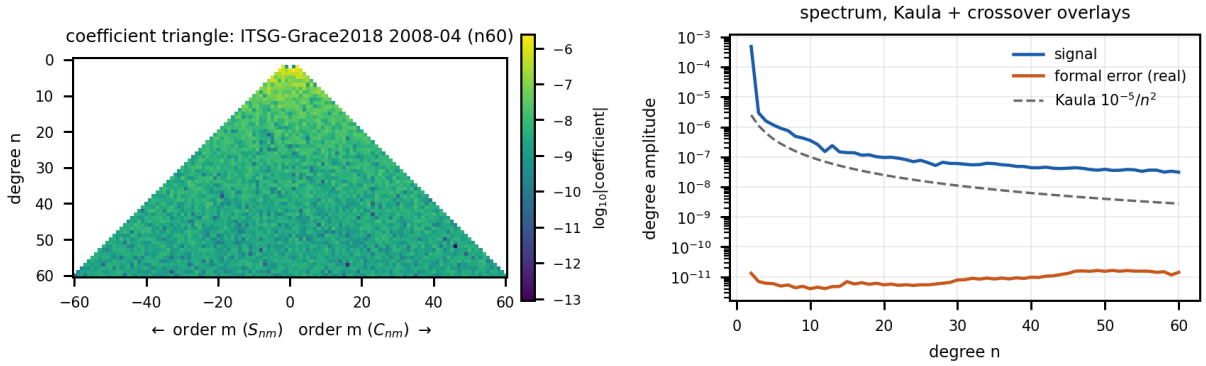
- GravIS inverts SEVEN drainage basins jointly (Sasgen et al. 2012); this recipe constrains one region against coarse neighbour boxes.
- The basin outline used here is decimated to 97.4% of the published area, and the neighbour outlines are rectangles.
- The iteration has not converged: step 3.2e-4 against Tol 1e-4 after 900 iterations, and the estimate was still drifting (–240.2 at 300, –242.5 at 900). Call it ±5 Gt/yr.
- Nothing here was tuned to close the gap. A 4% agreement with an independently produced product, reached by matching its documented processing step for step, is the result; making it smaller by adjusting a mask until the number matched would not be.

Demo gallery

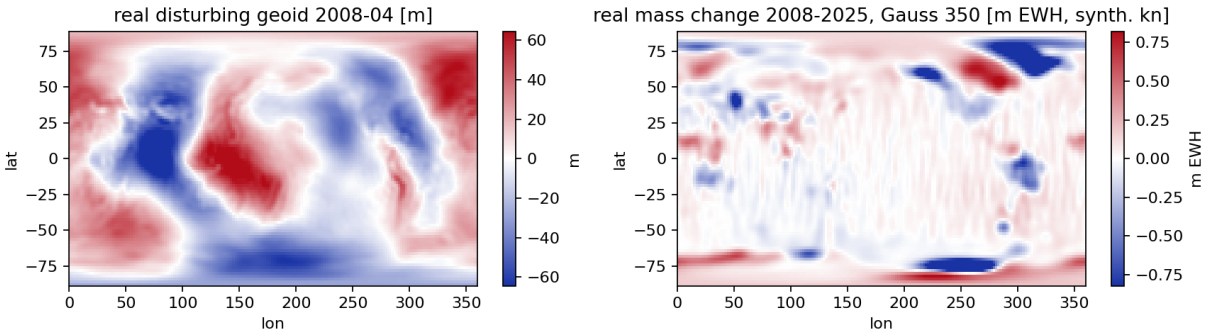
`demo_shAnalysis` is a case registry: `demo_shAnalysis("list")` prints the table below, `demo_shAnalysis("all")` runs everything, `demo_shAnalysis(["D04","D15"])` runs a selection, and Visible=false keeps figures hidden for smoke tests. Real files in tests/test_data (ITSG month, DDK3 Wbd) are used when present; every case falls back to synthetic data. Love numbers in the demos are SYNTHETIC — real work requires a real loading model.

ID	Case	Exercises
D01	Read & spectral diagnostics	shReadGFC, shDegreeRMS, triangle, spectrum
D02	Synthesis quantities & maps	shSynthesis, kernelFactors, plotSHMap, Height
D03	Normal field → geoid	normalFieldCS, subtractNormalField, toReference
D04	Filter comparison	gaussian, fan, destripe, DDK, triangle diff
D05	Climatology & basin series	climatology, basinKernel, plotBasinSeries
D06	tvANS pipeline	buildNoiseCov, tvANSFilter, basinDeconvolve
D07	Analysis (inverse)	shAnalysisGrid, Kaula
D08	Basin tools	basinKernel taper, deconvolve vs scaling
D09	Uncertainty	errorMap, mcPropagate, plotCovariance
D10	Load deformation	shSynthesisDeformation, Mode=points
D11	Gradient tensor	shSynthesisGradientTensor
D12	EOF analysis	eofAnalysis
D13	Trend breakpoints	trendBreaks, F-test
D14	Multi-center combination	combineCenters, VCE weights
D15	Sea-level fingerprint	seaLevelFingerprint, evalMask
D16	Export	writeGrid netCDF, writeAnimation MP4

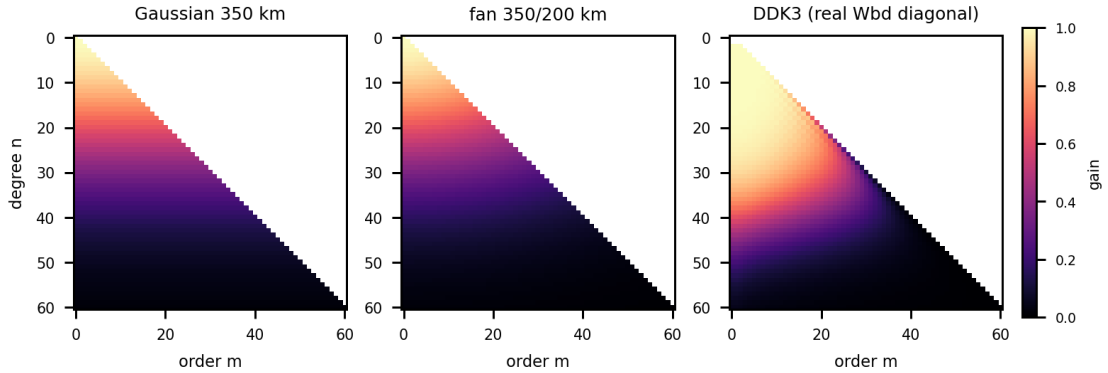
Note. Provenance of the figures below: rendered with the toolbox's Python validation port — the same algorithms that gate every MATLAB implementation, numerically equivalent by construction. D01/D02 use the REAL ITSG files shipped in tests/test_data (and D04 the real DDK3 Wbd binary); the remaining figures use synthetic data with known ground truth, because recovery demonstrations (VCE factors, EOF modes, F-test calibration) require a truth to recover. `shLowLevel.fetchITSG(years)` extends the real-data demos to full monthly series. The MATLAB demo cases produce the interactive originals.



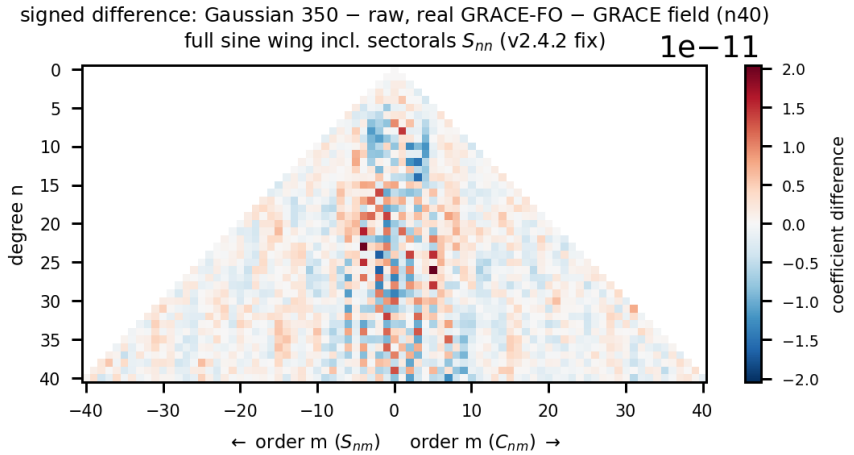
D01 — REAL DATA: the shipped ITSG-Grace2018 2008-04 month (rendered by the rebuilt Python port, v2.5). Coefficient triangle (apex up; C00/C20 removed for display) and the degree-amplitude spectrum with the REAL formal errors from the gfc file. At this n60 truncation the formal errors stay ~3 orders below the signal - the signal/error crossover (the resolution limit) lies beyond the truncation degree; Kaula decay as the yardstick.



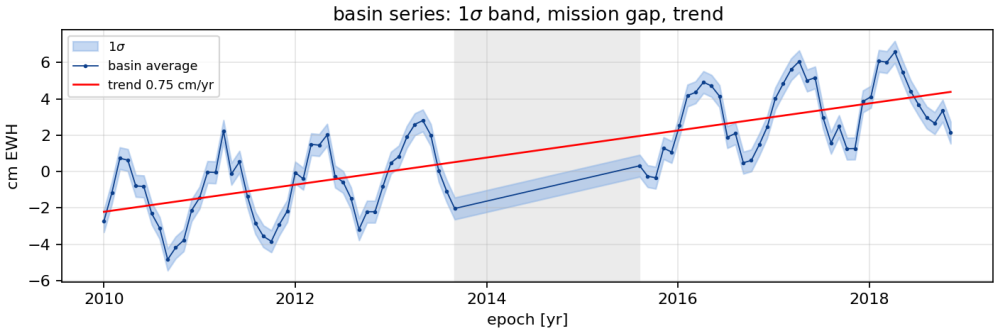
D02 — REAL DATA. Left: the disturbing geoid of 2008-04 after subtracting the WGS84 normal field computed from defining constants and rescaled to the ITSG (GM, R) — the classic ±100 m map (Indian Ocean low, New Guinea high). Right: the real 2008-2025 mass change (GRACE-FO minus GRACE, Gaussian 350 km, EWH): Greenland and West Antarctic mass loss stand out; residual meridional stripes at low latitudes are the real GRACE error structure. Honest caveats: synthetic Love numbers, no GIA correction, degree 1 absent — patterns are real, amplitudes approximate.



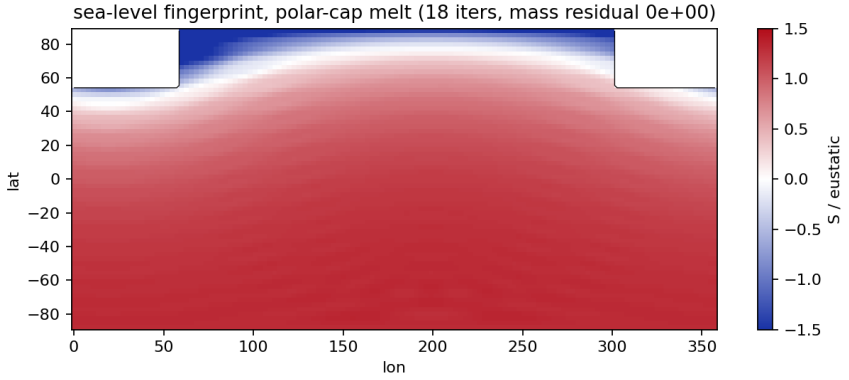
D04 — Filter gain over the (degree, order) triangle. Gaussian: isotropic in degree. Fan (Han): the order Gaussian additionally damps high orders — the striping direction. DDK3 (diagonal of the real Wbd blocks): anisotropic and order-dependent — near-zonal signal survives to n~35 while high orders are suppressed hard; exactly the structure a GRACE error covariance demands.



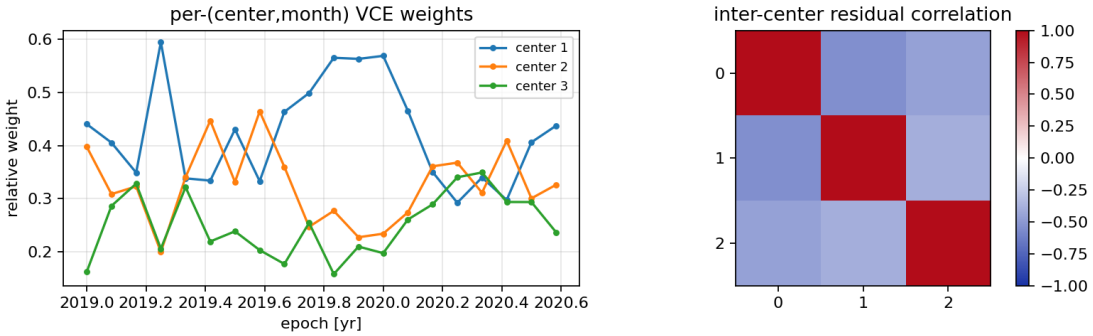
D04b — REAL DATA, v2.4.2 fix: the signed difference triangle (Gaussian 350 – raw) on the real GRACE-FO – GRACE field at n40 — the removed signal concentrates at the low orders where the real stripes and secular signal live. The sine wing now spans the full n ≥ m ≥ 1 support including the sectorals S_{nn}: before v2.4.2 the difference mode blanked the sectoral edge and every diff triangle rendered the left wing one column narrower than the right (regression-tested on the rendered CData since).



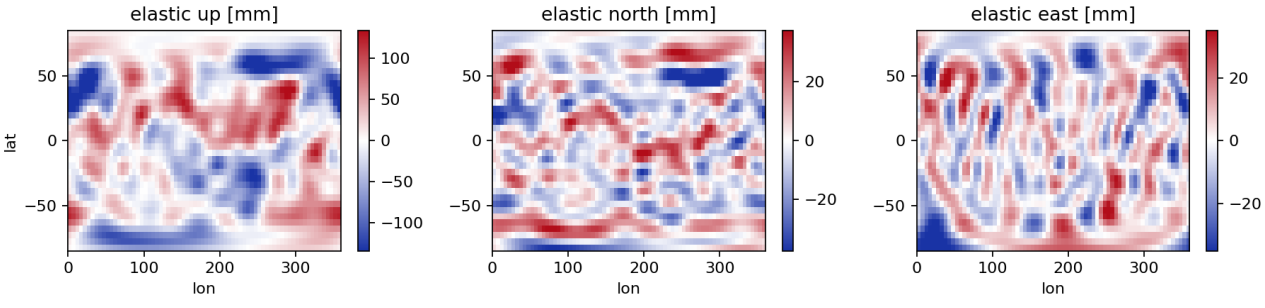
D05/D06 — The standard basin figure from plotBasinSeries: 1-sigma band, grey mission-gap patch, AR(1)-corrected trend annotation. With tvANS + basinDeconvolve the band comes from propagated posterior covariances instead of an empirical scatter.



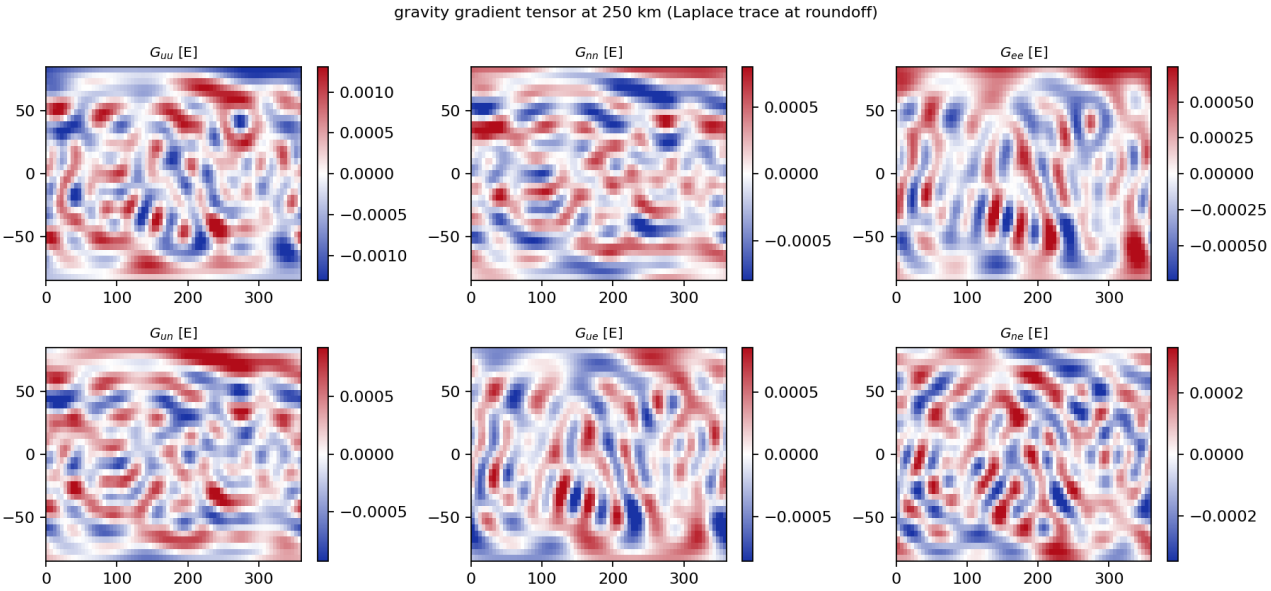
D15 — Sea-level fingerprint of a polar-cap melt, normalized by the eustatic mean: the near field FALLS (blue; lost self-attraction plus crustal rebound), the far field exceeds eustatic by ~20-30%. Mass conserved to machine precision — the printed residual is the built-in check.



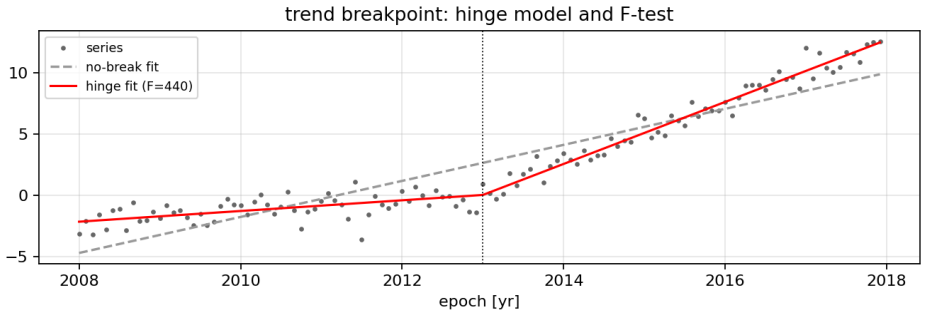
D14 — Multi-center VCE: per-(center, month) weights track the time-varying noise levels; the inter-center residual correlation matrix is the honesty diagnostic — VCE assumes independence, GRACE centers share data and background models.



D10 — Elastic load deformation: up, north, east from the same residual field. Horizontals are ~1/3 of the vertical (validation run: 0.38) and require the exact SH gradient (legendreALFDeriv); east is undefined at the poles.



D11 — All six gradient tensor components at 250 km in Eotvos. The Laplace invariant $G_{nn} + G_{ee} + G_{uu} = 0$ holds at roundoff (4e-15 of max|G| here) — a full-chain check of first and second Legendre derivatives, trig derivatives and frame terms at once.



D13 — Trend breakpoint: continuous piecewise-linear hinge model vs the no-break fit; the F-test (p via betainc, base MATLAB) quantifies the break. Null rejection is calibrated at the nominal level (5.1% at 5%).

Part IV — API reference

Generated from the source: the **arguments blocks are the ground truth** for input names, dimensions, classes and defaults (default 'required' = mandatory positional input); name-value options are passed as Name=Value. Conventions throughout: coefficient triangles are addressed C(n+1, m+1); vectors follow shLowLevel.shIndex ordering (MinDegree = 2 unless stated); latitudes are geocentric degrees; radii in km; epochs in decimal years. Every entry closes with a real-data example from the canonical GRACE chain; all examples are lint- and API-gated at build time.

shCoefficients class

Spherical harmonic (Stokes) coefficient set - single epoch. The single point of access for one gravity field: a GRACE/GRACE-FO monthly solution (GSM), a dealiasing product (GAA/GAB/GAC/GAD), a static model, or any derived field (difference, filtered, climatology component). Immutable value class: every operation returns a NEW object and appends to .history (full provenance).

property	size	type	default	notes
C	—	double	—	
S	—	double	—	
sigmaC	—	double	[]	
sigmaS	—	double	[]	
GM	1 x 1	double	3.986004415e14	
R	1 x 1	double	6378136.3	
epoch	1 x 1	double	NaN	
productType	1 x 1	string	"unknown"	
tideSystem	1 x 1	string	"unknown"	
name	1 x 1	string	""	
header	—	struct	struct()	
variableTerms	—	—	[]	
history	—	string	string.empty(0,1)	
nmax	—	—	—	

shCoefficients.shCoefficients

```
obj = g.shCoefficients(C, S, Name=Value)
```

Construct from coefficient matrices.

input	size	type	default	kind
C	—	double	0	positional
S	—	double	0	positional
SigmaC	—	double	[]	name-value
SigmaS	—	double	[]	name-value
GM	1 x 1	double	3.986004415e14	name-value
R	1 x 1	double	6378136.3	name-value
Epoch	1 x 1	double	NaN	name-value
ProductType	1 x 1	string	"unknown"	name-value
TideSystem	1 x 1	string	"unknown"	name-value
Name	1 x 1	string	""	name-value
Header	—	struct	struct()	name-value
VariableTerms	—	—	[]	name-value
History	—	string	string.empty(0,1)	name-value

output	description
obj	(1 x 1) shCoefficients immutable coefficient set with GM (3.986004415e14), R (6378136.3), epoch, sigmas, history

Example

```
g = shCoefficients(C, S, GM = 3.986004415e14, R = 6.3781363e6);
```

shCoefficients.plus

`out = g.plus(a, b)`

Coefficient addition (e.g. GSM + GAD background restore).

input	size	type	default	kind
a	—	—	required	positional
b	—	—	required	positional

output	description
out	(1 x 1) shCoefficients sum; sigmas RSS; epochs must match

Example

```
gsm = g + gad;                          % epoch-checked restore
```

shCoefficients.minus

`out = g.minus(a, b)`

Coefficient difference (e.g. monthly change gB - gA).

input	size	type	default	kind
a	—	—	required	positional
b	—	—	required	positional

output	description
out	(1 x 1) shCoefficients difference; sigmas RSS

Example

```
dg = g - gMean;                         % anomaly about the mean field
```

shCoefficients.uminus

`out = g.uminus(a)`

Unary minus: negate all coefficients.

input	size	type	default	kind
a	—	—	required	positional

output	description
out	(1 x 1) shCoefficients negated coefficients (sigmas kept)

Example

```
gNeg = -g;
```

shCoefficients.mtimes

`out = g.mtimes(a, b)`

Scalar scaling.

input	size	type	default	kind
a	—	—	required	positional
b	—	—	required	positional

output	description
out	(1 x 1) shCoefficients scalar-scaled; sigmas scale by |a|

Example

```
gHalf = 0.5 * g;                    % scalar scaling, sigmas follow
```

shCoefficients.compare

`[rep, h] = g.compare(other, varargin)`

Full comparison against another solution (v2.6.0). REP = g.compare(G2, ...) delegates to shLowLevel.compareSolutions with OBJ as the first solution; all options pass through (Plot=true for the 4-panel figure).

input	size	type	default	kind
other	—	—	required	positional
varargin	—	—	required	positional

output	description
rep	(1 x 1) struct shLowLevel.compareSolutions report
h	(1 x 1) graphics handle figure (Plot = true)

Example

```
rep = g.compare(gF, Names = ["raw", "G350"]);
```

shCoefficients.times

`out = g.times(a, b)`

Elementwise scaling .* (scalar or triangle matrix). W .* G scales every coefficient by the matching entry of the (nmax+1 x nmax+1) matrix W (per-coefficient weighting or tapering); a scalar factor delegates to mtimes. Sigmas scale by |factor| elementwise (v2.5.1).

input	size	type	default	kind
a	—	—	required	positional
b	—	—	required	positional

output	description
out	(1 x 1) shCoefficients elementwise-scaled coefficients

Example

```
gW = w .* g;                        % per-coefficient weighting
```

shCoefficients.truncate

`out = g.truncate(nmaxNew)`

Reduce the maximum degree.

input	size	type	default	kind
obj	—	—	required	positional
nmaxNew	1 x 1	double	required	positional

output	description
out	(1 x 1) shCoefficients truncated to the new nmax

Example

```
g60 = g.truncate(60);
```

shCoefficients.setCoefficient

`out = g.setCoefficient(n, m, Cval, Sval, Name=Value)`

Replace a single (n,m) coefficient pair.

input	size	type	default	kind
obj	—	—	required	positional
n	1 x 1	double	required	positional
m	1 x 1	double	required	positional
Cval	1 x 1	double	required	positional
Sval	1 x 1	double	NaN	positional
SigmaC	1 x 1	double	NaN	name-value
SigmaS	1 x 1	double	NaN	name-value

output	description
out	(1 x 1) shCoefficients copy with C/S(n+1, m+1) replaced

Example

```
g = g.setCoefficient(2, 0, c20slr);   % SLR C20 by hand
```

shCoefficients.applyTN14

`out = g.applyTN14(tn, Name=Value)`

Replace C20 (and C30 where provided) from a TN-14 table. OUT = obj.applyTN14(TN) with TN from shLowLevel.readTN14 (or a filename). Requires a finite obj.epoch; the nearest TN epoch within opts.Tolerance (default 0.05 yr) is used. opts.ReplaceC30: "auto" (default: replace when the table has a non-NaN C30 for that month), "never", "always".

input	size	type	default	kind
obj	—	—	required	positional
tn	—	—	required	positional
Tolerance	1 x 1	double	0.05	name-value
ReplaceC30	1 x 1	string ∈ ["auto","never","always"]	"auto"	name-value

output	description
out	(1 x 1) shCoefficients C20 (and C30 when available) replaced by the SLR values, sigmas updated

Example

```
g = g.applyTN14(tn14);               % C20/C30 replacement
```

shCoefficients.addDegree1

`out = g.addDegree1(tn, Name=Value)`

Insert TN-13 geocenter (degree-1) coefficients. OUT = obj.addDegree1(TN) with TN from shLowLevel.readTN13 (or a filename) sets C10, C11, S11 (with their sigmas) from the TN-13 record nearest to obj.epoch (within opts.Tolerance (0.05), default 0.05 yr). Without degree 1, EWH/mass grids are systematically biased - this completes the standard GSM + TN-14 + TN-13 processing chain.

input	size	type	default	kind
obj	—	—	required	positional
tn	—	—	required	positional
Tolerance	1 x 1	double	0.05	name-value

output	description
out	(1 x 1) shCoefficients degree-1 row set from the TN-13 record nearest to obj.epoch

Example

```
g = g.addDegree1(tn13);              % nearest epoch within 0.05 yr
```

shCoefficients.destripe

`out = g.destripe(Name=Value)`

Swenson & Wahr (2006) / P3M6 decorrelation. OUT = obj.destripe(minOrder=6, polyOrder=3, windowLength=[]) Formal errors are passed through UNCHANGED (the destriping operator is data-dependent; rigorous error propagation is not defined for it - use the tvANS filter on an shSeries for a filter with proper covariance semantics).

input	size	type	default	kind
obj	—	—	required	positional
minOrder	1 x 1	double	6	name-value
polyOrder	1 x 1	double	3	name-value
windowLength	—	double	[]	name-value

output	description
out	(1 x 1) shCoefficients Swenson-Wahr decorrelated

Example

```
g = g.destripe(minOrder = 6);
```

shCoefficients.gaussian

`out = g.gaussian(radiusKm)`

Jekeli (1981) isotropic smoothing; sigmas scaled by Wn.

input	size	type	default	kind
obj	—	—	required	positional
radiusKm	1 x 1	double	required	positional

output	description
out	(1 x 1) shCoefficients Gaussian-smoothed; sigmas scaled by Wn

Example

```
g = g.gaussian(350);                 % radius in km
```

shCoefficients.degreeRMS

`spec = g.degreeRMS(Name=Value)`

Degree variance/RMS/amplitude spectrum (+ error spectrum).

input	size	type	default	kind
obj	—	—	required	positional
n0	1 x 1	double	0	name-value

output	description
spec	struct: n, amp/rms/var, err degree spectrum incl. formal-error curve

Example

```
spec = g.degreeRMS();                % degree amplitude + error spectrum
```

shCoefficients.crossover

`[n0, nInterp] = g.crossover(Name=Value)`

Signal-vs-formal-error crossover degree.

input	size	type	default	kind
obj	—	—	required	positional
n0	1 x 1	double	2	name-value

output	description
nInterp	(1,1) double interpolated crossing degree

Example

```
nc = g.crossover();                  % needs formal sigmas
```

shCoefficients.spectrum

`h = g.spectrum(varargin)`

Plot the degree-amplitude spectrum. h = obj.spectrum(...)

input	size	type	default	kind
varargin	—	—	required	positional

output	description
h	(1 x 1) graphics handle spectrum plot

Example

```
g.spectrum(Kaula = true, MarkCrossover = true);
```

shCoefficients.triangle

`h = g.triangle(varargin)`

Plot the degree/order coefficient triangle. h = obj.triangle(...)

input	size	type	default	kind
varargin	—	—	required	positional

output	description
h	(1 x 1) graphics handle coefficient triangle plot

Example

```
g.triangle();                        % butterfly plot
```

shCoefficients.synthesis

`[grid, lat, lon] = g.synthesis(latVec, lonVec, Name=Value)`

Spatial synthesis (geoid, gravity, potential, EWH). [GRID, LAT, LON] = obj.synthesis(LATVEC, LONVEC, quantity ("geoid")=..., kn ([])=..., nmin (0)=..., nmax (NaN)=..., UseCache (true)=true). The Legendre functions are served from a verified process-wide cache (shLowLevel.legendreCached), so monthly time series on a fixed grid pay the recursion only once - no manual 'P' passing. EWH requires explicitly supplied load Love numbers kn (never

hardcoded).

input	size	type	default	kind
obj	—	—	required	positional
latVec	1 x :	double	required	positional
lonVec	1 x :	double	required	positional
quantity	1 x 1	string	"geoid"	name-value
kn	—	double	[]	name-value
hn	—	double	[]	name-value
Height	1 x 1	double	0	name-value
nmin	1 x 1	double	0	name-value
nmax	1 x 1	double	NaN	name-value
rho_ave	1 x 1	double	5517	name-value
rho_water	1 x 1	double	1000	name-value
UseCache	1 x 1	logical	true	name-value
Method	1 x 1	string ∈ ["auto","direct","fft"]	"auto"	name-value
LatType	1 x 1	string ∈ ["geocentric","geodetic"]	"geocentric"	name-value
Flattening	1 x 1	double	1/298.257223563	name-value
MaxMemGB	1 x 1	double	4	name-value

output	description
grid	(nlat x nlon) double synthesized field
lat	(1 x nlat) double geocentric latitudes
lon	(1 x nlon) double longitudes

Example

```
ewh = g.synthesis(-89:89, 0:359, quantity = "ewh", kn = kn);
```

shCoefficients.toReference

`out = g.toReference(Name=Value)`

Convert to another (GM (NaN), R (NaN)) reference (exact). OUT = g.toReference(GM=3.986004418e14, R=6378137.0) re-expresses the coefficients via shLowLevel.rescaleGMR; the physical field is invariant (Python-validated). Sigmas rescale identically; GM/R properties are updated.

input	size	type	default	kind
obj	—	—	required	positional
GM	1 x 1	double	NaN	name-value
R	1 x 1	double	NaN	name-value

output	description
out	(1 x 1) shCoefficients rescaled to the given GM/R

Example

```
g = g.toReference(GM = 3.986004415e14, R = 6.3781363e6);
```

shCoefficients.subtractNormalField

`out = g.subtractNormalField(Name=Value)`

Remove the ellipsoidal normal field. OUT = g.subtractNormalField() % WGS84 OUT = g.subtractNormalField(System ("WGS84")="GRS80") Computes the even zonals from the DEFINING constants

(shLowLevel.normalFieldCS, Heiskanen-Moritz closed form; validated against NIMA TR8350.2 to all published digits), rescales them from the ellipsoid's own (GM (NaN), a (NaN)) to the object's (GM, R) - the WGS84 GM and a DIFFER from the ICGEM values, skipping this step costs millimetres - and subtracts them including the deg...

input	size	type	default	kind
obj	—	—	required	positional
System	1 x 1	string ∈ ["WGS84","GRS80"]	"WGS84"	name-value
GM	1 x 1	double	NaN	name-value
a	1 x 1	double	NaN	name-value
f	1 x 1	double	NaN	name-value
omega	1 x 1	double	NaN	name-value

output	description
out	(1 x 1) shCoefficients disturbing field (even zonals of the normal field removed)

Example

```
g = g.subtractNormalField(System = "GRS80");
```

shCoefficients.map

`h = g.map(latDeg, lonDeg, Name=Value)`

Synthesize and plot in one call (shLowLevel.plotSHMap). H = g.map(-89:89, 0:359, quantity ("geoid")="ewh", kn ([])=kn) - all synthesis options plus the plotSHMap display options.

input	size	type	default	kind
obj	—	—	required	positional
latDeg	1 x :	double	-89:2:89	positional
lonDeg	1 x :	double	0:2:358	positional
quantity	1 x 1	string	"geoid"	name-value
kn	—	double	[]	name-value
hn	—	double	[]	name-value
Height	1 x 1	double	0	name-value
nmin	1 x 1	double	0	name-value
Projection	1 x 1	string	"plate"	name-value
Coast	1 x 1	logical	true	name-value
CLim	—	double	[]	name-value
Units	1 x 1	string	""	name-value
Title	1 x 1	string	""	name-value
ax	—	—	[]	name-value

output	description
h	(1 x 1) graphics handle synthesized map plot

Example

```
g.map(quantity = "geoid", Title = "Geoid 2008-04 [m]");
```

shCoefficients.fan

`out = g.fan(rDegKm, rOrdKm)`

Han fan filter (degree x order Gaussian) - shLowLevel.shFanFilter.

input	size	type	default	kind
obj	—	—	required	positional
rDegKm	1 x 1	double	required	positional
rOrdKm	1 x 1	double	required	positional

output	description
out	(1 x 1) shCoefficients fan-filtered

Example

```
g = g.fan(350, 200);
```

shCoefficients.deformation

`[up, north, east] = g.deformation(latDeg, lonDeg, Name=Value)`

Elastic load deformation up/north/east [m]. [UP, NORTH, EAST] = g.deformation(LAT, LON, kn=, hn=, ln=, Mode ("grid")="grid"|"points", nmin (1)=1, LatType ("geocentric")="geocentric") from the object's (residual!) coefficients - remove a mean field first. Love numbers are always user-supplied; see shLowLevel.shSynthesisDeformation for formulas and validation.

input	size	type	default	kind
obj	—	—	required	positional
latDeg	1 x :	double	required	positional
lonDeg	1 x :	double	required	positional
kn	—	double	required	name-value
hn	—	double	required	name-value
ln	—	double	required	name-value
nmin	1 x 1	double	1	name-value
Mode	1 x 1	string ∈ ["grid","points"]	"grid"	name-value
LatType	1 x 1	string ∈ ["geocentric","geodetic"]	"geocentric"	name-value
Flattening	1 x 1	double	1/298.257223563	name-value

output	description
up	(nlat x nlon | npts) double vertical deformation [m]
north	same size north component [m]
east	same size east component [m]

Example

```
[up, no, ea] = g.deformation(latPts, lonPts, kn = LN.kn, hn = LN.hn, ...
    ln = LN.ln, Mode = "points");
```

shCoefficients.applyDDK

`out = g.applyDDK(W)`

Apply a DDK (order-block-diagonal) decorrelation filter. OUT = obj.applyDDK(W) with W from shLowLevel.readDDK (struct, .mat, or documented ASCII exchange format). Degrees not covered by the filter pass through unchanged; sigmas are NOT propagated through the filter (set to NaN - use shLowLevel.mcPropagate for filtered uncertainties).

input	size	type	default	kind
obj	—	—	required	positional
W	—	—	required	positional

output	description
out	(1 x 1) shCoefficients DDK-decorrelated

Example

```
g = g.applyDDK("DDK3");
```

shCoefficients.evalAt

`out = g.evalAt(epoch)`

Evaluate gfct time-variable terms at an epoch.

input	size	type	default	kind
obj	—	—	required	positional
epoch	1 x 1	double	required	positional

output	description
out	(1 x 1) shCoefficients gfct variable terms evaluated at the epoch; static models pass through unchanged

Example

```
g2015 = g.evalAt(2015.5);             % gfct variable terms AT AN EPOCH
```

shCoefficients.write

`g.write(filename, Name=Value)`

Export to an ICGEM .gfc file (round-trip tested).

input	size	type	default	kind
obj	—	—	required	positional
filename	—	—	required	positional
Comment	1 x 1	string	""	name-value
Sidecar	1 x 1	logical	true	name-value

Example

```
g.write("out.gfc");
```

shCoefficients.vec

`x = g.vec(idx)`

Pack into a coefficient vector in shLowLevel.shIndex ordering.

input	size	type	default	kind
obj	—	—	required	positional
idx	1 x 1	struct	required	positional

output	description
x	(P x 1) double coefficients in idx ordering

Example

```
x = g10.vec(idx);                    % idx-ordered coefficient vector
```

shCoefficients.disp

`g.disp()`

Compact display: size, epoch, GM/R, processing history. disp(G) prints one header line with the model name, one contract line (nmax, epoch, product type, tide system, GM, R), whether formal sigmas are attached and how many gfct variable terms are carried, followed by the processing history. Called automatically when an shCoefficients object is shown without a semicolon.

Example

```
disp(g)                                 % degree, epoch, history
```

shCoefficients.read (static)

`obj = shCoefficients.read(filename, Name=Value)`

Read an ICGEM .gfc/.gfct/.gfc.gz file into an shCoefficients. G = shCoefficients.read(FILE) parses product type and mid-epoch from standard GRACE L2 filenames (GSM/GAA/GAB/GAC/GAD-2_...); override with Epoch (NaN)=..., ProductType ("")=... when needed.

input	size	type	default	kind
filename	—	—	required	positional
Epoch	1 x 1	double	NaN	name-value
ProductType	1 x 1	string	""	name-value

output	description
obj	(1 x 1) shCoefficients parsed gfc/gfct(.gz) with GM, R, sigmas, variable terms, Epoch= as given

Example

```
g = shCoefficients.read("ITSG-Grace2018_n60_2008-04.gfc", Epoch = 2008.29);
```

shCoefficients.fromVec (static)

`obj = shCoefficients.fromVec(x, idx, like)`

Unpack a shLowLevel.shIndex-ordered vector into an shCoefficients. G = shCoefficients.fromVec(X, IDX, LIKE) copies GM/R and other metadata from the template LIKE (an shCoefficients).

input	size	type	default	kind
x	: x 1	double	required	positional
idx	1 x 1	struct	required	positional
like	1 x 1	shCoefficients	required	positional

output	description
obj	(1 x 1) shCoefficients rebuilt from x with sizes/GM/R/epoch of the template

Example

```
g2 = shCoefficients.fromVec(x, idx, g10);   % sizes/GM/R from template
```

shCoefficients.analysis (static)

`[obj, info] = shCoefficients.analysis(grid, latVec, lonVec, nmax, Name=Value)`

Spherical harmonic analysis: gridded/scattered data -> Stokes coefficients (the inverse of synthesis).

input	size	type	default	kind
grid	—	double	required	positional
latVec	—	double	required	positional
lonVec	—	double	required	positional
nmax	1 x 1	double	required	positional
quantity	1 x 1	string	"geoid"	name-value

input	size	type	default	kind
Method	1 x 1	string	"auto"	name-value
Weights	1 x 1	string	"none"	name-value
Kaula	1 x 1	double	0	name-value
kn	—	double	[]	name-value
hn	—	double	[]	name-value
rho_ave	1 x 1	double	5517	name-value
rho_water	1 x 1	double	1000	name-value
GM	1 x 1	double	3.986004415e14	name-value
R	1 x 1	double	6378136.3	name-value
Epoch	1 x 1	double	NaN	name-value
Name	1 x 1	string	"analysis"	name-value
LatType	1 x 1	string ∈ ["geocentric","geodetic"]	"geocentric"	name-value
Flattening	1 x 1	double	1/298.257223563	name-value

output	description
obj	(1 x 1) shCoefficients Stokes coefficients estimated from the grid (exact on ring grids; Kaula for scattered points)

Example

```
g2 = shCoefficients.analysis(grid, lat, lon, 10, Kaula = 1e-10);
```

shSeries class

Time series of spherical harmonic coefficient sets. Wraps a stack of shCoefficients over epochs and provides everything with a time dimension: the mean field, the climatology (bias / trend / annual / semi-annual), GAX background restoration, the classic destripe+Gaussian chain applied per epoch, and the tvANS time-variable anisotropic Wiener filter with exact basin deconvolution. Immutable value class with provenance in .history.

property	size	type	default	notes
Cs	—	double	—	
Ss	—	double	—	
sigmaCs	—	double	[]	
sigmaSs	—	double	[]	
epochs	: x 1	double	—	
GM	1 x 1	double	3.986004415e14	
R	1 x 1	double	6378136.3	
productType	1 x 1	string	"unknown"	
names	: x 1	string	string.empty(0,1)	
history	—	string	string.empty(0,1)	
nmax	—	—	—	
nEpochs	—	—	—	

shSeries.shSeries

```
obj = ts.shSeries(in, Name=Value)
```

Construct from an shCoefficients array or raw stacks. ts = shSeries(objArray) ts = shSeries(Cs, Epochs ([])=..., Ss ([])=...) (raw-stack form via opts)

input	size	type	default	kind
in	—	—	required	positional
Ss	—	double	[]	name-value
Epochs	: x 1	double	[]	name-value
GM	1 x 1	double	3.986004415e14	name-value
R	1 x 1	double	6378136.3	name-value
ProductType	1 x 1	string	"unknown"	name-value
Names	: x 1	string	string.empty(0,1)	name-value
SigmaCs	—	double	[]	name-value
SigmaSs	—	double	[]	name-value
History	—	string	string.empty(0,1)	name-value

output	description
obj	(1 x 1) shSeries epoch-sorted monthly stack with sigmas and history

Example

```
ts = shSeries(Cs, Ss = Ss, Epochs = tYears);
```

shSeries.at

`g = ts.at(k)`

Extract epoch k as an shCoefficients.

input	size	type	default	kind
obj	—	—	required	positional
k	1 x 1	double	required	positional

output	description
g	(1 x 1) shCoefficients month k with epoch and sigmas

Example

```
g = ts.at(37);                          % single month as shCoefficients
```

shSeries.mean

`g = ts.mean(Name=Value)`

Mean field of the series. G = ts.mean (Omitnan=true default) returns an shCoefficients; its sigmas, when present, are the standard error of the mean.

input	size	type	default	kind
obj	—	—	required	positional
Omitnan	1 x 1	logical	true	name-value

output	description
g	(1 x 1) shCoefficients temporal mean field (omits NaN months)

Example

```
gM = ts.mean;                           % NaN-month tolerant
```

shSeries.climatology

`[clim, resid] = ts.climatology(Name=Value)`

Bias/trend/annual/semi-annual (+extra periods) fit. [CLIM, RESID] = ts.climatology(Robust=false, T0 (NaN)=mean(epochs), Periods (double.empty(1,0))=[], Weights ([])=[]) CLIM is an shClimatology; RESID the residual shSeries. Periods: extra periodic terms [years] - e.g. the GRACE tidal alias periods [161/365.25, 3.66, 7.48] (S2, K2, K1); without them, trend and semi-annual estimates absorb tidal aliasing. Weights: T x 1 per-epoch weights (e.g. 1./info.vce from the tvANS filter). CLIM carries per-coefficient 1-sigm...

input	size	type	default	kind
obj	—	—	required	positional
Robust	1 x 1	logical	false	name-value
T0	1 x 1	double	NaN	name-value
Periods	1 x :	double	double.empty(1,0)	name-value
Weights	: x 1	double	[]	name-value
ARCorrect	1 x 1	logical	false	name-value

output	description
clim	(1 x 1) shClimatology fitted bias/trend/annual/semiannual (+Periods=) with coefficient sigmas
resid	(1 x 1) shSeries residual series about the fit

Example

```
[clim, resid] = ts.climatology(Robust = true, ARCorrect = true);
```

shSeries.trendBreaks

`out = ts.trendBreaks(Name=Value)`

Piecewise-linear trends with break significance. OUT = ts.trendBreaks(Breaks=[2016.0], Periods (double.empty(1,0))=[], T0 (NaN)=, ARCorrect (false)=false) fits the standard climatology design PLUS hinge terms max(0, t - tb) per break epoch (continuous piecewise-linear trend) to every coefficient, and tests the break's significance with an F-test against the no-break model (p-values via betainc - base MATLAB, no toolbox). Python-validated: 5.1% null rejection at the 5% level; F = 23 for a 0.02/yr break in 0.05-si...

input	size	type	default	kind
obj	—	—	required	positional
Breaks	1 x :	double	required	name-value
Periods	1 x :	double	double.empty(1,0)	name-value
T0	1 x 1	double	NaN	name-value
ARCorrect	1 x 1	logical	false	name-value

output	description
out	struct: trends per segment (shCoefficients), F/p per break, segment epochs

Example

```
out = ts.trendBreaks(Breaks = 2021.0); % F-test on the hinge
```

shSeries.fan

`out = ts.fan(rDegKm, rOrdKm)`

Han fan filter on every epoch (degree x order Gaussian). OUT = ts.fan(RDEG, RORD) - see shLowLevel.shFanFilter.

input	size	type	default	kind
obj	—	—	required	positional
rDegKm	1 x 1	double	required	positional
rOrdKm	1 x 1	double	required	positional

output	description
out	(1 x 1) shSeries fan-filtered per month

Example

```
tsF = ts.fan(350, 200);
```

shSeries.applyTN14

`out = ts.applyTN14(tn, Name=Value)`

Replace C20 (and flagged C30) per month from TN-14. OUT = ts.applyTN14("TN-14.txt") - series-level convenience around shCoefficients.applyTN14 (v2.4.1; the per-month epoch selects the TN entry). All options pass through.

input	size	type	default	kind
obj	—	—	required	positional
tn	—	—	required	positional
Tolerance	1 x 1	double	0.05	name-value
ReplaceC30	1 x 1	string	"auto"	name-value

output	description
out	(1 x 1) shSeries C20/C30 replaced epoch-matched

Example

```
ts = ts.applyTN14(tn14);              % epoch-matched per month
```

shSeries.addDegree1

`out = ts.addDegree1(tn, Name=Value)`

Insert TN-13 geocenter degree-1 per month. OUT = ts.addDegree1("TN-13.txt") - series-level convenience around shCoefficients.addDegree1 (v2.4.1).

input	size	type	default	kind
obj	—	—	required	positional
tn	—	—	required	positional
Tolerance	1 x 1	double	0.05	name-value

output	description
out	(1 x 1) shSeries degree 1 completed epoch-matched

Example

```
ts = ts.addDegree1(tn13);
```

shSeries.removeAlias

`out = ts.removeAlias(Name=Value)`

Remove a tidal alias harmonic from every epoch. TS2 = ts.removeAlias() fits, per coefficient, a harmonic at the S2 tidal alias period (161 days) TOGETHER WITH bias, trend, annual and semi-annual, and subtracts only that harmonic. This is the correction GravIS applies to its Level-2B products (https://gravis.gfz.de/corrections): errors in the ocean-tide background model alias into the monthly fields at this period, and leaving them in puts a spurious ~161-day signal into every derived series.

input	size	type	default	kind
obj	—	—	required	positional
Period	1 x 1	double	161/365.25	name-value

input	size	type	default	kind
PhaseOffset	1 x 1	double	100	name-value
SplitEpoch	1 x 1	double	2018.0	name-value
T0	1 x 1	double	NaN	name-value

output	description
out	(1,1) shSeries copy with the alias harmonic removed from every epoch; the operation is appended to the history

Example

```
ts = ts.removeAlias();                % S2 161-day alias, 100 deg offset
ts = ts.removeAlias(SplitEpoch = 2018.4);
```

shSeries.removeGIA

`out = ts.removeGIA(gia, Name=Value)`

Subtract a glacial isostatic adjustment model. OUT = ts.removeGIA(GIA, T0 (NaN)=mean(epochs)) subtracts the linear GIA signal giaRate * (t - T0) from every epoch. GIA is an shCoefficients object holding RATE coefficients [1/yr] - read your model (ICE-6G_D, Caron et al., ...) from its gfc file with shCoefficients.read; models are ALWAYS user-supplied, never embedded. Trends computed after this are GIA-corrected; the model is treated as exact (its uncertainty, often the dominant trend error over Laurentia/Fennos...

input	size	type	default	kind
obj	—	—	required	positional
gia	1 x 1	shCoefficients	required	positional
T0	1 x 1	double	NaN	name-value

output	description
out	(1 x 1) shSeries GIA trend removed about the series mean epoch

Example

```
ts = ts.removeGIA(giaModel);          % e.g. ICE-6G trend gfc
```

shSeries.applyDDK

`out = ts.applyDDK(W)`

Apply a DDK decorrelation filter to every epoch. OUT = ts.applyDDK(W), W from shLowLevel.readDDK. Sigma stacks are invalidated (NaN) - the filter correlates coefficients.

input	size	type	default	kind
obj	—	—	required	positional
W	—	—	required	positional

output	description
out	(1 x 1) shSeries DDK-decorrelated per month

Example

```
ts = ts.applyDDK("DDK3");
```

shSeries.compare

`[rep, h] = ts.compare(others, varargin)`

Temporal comparison against other series (v2.6.0). REP = ts.compare(TS2) or ts.compare({TS2, TS3, ...}) delegates to shLowLevel.compareSeries with OBJ as the reference; all options pass through (Basin=, Names=, Plot=, ...).

input	size	type	default	kind
others	—	—	required	positional
varargin	—	—	required	positional

output	description
rep	(1 x 1) struct shLowLevel.compareSeries report
h	(1 x 1) graphics handle figure (Plot = true)

Example

```
rep = ts.compare({tsGFZ, tsJPL}, Names = ["ref", "GFZ", "JPL"]);
```

shSeries.restore

`out = ts.restore(gax, Name=Value)`

Add a background (GAA/GAB/GAC/GAD) series, epoch-matched. OUT = ts.restore(GAXSERIES, Tolerance=0.05, AllowMissing=false) Each epoch of OBJ is matched to the nearest GAX epoch within Tolerance [yr]. Missing matches error (or, with AllowMissing=true, leave those epochs unchanged with a note).

input	size	type	default	kind
obj	—	—	required	positional
gax	1 x 1	shSeries	required	positional
Tolerance	1 x 1	double	0.05	name-value
AllowMissing	1 x 1	logical	false	name-value

output	description
out	(1 x 1) shSeries background model added back epoch-matched (e.g. GSM + GAD)

Example

```
ts = ts.restore(tsGAD);               % GSM + GAD, epoch-matched
```

shSeries.minus

`out = ts.minus(a, b)`

Subtract a field (shCoefficients) or an epoch-matched series.

input	size	type	default	kind
a	—	—	required	positional
b	—	—	required	positional

output	description
out	(1 x 1) shSeries per-month difference (series or single field)

Example

```
dts = ts - ts.mean;
```

shSeries.destripe

`out = ts.destripe(Name=Value)`

Swenson & Wahr / P3M6, applied per epoch.

input	size	type	default	kind
obj	—	—	required	positional
minOrder	1 x 1	double	6	name-value

input	size	type	default	kind
polyOrder	1 x 1	double	3	name-value
windowLength	—	double	[]	name-value

output	description
out	(1 x 1) shSeries destriped per month

Example

```
ts = ts.destripe(minOrder = 6);
```

shSeries.gaussian

`out = ts.gaussian(radiusKm)`

Jekeli smoothing per epoch (vectorized over the stack).

input	size	type	default	kind
obj	—	—	required	positional
radiusKm	1 x 1	double	required	positional

output	description
out	(1 x 1) shSeries Gaussian-smoothed per month

Example

```
ts = ts.gaussian(300);
```

shSeries.truncate

`out = ts.truncate(nmaxNew)`

Reduce the maximum degree of the whole series.

input	size	type	default	kind
obj	—	—	required	positional
nmaxNew	1 x 1	double	required	positional

output	description
out	(1 x 1) shSeries truncated per month

Example

```
ts8 = ts.truncate(8);                % keep low degrees
```

shSeries.dropNaN

`out = ts.dropNaN()`

Remove epochs containing any non-finite coefficient. OUT = ts.dropNaN() drops every epoch whose C or S stack has a NaN/Inf anywhere - typically placeholder months across the GRACE / GRACE-FO gap (2017.5-2018.5) or corrupt reads. The remaining series is irregularly sampled, which climatology and the tvANS filter handle natively (least-squares fit on the actual epochs; per-month VCE).

output	description
out	(1 x 1) shSeries gap months removed

Example

```
ts = ts.dropNaN;                     % remove gap months
```

shSeries.select

`out = ts.select(which)`

Subset the series by index, mask, or time range. OUT = ts.select(WHICH) with WHICH one of logical (1,T) keep masked epochs indices keep listed epochs (in the given order) [tMin tMax] keep epochs with tMin <= t <= tMax (two-element non-integer-valued row => range; use explicit logical/index forms otherwise)

input	size	type	default	kind
obj	—	—	required	positional
which	—	—	required	positional

output	description
out	(1 x 1) shSeries months selected by logical/index mask

Example

```
ts = ts.select(ts.epochs > 2018.5);    % GRACE-FO era only
```

shSeries.filter

`[out, op, info] = ts.filter(method, Name=Value)`

Optimal time-variable filtering of the series. [TSF, OP, INFO] = ts.filter("tvANS", Constraints ([])=..., NoiseCov ([])=..., SignalMode ("isotropic")="isotropic"|"inhomogeneous", NIterSignal (3)=3, Robust (false)=false, MinDegree (2)=2) runs the tvANS chain (see shLowLevel.tvANSFilter): deterministic-model separation, empirical or supplied noise covariance, monthly VCE scaling, data-driven signal covariance, per-month anisotropic Wiener filter, optional hard constraints. OP is the stored linear operator - requir...

input	size	type	default	kind
obj	—	—	required	positional
method	1 x 1	string	required	positional
NoiseCov	—	double	[]	name-value
Constraints	—	double	[]	name-value
SignalMode	1 x 1	string	"isotropic"	name-value
NIterSignal	1 x 1	double	3	name-value
Robust	1 x 1	logical	false	name-value
MinDegree	1 x 1	double	2	name-value
Blocks	1 x 1	string ∈ ["auto","on","off"]	"auto"	name-value
Shrinkage	—	double	[]	name-value
VCEMinDegree	—	double	[]	name-value
VCEBands	1 x :	double	[]	name-value

output	description
out	(1 x 1) shSeries tvANS-filtered series with sigmaCs/Ss posterior stacks (exact incl. constraints, v2.5)
op	struct stored linear operator (V/Ut/lam/s or blocks, model, detLeverage/detResVar) for deconvolution and resolution maps
info	struct: sigmaXfres (P x T), sigmaNote, VCE diagnostics

Example

```
[tsF, op, info] = ts.filter("tvANS", Constraints = Ac, Blocks = "off");
```

shSeries.basinAverage

`[avg, out] = ts.basinAverage(B, Name=Value)`

Basin averages, naive or exactly deconvolved. AVG = ts.basinAverage(B) - naive averages B'*x ./ diag(B'*B), K x T, with B (P x K) basin kernels in shLowLevel.shIndex ordering (build from a grid mask via shLowLevel.synthesisMatrix: b = Y' * (w .* mask)). [AVG, OUT] = ts.basinAverage(B, Deconvolve (false)=true, Op (struct())=op) removes filter attenuation and inter-basin leakage exactly using the stored operator from ts.filter (see shLowLevel.basinDeconvolve). OUT.sigma (K x T) is the 1-sigma noise uncertainty of ...

input	size	type	default	kind
obj	—	—	required	positional
B	—	double	required	positional
Deconvolve	1 x 1	logical	false	name-value
Op	—	struct	struct()	name-value
Ridge	1 x 1	double	0	name-value

output	description
avg	(K x T) double basin averages (deconvolved when Deconvolve=true)
out	struct: sigma (K x T), c, attn, condA (deconvolution path)

Example

```
[avg, out] = tsF.basinAverage(B, Deconvolve = true, Op = op);
```

shSeries.disp

`ts.disp()`

Compact display: months, span, gaps, processing history. disp(TS) prints the product type, the number of epochs, the maximum degree and the covered epoch range, followed by the processing history. Called automatically when an shSeries object is shown without a semicolon.

Example

```
disp(ts)
```

shSeries.read (static)

`obj = shSeries.read(files, Name=Value)`

Read a series of gfc files (pattern or list), sorted by epoch. TS = shSeries.read("GSM-2_*.gfc") or shSeries.read(fileList).

input	size	type	default	kind
files	—	—	required	positional
Truncate	1 x 1	double	NaN	name-value

output	description
obj	(1 x 1) shSeries wildcard-read, epoch-sorted stack

Example

```
ts2 = shSeries.read("ITSG-Grace*_n60_*.gfc");   % wildcard, epoch-sorted
```

shSeries.fromFolder (static)

`obj = shSeries.fromFolder(folder, Name=Value)`

Read every matching gfc file in a folder as a series. TS = shSeries.fromFolder("itsg_series") TS = shSeries.fromFolder(dest, Pattern="*n96*.gfc*", Truncate (NaN)=60) Convenience wrapper around shSeries.read (sorted by epoch); pairs with shLowLevel.fetchITSG, which downloads ITSG monthly solutions into tests/test_data/itsg_series on demand.

input	size	type	default	kind
folder	1 x 1	string	required	positional
Pattern	1 x 1	string	"*.gfc*"	name-value
Truncate	1 x 1	double	NaN	name-value

output	description
obj	(1 x 1) shSeries all gfc(.gz) files of a folder, epoch-sorted

Example

```
ts = shSeries.fromFolder(shLowLevel.dataFolder() + "/itsg_series");
```

shClimatology class

Deterministic climatology of an SH series. Bias + trend + annual + semi-annual model per Stokes coefficient, as fitted by shSeries.climatology (least squares or Huber-robust): x(t) = bias + trend*(t-t0) + cosA*cos(2*pi*(t-t0)) + sinA*sin(...) + cosSA*cos(4*pi*(t-t0)) + sinSA*sin(4*pi*(t-t0)) + sum_k extra cos/sin terms of period p_k (v2.1) All phases and the bias refer to the reference epoch t0. Extra periods (Periods= option) typically hold the GRACE tidal alias terms S2 (161 d), K2 (3.66 yr), K1 (7.48 yr); access them via periodic(k). Per-coefficient 1-sigma significance from the fit is sto

property	size	type	default	notes
t0	1 x 1	double	NaN	
GM	1 x 1	double	3.986004415e14	
R	1 x 1	double	6378136.3	
biasC	—	double	—	
biasS	—	double	—	
trendC	—	double	—	
trendS	—	double	—	
cosAnnC	—	double	—	
cosAnnS	—	double	—	
sinAnnC	—	double	—	
sinAnnS	—	double	—	
cosSemiC	—	double	—	
cosSemiS	—	double	—	
sinSemiC	—	double	—	
sinSemiS	—	double	—	
periods	1 x :	double	double.empty(1,0)	extra periods [yr] (v2.1)
extraCosC	—	double	[]	n1 x n1 x K stacks for the extra periods
extraCosS	—	double	[]	
extraSinC	—	double	[]	
extraSinS	—	double	[]	
coefSigma	—	double	[]	(6+2K) x 2*Nc per-coefficient 1-sigma (v2.1)
history	—	string	string.empty(0,1)	
nmax	—	—	—	

shClimatology.withNote

`obj = clim.withNote(txt)`

Append a provenance note to the history (v2.2). OBJ = clim.withNote(TXT) - the only supported way to annotate from outside the class (history is SetAccess=private).

input	size	type	default	kind
obj	—	—	required	positional
txt	—	—	required	positional

output	description
obj	(1 x 1) shClimatology copy with the note appended to its history

Example

```
clim = clim.withNote("GIA: ICE-6G_D removed");
```

shClimatology.removeGIA

`out = clim.removeGIA(gia)`

Subtract a GIA rate model from the fitted trend. OUT = clim.removeGIA(GIA): GIA is an shCoefficients holding RATE coefficients [1/yr] (user-supplied model file via shCoefficients.read). Only trendC/trendS change; trend sigmas are kept - the model is treated as exact (documented limitation; GIA model spread often dominates regional trend error budgets - compare several models).

input	size	type	default	kind
obj	—	—	required	positional
gia	1 x 1	shCoefficients	required	positional

output	description
out	(1,1) shClimatology copy with the GIA trend removed from the trend component; history appended

Example

```
clim = clim.removeGIA(giaModel);
```

shClimatology.eval

`g = clim.eval(epoch)`

Evaluate the full climatology at an epoch -> shCoefficients.

input	size	type	default	kind
obj	—	—	required	positional
epoch	1 x 1	double	required	positional

output	description
g	(1 x 1) shCoefficients model field evaluated at the epoch

Example

```
g2020 = clim.eval(2020.0);            % model field at an epoch
```

shClimatology.bias

`g = clim.bias()`

Field at the reference epoch t0.

output	description
g	(1 x 1) shCoefficients bias component with sigmas

Example

```
g0 = clim.bias();
```

shClimatology.trend

`g = clim.trend()`

Linear trend field, units of the coefficients per year.

output	description
g	(1 x 1) shCoefficients trend component [units/yr] with sigmas

Example

```
gT = clim.trend();                    % [units/yr], sigmas attached
```

shClimatology.cosAnnual

`g = clim.cosAnnual()`

Cosine annual component (sigmas attached).

output	description
g	(1 x 1) shCoefficients cosine annual component

Example

```
gc = clim.cosAnnual();
```

shClimatology.sinAnnual

`g = clim.sinAnnual()`

Sine annual component (sigmas attached).

output	description
g	(1 x 1) shCoefficients sine annual component

Example

```
gs = clim.sinAnnual();
```

shClimatology.cosSemiannual

`g = clim.cosSemiannual()`

Cosine semiannual component (sigmas attached).

output	description
g	(1 x 1) shCoefficients cosine semiannual component

Example

```
gc2 = clim.cosSemiannual();
```

shClimatology.sinSemiannual

`g = clim.sinSemiannual()`

Sine semiannual component (sigmas attached).

output	description
g	(1 x 1) shCoefficients sine semiannual component

Example

```
gs2 = clim.sinSemiannual();
```

shClimatology.periodic

`[gc, gs] = clim.periodic(k)`

Cos/sin components of the k-th extra period (v2.1). [GC, GS] = clim.periodic(K) returns shCoefficients for the cos and sin terms of period obj.periods(K) [yr] - e.g. the S2/K2/K1 tidal alias terms when fitted via ts.climatology(Periods=[161/365.25, 3.66, 7.48]).

input	size	type	default	kind
obj	—	—	required	positional
k	1 x 1	double	required	positional

output	description
gc	(1 x 1) shCoefficients cosine component of the k-th extra period
gs	(1 x 1) shCoefficients sine component

Example

```
[gc, gs] = clim.periodic(1);          % extra Periods= component
```

shClimatology.amplitudeMap

`[A, lat, lon, phase] = clim.amplitudeMap(which, latVec, lonVec, Name=Value)`

Pointwise seasonal amplitude (and phase) map. [A, LAT, LON, PHASE] = clim.amplitudeMap(WHICH, ... with WHICH "annual" | "semiannual" | k (index into clim.periods, v2.1), LATVEC, LONVEC, quantity ("geoid")=..., kn ([])=...) synthesizes the cos and sin component fields and combines them pointwise: A = sqrt(fc.^2 + fs.^2) PHASE = atan2(fs, fc) [rad, relative to t0] (amplitude of a harmonic is NOT a linear functional of the coefficients, so it must be formed in the space domain).

input	size	type	default	kind
obj	—	—	required	positional
which	1 x 1	—	required	positional
latVec	1 x :	double	required	positional
lonVec	1 x :	double	required	positional
quantity	1 x 1	string	"geoid"	name-value
kn	—	double	[]	name-value

output	description
A	(nlat x nlon) double amplitude of the harmonic in the requested quantity
lat	(1 x nlat) double latitudes
lon	(1 x nlon) double longitudes
phase	(nlat x nlon) double phase [rad] of the harmonic

Example

```
[A, lat, lon] = clim.amplitudeMap("annual", -89:89, 0:359, quantity = "ewh", kn = kn);
```

shClimatology.disp

`clim.disp()`

Compact display: fitted components, reference epoch, note. disp(CLIM) prints the maximum degree and the reference epoch t0, the list of fitted components (bias, trend, annual and semi-annual cosine/sine, plus any extra Periods) and the processing history including the GIA note. Called automatically when an shClimatology object is shown without a semicolon.

Example

```
disp(clim)
```

shClimatology.fromCoef (static)

```
obj = shClimatology.fromCoef(coef, t0, series, Name=Value)
```

Build from shLowLevel.fitDeterministicModel output (internal). OBJ = shClimatology.fromCoef(COEF, T0, SERIES, Periods (double.empty(1,0))=[], CoefSigma=[]) with COEF (6+2K) x (2*Nc) over the flattened [C(:); S(:)] stack of SERIES; rows 7:end are the cos/sin pairs of the K extra Periods [yr].

input	size	type	default	kind
coef	—	double	required	positional
t0	1 x 1	double	required	positional
series	1 x 1	shSeries	required	positional
Periods	1 x :	double	double.empty(1,0)	name-value
CoefSigma	—	double	[]	name-value

output	description
obj	(1 x 1) shClimatology rebuilt from a coefficient table with the series' layout

Example

```
clim = shClimatology.fromCoef(coef, t0, ts);
```

Package +shLowLevel

Readers, writers & data management

shLowLevel.shReadGFC

```
model = shLowLevel.shReadGFC(filename)
```

Read an ICGEM ASCII .gfc gravity field model file. MODEL = SHREADGFC(FILENAME) parses header key:value pairs and the gfc/gfct data block of a standard ICGEM-format file (GRACE/GRACE-FO monthly solutions, static models, GOCE, EGM2008, etc.). Gzip- compressed files (.gfc.gz) are transparently decompressed.

input	size	type	default	kind
filename	—	—	required	positional

output	description
model	(1,1) struct fields: C/S/sigmaC/sigmaS (nmax+1 x nmax+1 double), GM/R (1,1 double), nmax (1,1 double), tide (string), variable terms (trend/annual/semiannual) when present

Example

```
g = shLowLevel.shReadGFC("ITSG-Grace2018_n60_2008-04.gfc");
```

shLowLevel.shEvalGFCT

```
[Ct, St] = shLowLevel.shEvalGFCT(model, epoch)
```

Evaluate time-variable Stokes coefficients at a given epoch. [CT, ST] = SHEVALGFCT(MODEL, EPOCH) reconstructs C(t) = C0 + trnd*(t-t0) + sum_k [acos_k*cos(2*pi*(t-t0)) ... + asin_k*sin(2*pi*(t-t0))] (and analogously for S) from MODEL.C / MODEL.S (constant part) and MODEL.variableTerms (from SHREADGFC), at EPOCH (same units/convention as the t0 values in the file -- typically decimal years).

input	size	type	default	kind
model	—	—	required	positional
epoch	—	—	required	positional

output	description
Ct	(nmax+1 x nmax+1) double cosine coefficients at the epoch
St	(nmax+1 x nmax+1) double sine coefficients at the epoch

Example

```
[C, S] = shLowLevel.shEvalGFCT(model, 2015.5);   % piecewise gfct at an epoch
```

shLowLevel.readTN13

`tn = shLowLevel.readTN13(filename)`

Read a GRACE/GRACE-FO TN-13 geocenter (degree-1) file. TN = shLowLevel.readTN13(FILENAME) parses the official TN-13 product (JPL/CSR/GFZ, e.g. TN-13_GEOC_CSR_RL06.txt): data lines GRCOF2 n m Clm Slm Clm_sig Slm_sig begin end with n=1, m=0/1 and begin/end epochs yyyymmdd.hhmm. Line pairs (m=0, m=1) sharing the same interval are combined into one monthly record with the interval mid-point as epoch.

input	size	type	default	kind
filename	—	—	required	positional

output	description
tn	struct:
epoch	(K,1) double interval mid-points [decimal years]
C10, C11, S11	(K,1) double degree-1 Stokes coefficients sigC10, sigC11, sigS11 (K,1) double 1-sigma t0, t1 (K,1) double interval bounds [decimal years]

Example

```
tn13 = shLowLevel.readTN13("TN-13_GEOC_CSR_RL06.3.txt");
```

shLowLevel.readTN14

`tn = shLowLevel.readTN14(filename)`

Parse a TN-14 style SLR C20/C30 replacement table. TN = shLowLevel.readTN14(FILENAME) reads the CSR/GFZ "TN-14" technical-note text format: an arbitrary text header followed by data lines with 10 whitespace-separated numeric columns MJD_begin yearfrac_begin C20 C20-mean*1e10 sigC20*1e10 ... C30 C30-mean*1e10 sigC30*1e10 MJD_end yearfrac_end C30 columns contain 'NaN' for months where no SLR C30 is provided (pre GRACE-FO accelerometer degradation).

input	size	type	default	kind
filename	—	—	required	positional

output	description
tn	(1,1) struct fields: t (T x 1 double, decimal years), C20/C30 (T x 1 double), sigmaC20/sigmaC30 (T x 1 double)

Example

```
tn14 = shLowLevel.readTN14("TN-14_C30_C20_SLR_GSFC.txt");
```

shLowLevel.readSINEX

`snx = shLowLevel.readSINEX(filename, Name=Value)`

Read gravity-field SINEX files (solutions, covariances, NEQs). SNX = shLowLevel.readSINEX(FILENAME) parses the SINEX blocks used by gravity-field solution providers (ITSG, COST-G, ...): +SOLUTION/ESTIMATE parameter list: type CN/SN, degree, order, value, std +SOLUTION/MATRIX_ESTIMATE L|U COVA covariance (lower/upper, up to 3 values per line) +SOLUTION/NORMAL_EQUATION_MATRIX L|U normal-equation matrix .gz files are gunzipped transparently.

input	size	type	default	kind
filename	—	—	required	positional
Index	—	struct	struct([])	name-value
Output	1 x 1	string ∈ ["raw","covariance"]	"raw"	name-value
Only	1 x 1	string ∈ ["all","estimate"]	"all"	name-value

output	description
snx	(1,1) struct fields: x (P x 1 double), idx (struct), Cxx (P x P double, [] unless Only="full"), epoch (1,1 double), N/b (normal equations when present)

Example

```
snx = shLowLevel.readSINEX("ITSG-Grace2018_n96_2008-04_head12.snx");
fprintf("%s with %d params\n", snx.kind, numel(snx.x))
```

shLowLevel.readDDK

`W = shLowLevel.readDDK(source, Name=Value)`

Load a DDK (Kusche) anisotropic decorrelation filter. W = shLowLevel.readDDK(SOURCE) loads a DDK filter into the toolbox's order-block container, ready for shLowLevel.applyDDK / g.applyDDK / ts.applyDDK. The DDK filters (Kusche 2007; Kusche et al. 2009, DDK1..DDK8) are the standard published anisotropic alternative to Gaussian smoothing and the usual benchmark against tvANS.

input	size	type	default	kind
source	—	—	required	positional
Nmax	—	double	[]	name-value

output	description
W	struct: nmax (1 x 1), name, blocks (1 x 241) struct array with m, cs, n (degrees), M (square block matrix)

Example

```
W = shLowLevel.readDDK("DDK3");   % DDK3 ships with the toolbox; others via shLowLevel.fetchDDK
```

shLowLevel.readMascon

`mas = shLowLevel.readMascon(filename)`

Read a GRACE/GRACE-FO mascon netCDF (JPL / CSR / GSFC style). MAS = shLowLevel.readMascon(FILENAME) reads gridded mascon solutions for comparison against the SH chain (base MATLAB: ncread/ncinfo, no toolboxes). Variable names are auto-detected among the layouts used by JPL RL06 ('lwe_thickness'), CSR RL06 ('lwe_thickness'), and generic 'lwe'/'water_thickness'; time units "days since YYYY-MM-DD" are converted to decimal years.

input	size	type	default	kind
filename	—	—	required	positional

output	description
mas	struct: lat (I x 1), lon (J x 1), t (T x 1 decimal years), ewh (I x J x T) [m], units/meta from the netCDF

Example

```
M = shLowLevel.readMascon("CSR_..._Mascons.nc");   % lat/lon/time EWH stack
```

shLowLevel.readLoveNumbers

`[LN, info] = shLowLevel.readLoveNumbers(filename, Name=Value)`

Read load Love numbers from a user-supplied table file. LN = shLowLevel.readLoveNumbers(FILE) parses a plain-text load-Love-number table (comment lines starting with %, #, ! or // are skipped; an optional header line naming the columns is recognized). The toolbox never hardcodes Love numbers - this reader turns YOUR loading-model table (PREM, Gutenberg-Bullen, ak135, ...) into the kn/hn/ln vectors the synthesis and deformation routines expect.

input	size	type	default	kind
filename	—	—	required	positional
Columns	1 x 1	string	"auto"	name-value
MaxDegree	—	double	[]	name-value
Interp	1 x 1	string ∈ ["none", "pchip"]	"none"	name-value
InFrame	1 x 1	string ∈ ["", "CE", "CF", "CM"]	""	name-value
OutFrame	1 x 1	string ∈ ["", "CE", "CF", "CM"]	""	name-value

output	description
LN	struct: n (D x 1) degrees, ascending kn (D x 1) potential load Love numbers k'_n hn (D x 1) vertical (NaN if absent from file) ln (D x 1) horizontal (NaN if absent from file)
info	struct: source, columns, frame, interpolated (logical mask)

Example

```
LN = shLowLevel.readLoveNumbers("prem_load.txt", Columns="n h l k", ...
         MaxDegree=96, Interp="pchip", InFrame="CE", OutFrame="CF");
[up, no, ea] = g.deformation(latPts, lonPts, kn=LN.kn, hn=LN.hn, ...
         ln=LN.ln, Mode="points");
```

shLowLevel.listICGEM

`[T, info] = shLowLevel.listICGEM(Name=Value)`

List gravity field models available at ICGEM (GFZ). T = shLowLevel.listICGEM() returns a table of the STATIC global models from https://icgem.gfz.de/tom_longtime (name, year, max degree, data sources, direct .gfc download URL) - feed a row's name into shLowLevel.fetchICGEM to download.

input	size	type	default	kind
Type	1 x 1	string ∈ ["static","temporal"]	"static"	name-value
Series	1 x 1	string	""	name-value
Source	1 x 1	string	""	name-value
Timeout	1 x 1	double	60	name-value

output	description
T	table: static: name, year, degree, data, url temporal: group, center, series, path, url, zip temporal+Series: name, url
info	struct: source, retrieved, note

Example

```
T = shLowLevel.listICGEM(Type = "temporal", Series = "ITSG-Grace2018");
```

shLowLevel.fetchITSG

`[files, info] = shLowLevel.fetchITSG(months, Name=Value)`

Download ITSG monthly OR daily solutions from TU Graz. FILES = shLowLevel.fetchITSG(2019:2020) downloads all available monthly GSM solutions for the given years into dataFolder/itsg_series (websave, base MATLAB). Already-present files are skipped unless Update (false)=true, which re-downloads them with a safe swap (the fresh file is parse-verified before it replaces the old one); months that do not exist on the server (mission gap 2017-07..2018-05, intra-mission dropouts) are reported in INFO.missing, not errors.

input	size	type	default	kind
months	—	—	""	positional
Dest	1 x 1	string	""	name-value
Nmax	1 x 1	double	NaN	name-value
Product	1 x 1	string ∈ ["monthly", "daily"]	"monthly"	name-value
Timeout	1 x 1	double	60	name-value
Proxy	1 x 1	string	""	name-value
Update	1 x 1	logical	false	name-value
Quiet	1 x 1	logical	false	name-value
Release	1 x 1	string	""	name-value
BaseURL	1 x 1	string	"https://ftp.tugraz.at/pub/ITSG/GRACE"	name-value
Catalog	1 x :	double	[]	name-value

output	description
files	(1,:) string paths of all present files (new + existing)
info	struct: fetched, updated, skipped, missing (string arrays), url

Example

```
shLowLevel.fetchITSG(["2008-04", "2025-12"], Nmax = 96);
```

shLowLevel.fetchICGEM

`[file, info] = shLowLevel.fetchICGEM(model, Name=Value)`

Download a static gravity field model from ICGEM. FILE = shLowLevel.fetchICGEM("EGM2008") resolves the model name against shLowLevel.listICGEM (case-insensitive exact match, else unique prefix) and downloads the .gfc into <dataFolder>/icgem. Existing files are skipped. Load with shCoefficients.read(FILE).

input	size	type	default	kind
model	—	—	required	positional
Dest	1 x 1	string	""	name-value
Timeout	1 x 1	double	300	name-value
List	—	table	table()	name-value
Proxy	1 x 1	string	""	name-value
Update	1 x 1	logical	false	name-value
Quiet	1 x 1	logical	false	name-value
Pause	1 x 1	double	3	name-value
Retries	1 x 1	double	3	name-value
Type	1 x 1	string ∈ ["static", "temporal"]	"static"	name-value
Files	1 x 1	string	"*.gfc*"	name-value
Mode	1 x 1	string ∈ ["auto", "archive", "files"]	"auto"	name-value
FileList	—	table	table()	name-value

output	description
file	string local path(s): one per model, or every file of the series in temporal mode (bulk selections are concatenated)
info	struct: url, skipped (true if already present) and mode (1,1) string reporting the path that actually ran - "model" (single static model) | "archive" (whole-series ZIP in one request) | "files" (per-file fallback) | "present" (nothing to do); a script can tell a fresh download from a no-op by this field

Example

```
f = shLowLevel.fetchICGEM("GOCO06s");       % static model by name
```

shLowLevel.fetchDDK

`[files, info] = shLowLevel.fetchDDK(which, Name=Value)`

Download DDK filter matrices (Wbd binaries) on demand. FILES = shLowLevel.fetchDDK(1:8) fetches the released anisotropic DDK decorrelation filters DDK1..DDK8 (Kusche 2007; Kusche et al. 2009; ~9.8 MB each) from the MIT-licensed strawpants/GRACE-filter repository into <dataFolder>/DDK. Existing files are skipped unless Update (false)=true, which re-downloads them with a safe swap (the fresh file is parse-verified by shLowLevel.readDDK before replacing the old one). DDK3 additionally ships inside the toolbox (test...

input	size	type	default	kind
which	1 x :	double	required	positional
Dest	1 x 1	string	""	name-value
Timeout	1 x 1	double	120	name-value
Proxy	1 x 1	string	""	name-value
Update	1 x 1	logical	false	name-value
Quiet	1 x 1	logical	false	name-value

output	description
files	(1,:) string present file paths (new + existing)
info	struct: fetched, updated, skipped, names, url

Example

```
shLowLevel.fetchDDK(1:8);                   % ~80 MB, skips existing
```

shLowLevel.fetchTN

`[files, info] = shLowLevel.fetchTN(Name=Value)`

Download the TN-13/TN-14 low-degree completion files on demand. FILES = shLowLevel.fetchTN() fetches the GRACE/GRACE-FO technical notes that complete the standard processing chain into <dataFolder>/TN: TN-14 GSFC SLR C20/C30 replacement (shLowLevel.readTN14/applyTN14) TN-13 geocenter (degree 1) per provider (shLowLevel.readTN13/addDegree1) Source: the GFZ ISDC open document server (URLs verified 2026-08-07; the CSR/JPL/GFZ RL06.3 files and the GSFC TN-14 also ship as test fixtures, so the toolbox works ...

input	size	type	default	kind
Providers	1 x :	string ∈ ["GFZ", "CSR", "JPL"]	["GFZ", "CSR", "JPL"]	name-value
TN14	1 x 1	logical	true	name-value
Release	1 x 1	string	"RL06.3"	name-value
Update	1 x 1	logical	false	name-value
Dest	1 x 1	string	""	name-value
BaseURL	1 x 1	string	""	name-value
Proxy	1 x 1	string	""	name-value
Timeout	1 x 1	double	60	name-value
Quiet	1 x 1	logical	false	name-value

output	description
files	(1,:) string verified present file paths (new + updated + kept existing)
info	(1,1) struct fetched / updated / skipped / failed (string arrays of file names), urls, dest

Example

```
    shLowLevel.fetchTN(Update = true);           % monthly refresh, safe swap
    tn13 = shLowLevel.readTN13(fullfile(shLowLevel.dataFolder(), "TN", ...
              "TN-13_GEOC_CSR_RL06.3.txt"));
    g1 = g.applyTN14(tn14).addDegree1(tn13);
```

shLowLevel.dataFolder

`folder = shLowLevel.dataFolder(newFolder)`

Get or set the persistent user data folder of the toolbox. F = shLowLevel.dataFolder() returns the current data folder (created on demand). Default: <toolbox root>/data. F = shLowLevel.dataFolder(PATH) sets and persists a user folder (MATLAB preference, survives sessions: setpref('shAnalysis','dataFolder')). F = shLowLevel.dataFolder("reset") back to the default.

input	size	type	default	kind
newFolder	1 x 1	string	""	positional

output	description
folder	(1 x 1) string current data folder (created on demand)

Example

```
shLowLevel.dataFolder("D:/grace_data")      % shared download location
```

shLowLevel.ddkNames

`names = shLowLevel.ddkNames()`

Released Wbd filenames for DDK1..DDK8 (strong -> weak). NAMES = shLowLevel.ddkNames() - single source of truth for the mapping (verified against the strawpants/GRACE-filter repository listing). Claude (Fable 5), 2026-08-07 (v2.4.1).

output	description
names	(1 x 8) string Wbd file names for DDK1..DDK8

Example

```
names = shLowLevel.ddkNames();              % Wbd file names DDK1..8
```

shLowLevel.parseGraceFilename

`meta = shLowLevel.parseGraceFilename(filename)`

Product type and mid-epoch from a GRACE L2 filename. META = shLowLevel.parseGraceFilename(FILENAME) parses the standard GRACE / GRACE-FO Level-2 naming convention PPP-2_YYYYDDD-YYYYDDD_*.gfc[.gz] e.g. GSM-2_2024032-2024060_GRFO_UTCSR_BA01_0600.gfc, where PPP is one of GSM, GAA, GAB, GAC, GAD and YYYYDDD are year + day-of-year of the coverage span. Best-effort: fields are NaN/"unknown" when the pattern does not match (static models, renamed files); callers can always override via explicit options.

input	size	type	default	kind
filename	—	—	required	positional

output	description
meta	(1,1) struct fields: center/product/release (string), epoch/year/month (double), nmax (double, NaN if absent)

Example

```
meta = shLowLevel.parseGraceFilename("GSM-2_2008095-2008120_...gz");
```

shLowLevel.icgemDate2Year

`y = shLowLevel.icgemDate2Year(d)`

Convert ICGEM/GRACE yyyymmdd(.hhmm) epochs to decimal years. Y = shLowLevel.icgemDate2Year(D) converts date codes of the form yyyymmdd or yyyymmdd.hhmm (as used in ICGEM gfct files, TN-13/TN-14) to decimal years, using day-of-year / (days in year) with proper leap years. Values that do not look like date codes (< 1.8e7, e.g. already-decimal years in legacy synthetic files) are returned UNCHANGED - this rule keeps pre-v2.1 files evaluating identically.

input	size	type	default	kind
d	—	double	required	positional

output	description
t	(same size as input) double decimal years

Example

```
t = shLowLevel.icgemDate2Year(20080415.0000);   % -> 2008.29
```

shLowLevel.writeGFC

`shLowLevel.writeGFC(filename, C, S, GM, R, Name=Value)`

Write Stokes coefficients to an ICGEM ASCII .gfc file. shLowLevel.writeGFC(FILENAME, C, S, GM, R) writes a static ICGEM-format gravity field file readable by shLowLevel.shReadGFC (round-trip tested to full double precision: coefficients are written with %22.14e).

input	size	type	default	kind
filename	—	—	required	positional
C	—	double	required	positional
S	—	double	required	positional
GM	1 x 1	double	required	positional
R	1 x 1	double	required	positional
SigmaC	—	double	[]	name-value
SigmaS	—	double	[]	name-value
ModelName	—	—	"shAnalysis_export"	name-value
TideSystem	—	—	"unknown"	name-value
Comment	—	—	""	name-value
Sidecar	1 x 1	logical	true	name-value

Example

```
shLowLevel.writeGFC("out.gfc", g.C, g.S, g.GM, g.R, Comment = "DDK3 filtered");
```

shLowLevel.writeGrid

`shLowLevel.writeGrid(filename, grid, latDeg, lonDeg, Name=Value)`

Export gridded fields to CF-style netCDF (base MATLAB). shLowLevel.writeGrid('ewh.nc', GRID, LAT, LON, Name ("field")="lwe_thickness", Units ("")="m", Epoch=2010.29) writes a 2-D field or a 3-D stack (nlat x nlon x T with Epochs ([])=) using nccreate/ncwrite - directly readable back by shLowLevel.readMascon (roundtrip-tested), by GMT, panoply, xarray, etc. Time is stored as "days since 2002-01-01" like the mascon products.

input	size	type	default	kind
filename	—	—	required	positional
grid	—	double	required	positional
latDeg	: x 1	double	required	positional
lonDeg	: x 1	double	required	positional
Name	1 x 1	string	"field"	name-value

input	size	type	default	kind
Units	1 x 1	string	""	name-value
Epochs	: x 1	double	[]	name-value
LatType	1 x 1	string	"geocentric"	name-value
Description	1 x 1	string	""	name-value
Sidecar	1 x 1	logical	true	name-value

Example

```
shLowLevel.writeGrid("ewh.nc", grid, lat, lon);   % netCDF/geotiff/ascii
```

shLowLevel.writeAnimation

`shLowLevel.writeAnimation(ts, filename, Name=Value)`

Export a monthly series as an MP4 map animation. shLowLevel.writeAnimation(TS, 'ewh.mp4', quantity="ewh", kn ([])=kn) renders one shLowLevel.plotSHMap frame per epoch into an MPEG-4 via VideoWriter (base MATLAB), with a FIXED color scale over all epochs (robust 98th percentile across the whole stack - the single most common animation mistake is a per-frame scale).

input	size	type	default	kind
ts	1 x 1	shSeries	required	positional
filename	—	—	required	positional
quantity	1 x 1	string	"ewh"	name-value
kn	—	double	[]	name-value
hn	—	double	[]	name-value
lat	1 x :	double	-89:2:89	name-value
lon	1 x :	double	0:2:358	name-value
FrameRate	1 x 1	double	4	name-value
Sidecar	1 x 1	logical	true	name-value
CLim	—	double	[]	name-value
Coast	1 x 1	logical	true	name-value
Units	1 x 1	string	""	name-value
Projection	1 x 1	string ∈ ["plate","hammer"]	"plate"	name-value

Example

```
shLowLevel.writeAnimation(ts, "ewh_2019_2022.avi", quantity = "ewh", kn = kn);   % .avi portable, .mp4 on Windows/m
```

Legendre, spectral tools & indexing

shLowLevel.legendreALF

`P = shLowLevel.legendreALF(nmax, lat)`

Fully normalized (4-pi) associated Legendre functions Pbar_nm. P = LEGENDREALF(NMAX, LAT) computes fully normalized associated Legendre functions up to degree/order NMAX for each latitude in LAT (radians, geocentric, scalar or vector), using the scaled forward column recursion of Holmes & Featherstone (2002): the u^m factor of each column is carried separately with a 1e-280 scale factor, which keeps the recursion overflow/underflow-free up to at least degree 2190 (EGM2008/XGM2019e) at all latitudes including nea...

input	size	type	default	kind
nmax	—	—	required	positional
lat	—	—	required	positional

output	description
P	(NMAX+1)x(NMAX+1)xnumel(LAT) double P(n+1,m+1,k) = Pbar_{n,m}(sin(lat(k))), upper triangle 0

Example

```
P = shLowLevel.legendreALF(96, deg2rad(-89:89));   % stable to n=2190
```

shLowLevel.legendreALFDeriv

`[P, D, D2] = shLowLevel.legendreALFDeriv(nmax, latRad)`

Fully normalized ALFs and their latitude derivatives. [P, D] = shLowLevel.legendreALFDeriv(NMAX, LATRAD) returns the 4pi-fully- normalized associated Legendre functions P (via shLowLevel.legendreALF, scaled Holmes-Featherstone, stable to degree 2190) and their exact first derivatives with respect to LATITUDE phi,

input	size	type	default	kind
nmax	1 x 1	double	required	positional
latRad	1 x :	double	required	positional

output	description
P	(nmax+1, nmax+1, nlat) double Pbar_nm(sin lat)
D	(nmax+1, nmax+1, nlat) double dPbar_nm/dphi [1/rad] D2 (nmax+1, nmax+1, nlat) double d2Pbar_nm/dphi2 [1/rad^2]: the SAME column stencil applied to D - exact, because the first-derivative identity holds degree-wise as a function of phi and differentiates term by term (Python-validated vs numerical second differences, rel ~2e-7 = FD floor; v2.4)

Example

```
[P, dP] = shLowLevel.legendreALFDeriv(60, theta);   % for deformation/tensor
```

shLowLevel.legendreCached

`P = shLowLevel.legendreCached(nmax, latRad)`

Cached fully normalized Legendre functions. P = shLowLevel.legendreCached(NMAX, LATRAD) returns shLowLevel.legendreALF(NMAX, LATRAD), serving repeated requests for the same (NMAX, LATRAD) from a process-wide cache. Cache hits are verified with an EXACT isequal comparison of the stored latitude vector -- a mismatched latVec can never silently return the wrong P (this removes the documented footgun of shSynthesis' raw 'P' passthrough).

input	size	type	default	kind
nmax	—	—	required	positional
latRad	—	—	required	positional

output	description
P	(nmax+1)x(nmax+1)xnumel(latRad) double

Example

```
shLowLevel.legendreCached("clear")          % drop the synthesis cache
```

shLowLevel.ylm

`Y = shLowLevel.ylm(latRad, lonRad, idx)`

4-pi normalized SH basis vectors at arbitrary points. Y = shLowLevel.ylm(LATRAD, LONRAD, IDX) evaluates the basis at nPts points (geocentric latitude/longitude in radians, equal-length vectors), so that f(points) = Y * x for a coefficient vector x in IDX ordering. Vectorized over points via one shLowLevel.legendreALF call.

input	size	type	default	kind
latRad	: x 1	double	required	positional

input	size	type	default	kind
lonRad	: x 1	double	required	positional
idx	1 x 1	struct	required	positional

output	description
Y	(nPts x idx.P) double

Example

```
Y = shLowLevel.ylm(deg2rad(latPts), deg2rad(lonPts), idx);   % design rows, idx order
```

shLowLevel.shIndex

`idx = shLowLevel.shIndex(Lmax, Name=Value)`

Canonical order-major index for SH coefficient vectors. idx = shLowLevel.shIndex(Lmax) builds the index table used by all shLowLevel functions: order-major ordering, C before S per (n,m), degrees n = MinDegree..Lmax (default MinDegree = 2; degree 0/1 and the SLR-C20 replacement are assumed to be handled upstream).

input	size	type	default	kind
Lmax	1 x 1	double	required	positional
MinDegree	1 x 1	double	2	name-value

output	description
idx	struct: n/m/cs (P x 1), P (1 x 1), nmax, MinDegree - the canonical vector ordering of the toolbox

Example

```
idx = shLowLevel.shIndex(60);               % MinDegree = 2 default
```

shLowLevel.vecFromCS

`x = shLowLevel.vecFromCS(cnm, snm, idx)`

Pack lower-triangular Cnm/Snm matrices into a shLowLevel vector. x = shLowLevel.vecFromCS(cnm, snm, idx) with cnm, snm of size (Lmax+1)x(Lmax+1), cnm(n+1,m+1) = Cnm, snm(n+1,m+1) = Snm.

input	size	type	default	kind
cnm	—	double	required	positional
snm	—	double	required	positional
idx	1 x 1	struct	required	positional

output	description
x	(P x 1) double coefficients gathered in idx ordering

Example

```
x = shLowLevel.vecFromCS(g10.C, g10.S, idx);     % triangle -> idx-ordered vector
```

shLowLevel.csFromVec

`[cnm, snm] = shLowLevel.csFromVec(x, idx)`

Unpack a shLowLevel vector into lower-triangular Cnm/Snm matrices. [cnm, snm] = shLowLevel.csFromVec(x, idx). Coefficients outside the index range (n < MinDegree) are zero.

input	size	type	default	kind
x	: x 1	double	required	positional
idx	1 x 1	struct	required	positional

output	description
cnm	(nmax+1 x nmax+1) double cosine coefficients, C(n+1, m+1)
snm	(nmax+1 x nmax+1) double sine coefficients, S(:, 1) = 0

Example

```
[C, S] = shLowLevel.csFromVec(x, idx);          % vector -> triangle matrices
```

shLowLevel.shDegreeRMS

`spec = shLowLevel.shDegreeRMS(C, S, varargin)`

Spectral (degree-domain) analysis of spherical harmonic coefficients. SPEC = SHDEGREERMS(C, S) computes, per degree n = 0..nmax: .degVariance(n+1) = sum_m (Cnm^2 + Snm^2) .degRMS(n+1) = sqrt(degVariance) .degAmplitude(n+1) = R * degRMS [geoid-height-equivalent, m] .cumAmplitude(n+1) = R * sqrt(cumsum(degVariance)) [cumulative, m]

input	size	type	default	kind
C	—	—	required	positional
S	—	—	required	positional
varargin	—	—	required	positional

output	description
spec	struct: degree (nmax+1 x 1), degVariance/degRMS/ degAmplitude, cumVariance/cumRMS/cumAmplitude, R, domain; errVariance/errRMS/errAmplitude when sigmas are supplied

Example

```
spec = shLowLevel.shDegreeRMS(g.C, g.S);          % degree-amplitude spectrum struct
```

shLowLevel.shOrderRMS

`spec = shLowLevel.shOrderRMS(C, S, Name=Value)`

Order-domain spectral diagnostics (striping direction). SPEC = shLowLevel.shOrderRMS(C, S) sums coefficient power per ORDER m - the complementary view to shLowLevel.shDegreeRMS and the natural axis for GRACE striping (correlated errors live at high orders):

input	size	type	default	kind
C	—	double	required	positional
S	—	double	required	positional
R	1 x 1	double	6378136.3	name-value
n0	1 x 1	double	0	name-value
sigmaC	—	double	[]	name-value
sigmaS	—	double	[]	name-value

output	description
spec	struct: order, ordVariance, ordRMS, ordAmplitude, cumVariance, cumRMS, cumAmplitude, (err*), R, domain = 'order' (consumed by plotSHSpectrum)

Example

```
spec = shLowLevel.shOrderRMS(g.C, g.S);          % stripes live at high orders
```

shLowLevel.shSpectralCrossover

`[nCrossover, degreeInterp] = shLowLevel.shSpectralCrossover(spec)`

Degree at which the error spectrum first exceeds the signal (degree-amplitude) spectrum -- a standard guide for choosing a truncation degree or Gaussian filter radius.

input	size	type	default	kind
spec	—	—	required	positional

output	description
nCrossover	(1,1) double first degree with err >= signal (NaN if none)
degreeInterp	(1,1) double linearly interpolated crossing degree

Example

```
spec = shLowLevel.shDegreeRMS(g.C, g.S, 'sigmaC', g.sigmaC, 'sigmaS', g.sigmaS);
nc = shLowLevel.shSpectralCrossover(spec);   % signal/error crossover degree
```

shLowLevel.gaussLegendre

`[x, w] = shLowLevel.gaussLegendre(n)`

Gauss-Legendre nodes and weights on [-1, 1]. [X, W] = shLowLevel.gaussLegendre(N) via Golub-Welsch (symmetric eigenvalue problem of the Jacobi matrix). Exact for polynomials up to degree 2N-1: integral f(t) dt = sum(W .* f(X)).

input	size	type	default	kind
n	1 x 1	double	required	positional

output	description
x	(n,1) double nodes, ascending
w	(n,1) double weights, sum(w) = 2

Example

```
[xNodes, w] = shLowLevel.gaussLegendre(61);   % quadrature nodes/weights
```

Synthesis & analysis

shLowLevel.shSynthesis

`[grid, lat, lon, P] = shLowLevel.shSynthesis(C, S, GM, R, latVec, lonVec, varargin)`

Spherical harmonic synthesis of a global/regional grid. [GRID, LAT, LON] = SHSYNTHESIS(C, S, GM, R, LATVEC, LONVEC) synthesizes the disturbing-potential-derived quantity (default: geoid height) on the grid LATVEC x LONVEC (degrees) from fully normalized (ICGEM) Stokes coefficients C, S at the reference sphere r = R.

input	size	type	default	kind
C	—	—	required	positional
S	—	—	required	positional
GM	—	—	required	positional
R	—	—	required	positional
latVec	—	—	required	positional
lonVec	—	—	required	positional
varargin	—	—	required	positional

output	description
grid	(nlat x nlon) double synthesized quantity
lat	(1, nlat) double latitude grid [deg]
lon	(1, nlon) double longitude grid [deg]

output	description
P	(1,1) struct cached Legendre functions (pass back in for reuse)

Example

```
grid = shLowLevel.shSynthesis(g.C, g.S, g.GM, g.R, -89:89, 0:359);
```

shLowLevel.synthesisMatrix

`[Y, w, grid] = shLowLevel.synthesisMatrix(idx, Name=Value)`

Dense SH synthesis matrix on a Gauss-Legendre grid. [Y, W, GRID] = shLowLevel.synthesisMatrix(IDX) returns Y (Ngrid x P) with f = Y*x for coefficient vectors x in IDX ordering (shLowLevel.shIndex), and quadrature weights W (Ngrid x 1), normalized such that the discrete analysis A = Y' * diag(W) is EXACT for band-limited fields: A*Y = eye(P) to machine precision (asserted in the test suite).

input	size	type	default	kind
idx	1 x 1	struct	required	positional
NLat	1 x 1	double	idx.Lmax + 1	name-value
NLon	1 x 1	double	2*idx.Lmax + 2	name-value

output	description
Y	(NLat*NLon x idx.P) double synthesis matrix
w	(NLat*NLon x 1) double quadrature weights, sum(w) = 1
grid	struct: theta, lambda [rad], latDeg (NLat x 1), lonDeg (NLon x 1)

Example

```
[Y, w] = shLowLevel.synthesisMatrix(idx);    % quadrature-consistent pair
```

shLowLevel.shSynthesisDeformation

`[up, north, east] = shLowLevel.shSynthesisDeformation(C, S, R, latDeg, lonDeg, Name=Value)`

Elastic load deformation: up / north / east. [UP, NORTH, EAST] = shLowLevel.shSynthesisDeformation(C, S, R, LAT, LON, kn (""")=..., hn (""")=..., ln (""")=...) synthesizes the three components of the elastic surface deformation caused by the load expressed in the (residual) Stokes coefficients C, S (Wahr et al. 1998; Kusche & Schrama 2005 - the GRACE <-> GNSS loading comparison):

input	size	type	default	kind
C	—	double	required	positional
S	—	double	required	positional
R	1 x 1	double	required	positional
latDeg	1 x :	double	required	positional
lonDeg	1 x :	double	required	positional
kn	—	double	required	name-value
hn	—	double	required	name-value
ln	—	double	required	name-value
nmin	1 x 1	double	1	name-value
Mode	1 x 1	string ∈ ["grid","points"]	"grid"	name-value

output	description
up	(nlat x nlon | npts) double vertical elastic deformation [m]
north	same size horizontal north component [m]
east	same size horizontal east component [m]

Example

```
[up, no, ea] = shLowLevel.shSynthesisDeformation(g.C, g.S, g.R, latPts, lonPts, kn = LN.kn, hn = LN.hn, ln = LN.ln
```

shLowLevel.shSynthesisGradientTensor

`[comps, info] = shLowLevel.shSynthesisGradientTensor(C, S, GM, R, latDeg, lonDeg, Name=Value)`

Full gravity gradient tensor in the local NEU frame. [COMPS, INFO] = shLowLevel.shSynthesisGradientTensor(C, S, GM, R, LAT, LON, Height (250e3)=250e3) synthesizes all six independent components of the disturbing gravity gradient tensor (GOCE-style diagnostics) in the local north-east-up frame at radius r = R + Height:

input	size	type	default	kind
C	—	double	required	positional
S	—	double	required	positional
GM	1 x 1	double	required	positional
R	1 x 1	double	required	positional
latDeg	1 x :	double	required	positional
lonDeg	1 x :	double	required	positional
Height	1 x 1	double	250e3	name-value
nmin	1 x 1	double	2	name-value

output	description
comps	struct of (nlat,nlon) double [1/s^2]: uu, nn, ee, un, ue, ne (symmetric tensor; 1 Eotvos = 1e-9 1/s^2). Components involving east are NaN at |lat| = 90.
info	struct: maxTraceResidual (max |trace|/max|G| over the grid, excluding poles) - the built-in Laplace self-check

Example

```
T = shLowLevel.shSynthesisGradientTensor(g.C, g.S, g.GM, g.R, lat, lon, Height = 450e3);
```

shLowLevel.shAnalysisGrid

`[C, S, info] = shLowLevel.shAnalysisGrid(grid, latVec, lonVec, nmax, Name=Value)`

Spherical harmonic analysis: estimate Stokes coefficients from gridded or scattered data (the inverse of shLowLevel.shSynthesis).

input	size	type	default	kind
grid	—	double	required	positional
latVec	—	double	required	positional
lonVec	—	double	required	positional
nmax	1 x 1	double	required	positional
Method	1 x 1	string ∈ ["auto","rings","ls"]	"auto"	name-value
quantity	—	—	'geoid'	name-value
GM	1 x 1	double	3.986004415e14	name-value
R	1 x 1	double	6378136.3	name-value
kn	—	double	[]	name-value
hn	—	double	[]	name-value
rho_ave	1 x 1	double	5517	name-value
rho_water	1 x 1	double	1000	name-value
Weights	1 x 1	string ∈ ["none","coslat"]	"none"	name-value
Kaula	1 x 1	double	0	name-value

input	size	type	default	kind
ChunkSize	1 x 1	double	2000	name-value
LatType	1 x 1	string ∈ ["geocentric","geodetic"]	"geocentric"	name-value
Flattening	1 x 1	double	1/298.257223563	name-value

output	description
C	(nmax+1 x nmax+1) double estimated cosine coefficients
S	(nmax+1 x nmax+1) double estimated sine coefficients
info	(1,1) struct fields: condest (1,1 double), nObs (1,1 double), kaula (1,1 double, NaN when unregularized)

Example

```
g2 = shLowLevel.shAnalysisGrid(grid, lat, lon, 10);   % exact on ring grids
```

shLowLevel.kernelFactors

`kernel = shLowLevel.kernelFactors(quantity, nmax, GM, R, Name=Value)`

Degree-dependent spectral factors for SH synthesis/analysis. KERNEL = shLowLevel.kernelFactors(QUANTITY, NMAX, GM, R, kn ([])=..., ...) returns the (NMAX+1)x1 factor f_n such that a field of QUANTITY is q(lat,lon) = sum_n f_n * sum_m Pbar_nm (C_nm cos + S_nm sin). Shared by shLowLevel.shSynthesis (multiply) and shLowLevel.shAnalysisGrid (divide); the single source of truth for the quantity definitions.

input	size	type	default	kind
quantity	—	—	required	positional
nmax	1 x 1	double	required	positional
GM	1 x 1	double	required	positional
R	1 x 1	double	required	positional
kn	—	double	[]	name-value
hn	—	double	[]	name-value
rho_ave	1 x 1	double	5517	name-value
rho_water	1 x 1	double	1000	name-value
Height	1 x 1	double	0	name-value

output	description
kernel	(nmax+1,1) double (zeros where the quantity carries no information, e.g. n=1 for gravity_anomaly)

Example

```
kf = shLowLevel.kernelFactors("ewh", 60, GM, R, kn = kn);
```

shLowLevel.normalFieldCS

`[Cn, info] = shLowLevel.normalFieldCS(nmax, Name=Value)`

Even zonal harmonics of an ellipsoidal normal field. [CN, INFO] = shLowLevel.normalFieldCS(NMAX, System="WGS84") returns the fully normalized even zonal coefficients Cbar_{2n,0} of the normal gravity field, COMPUTED from the defining constants (GM (NaN), a (NaN), f (NaN), omega (NaN)) via the closed-form theory (Heiskanen & Moritz 1967, sect. 2-9) - no coefficient tables anywhere:

input	size	type	default	kind
nmax	1 x 1	double	required	positional
System	1 x 1	string ∈ ["WGS84","GRS80"]	"WGS84"	name-value
GM	1 x 1	double	NaN	name-value
a	1 x 1	double	NaN	name-value

input	size	type	default	kind
f	1 x 1	double	NaN	name-value
omega	1 x 1	double	NaN	name-value

output	description
Cn	(nmax+1,1) double Cbar_{n,0} (zero at odd/omitted degrees; Cn(1) = 1 represents the central GM/r term - subtract it ONLY together with a matching degree-0 of the model, see subtractNormalField)
info	struct: GM, a, f, omega, J2, e2, m, q0, system

Example

```
[Cn, info] = shLowLevel.normalFieldCS(96, System = "WGS84");
```

shLowLevel.rescaleGMR

`[C2, S2, sig] = shLowLevel.rescaleGMR(C, S, GM1, R1, GM2, R2, sigmaC, sigmaS)`

Convert Stokes coefficients between (GM, R) reference values. [C2, S2] = shLowLevel.rescaleGMR(C, S, GM1, R1, GM2, R2) re-expresses coefficients given w.r.t. (GM1, R1) in the reference (GM2, R2):

input	size	type	default	kind
C	—	double	required	positional
S	—	double	required	positional
GM1	1 x 1	double	required	positional
R1	1 x 1	double	required	positional
GM2	1 x 1	double	required	positional
R2	1 x 1	double	required	positional
sigmaC	—	double	[]	positional
sigmaS	—	double	[]	positional

output	description
C2, S2	(n1,n1) double
sig	struct with fields C, S (empty if no sigmas given)

Example

```
[C2, S2] = shLowLevel.rescaleGMR(g.C, g.S, g.GM, g.R, GMref, Rref);
```

shLowLevel.geodetic2geocentric

`latGC = shLowLevel.geodetic2geocentric(latGD, Name=Value)`

Convert geodetic to geocentric latitude. LATGC = shLowLevel.geodetic2geocentric(LATGD) converts geodetic (ellipsoidal) latitudes to geocentric latitudes via tan(latGC) = (1 - f)^2 * tan(latGD) All spherical-harmonic routines in this toolbox (synthesis, analysis, Legendre functions) expect GEOCENTRIC latitude; user grids from maps, GIS or mascon products are almost always GEODETIC. Forgetting this conversion biases mid-latitude values by up to ~0.19 deg of latitude (~21 km) - a classic silent error source.

input	size	type	default	kind
latGD	—	double	required	positional
Flattening	1 x 1	double	1/298.257223563	name-value

output	description
latGC	double array geocentric latitudes [deg], poles/equator exact

Example

```
latGc = shLowLevel.geodetic2geocentric(latGd);   % before synthesis
```

shLowLevel.geocentric2geodetic

`latGD = shLowLevel.geocentric2geodetic(latGC, Name=Value)`

Convert geocentric to geodetic latitude. LATGD = shLowLevel.geocentric2geodetic(LATGC) inverts shLowLevel.geodetic2geocentric: tan(latGD) = tan(latGC) / (1 - f)^2.

input	size	type	default	kind
latGC	—	double	required	positional
Flattening	1 x 1	double	1/298.257223563	name-value

output	description
latGD	double array geodetic latitudes [deg]

Example

```
latGd = shLowLevel.geocentric2geodetic(latGc);   % WGS84
```

shLowLevel.evalMask

`[mask, grid] = shLowLevel.evalMask(idx, region, Name=Value)`

Evaluate a region definition on the toolbox quadrature grid. [MASK, GRID] = shLowLevel.evalMask(IDX, REGION) evaluates REGION on the Gauss-Legendre quadrature grid of shLowLevel.synthesisMatrix(IDX) and returns MASK (Ngrid x 1, in [0,1]) plus the GRID struct. Shared by shLowLevel.basinKernel and shLowLevel.slepianBasis.

input	size	type	default	kind
idx	1 x 1	struct	required	positional
region	—	—	required	positional
BufferKm	1 x 1	double	0	name-value
R	1 x 1	double	6378136.3	name-value
OverSample	1 x 1	double	2	name-value

output	description
mask	(Ngrid,1) double in [0,1]
grid	struct from shLowLevel.synthesisMatrix (latDeg, lonDeg per ring)

Example

```
[mask, area] = shLowLevel.evalMask(idx, capSpec);   % quadrature-grid mask
```

Filtering & operators

shLowLevel.shGaussianWeights

`Wn = shLowLevel.shGaussianWeights(nmax, radius_km, varargin)`

Degree-domain weights for isotropic Gaussian smoothing. WN = SHGAUSSIANWEIGHTS(NMAX, RADIUS_KM) computes the Jekeli (1981) Gaussian averaging function weights Wn, n = 0..NMAX, for an averaging radius RADIUS_KM [km] (half-width at which the spatial weighting function drops to 1/2). Standard GRACE/GRACE-FO mass-change smoothing.

input	size	type	default	kind
nmax	—	—	required	positional
radius_km	—	—	required	positional
varargin	—	—	required	positional

output	description
Wn	(nmax+1 x 1) double Jekeli (1981) weights, W(1) = 1, monotone non-increasing

Example

```
w = shLowLevel.shGaussianWeights(60, 350);   % Jekeli recursion
```

shLowLevel.shGaussianFilter

`[Cf, Sf, Wn] = shLowLevel.shGaussianFilter(C, S, radius_km, varargin)`

Apply isotropic Gaussian smoothing to Stokes coefficients. [CF, SF] = SHGAUSSIANFILTER(C, S, RADIUS_KM) multiplies each degree n row of C, S by the Jekeli (1981) weight Wn(n), see SHGAUSSIANWEIGHTS. Standard pre-processing step for GRACE/GRACE-FO mass-change maps before spatial synthesis, used to suppress high-degree noise.

input	size	type	default	kind
C	—	—	required	positional
S	—	—	required	positional
radius_km	—	—	required	positional
varargin	—	—	required	positional

output	description
Cf	(nmax+1 x nmax+1) double smoothed cosine coefficients
Sf	(nmax+1 x nmax+1) double smoothed sine coefficients
Wn	(nmax+1 x 1) double Jekeli degree weights applied

Example

```
[Cf, Sf] = shLowLevel.shGaussianFilter(g.C, g.S, 350);
```

shLowLevel.shFanFilter

`[Cf, Sf] = shLowLevel.shFanFilter(C, S, rDegKm, rOrdKm, Name=Value)`

Han fan filter: separable degree x order Gaussian smoothing. [CF, SF] = shLowLevel.shFanFilter(C, S, RDEGKM, RORDKM) applies the fan filter of Han et al. (2005): the product of a Jekeli Gaussian in DEGREE and one in ORDER, W_nm = W_n(rDeg) * W_m(rOrd), damping high orders (the striping direction) harder than an isotropic Gaussian of equal degree radius - the cheap anisotropic benchmark between Jekeli and DDK/tvANS. Common choice: rOrd = rDeg (original fan) or rOrd < rDeg for stronger destriping.

input	size	type	default	kind
C	—	double	required	positional
S	—	double	required	positional
rDegKm	1 x 1	double	required	positional
rOrdKm	1 x 1	double	required	positional
R	1 x 1	double	6378136.3	name-value

output	description
Cf, Sf	(n1,n1) double filtered coefficients

Example

```
[C2, S2] = shLowLevel.shFanFilter(g.C, g.S, 350, 200);
```

shLowLevel.shDestripe

`[Cf, Sf] = shLowLevel.shDestripe(C, S, varargin)`

Remove correlated (north-south striping) noise from Stokes coefficients, following Swenson & Wahr (2006).

input	size	type	default	kind
C	—	—	required	positional
S	—	—	required	positional
varargin	—	—	required	positional

output	description
Cf	(nmax+1 x nmax+1) double destriped cosine coefficients
Sf	(nmax+1 x nmax+1) double destriped sine coefficients

Example

```
[Cf, Sf] = shLowLevel.shDestripe(g.C, g.S, minOrder = 6);
```

shLowLevel.applyDDK

`[Cf, Sf] = shLowLevel.applyDDK(C, S, W)`

Apply an order-block-diagonal (DDK-style) filter to coefficients. [CF, SF] = shLowLevel.applyDDK(C, S, W) applies the filter container from shLowLevel.readDDK: within each (order m, cs) block, the degree vector c(n1..n2) is replaced by W.blocks(k).M * c. Degrees present in C but not covered by any block pass through UNCHANGED (typically n < 2). User-facing wrappers: g.applyDDK(W), ts.applyDDK(W).

input	size	type	default	kind
C	—	double	required	positional
S	—	double	required	positional
W	1 x 1	struct	required	positional

output	description
Cf	(nmax+1 x nmax+1) double DDK-filtered cosine coefficients
Sf	(nmax+1 x nmax+1) double DDK-filtered sine coefficients

Example

```
W = shLowLevel.readDDK("DDK3", Nmax = 60);
[Cf, Sf] = shLowLevel.applyDDK(g.C, g.S, W);
```

shLowLevel.tvANSFilter

`[Xf, op, info] = shLowLevel.tvANSFilter(X, tYears, idx, Name=Value)`

Time-variable anisotropic Wiener filtering of a SH series. [XF, OP, INFO] = shLowLevel.tvANSFilter(X, TYEARS, IDX) runs the tvANS chain on the coefficient series X (P x T, IDX ordering, shLowLevel.shIndex):

input	size	type	default	kind
X	—	double	required	positional
tYears	: x 1	double	required	positional
idx	1 x 1	struct	required	positional
NoiseCov	—	double	[]	name-value
Constraints	—	double	[]	name-value
SignalMode	—	— ∈ {'isotropic','inhomogeneous'}	'isotropic'	name-value
NIterSignal	1 x 1	double	3	name-value
Robust	1 x 1	logical	false	name-value
Shrinkage	1 x 1	double	0.1	name-value
VCEMinDegree	1 x 1	double	round(2/3 * idx.Lmax)	name-value
VCEBands	1 x :	double	[]	name-value

input	size	type	default	kind
Blocks	—	— ∈ {'auto','on','off'}	'auto'	name-value

output	description
Xf	(P,T) double filtered series (deterministic part restored)
op	struct filter operator: Ut, V, lam, s, Ac, SA, M, idx, tYears, model, Xres, Xfres (use with shLowLevel.opApply / shLowLevel.basinDeconvolve / shLowLevel.resolutionMap)
info	struct noise/signal/VCE diagnostics; additionally info.sigmaXfres (P x T): 1-sigma posterior uncertainty of the filtered residual per coefficient/month, diag((I-W_t)S) = (V.^2)*(s_t*lam./(lam+s_t)) (validated to 3.5e-16 against the direct matrix product). With constraints (v2.5) this is the EXACT diagonal of (W-I)S(W-I)' + s*W*N*W' - the earlier unconstrained formula underestimated along constrained directions.

Example

```
[Xf, op, info] = shLowLevel.tvANSFilter(X, tYears, idx);
sigma = info.sigmaXfres;              % exact posterior (v2.5 incl. constraints)
```

shLowLevel.buildNoiseCov

`[N, info] = shLowLevel.buildNoiseCov(Xres, idx, Name=Value)`

Monthly-solution noise covariance (shape matrix). [N, info] = shLowLevel.buildNoiseCov(Xres, idx) estimates the noise covariance from the residual coefficient series Xres (P x T, residuals about the deterministic model). Two modes:

input	size	type	default	kind
Xres	—	double	required	positional
idx	1 x 1	struct	required	positional
Mode	—	— ∈ {'empirical','full'}	'empirical'	name-value
FullCov	—	double	[]	name-value
Shrinkage	1 x 1	double	0.1	name-value
Assemble	—	— ∈ {'full','blocks'}	'full'	name-value

output	description
N	(P x P) double or block struct noise covariance (order/parity block-diagonal for Assemble ('full')='blocks')
info	struct: mode, shrinkage, blocks metadata

Example

```
N = shLowLevel.buildNoiseCov(Xres, idx);          % empirical order/parity blocks
```

shLowLevel.buildSignalCov

`[S, info] = shLowLevel.buildSignalCov(Xres, N, idx, Name=Value)`

Iterative, data-driven signal covariance. [S, info] = shLowLevel.buildSignalCov(Xres, N, idx) estimates the signal covariance from the residual series itself - no hydrology prior.

input	size	type	default	kind
Xres	—	double	required	positional
N	—	—	required	positional
idx	1 x 1	struct	required	positional
Mode	—	— ∈ {'isotropic','inhomogeneous'}	'isotropic'	name-value
NIter	1 x 1	double	3	name-value
FloorRel	1 x 1	double	1e-8	name-value

input	size	type	default	kind
MapSmooth	1 x 1	logical	true	name-value

output	description
S	(P x P) double signal covariance in idx ordering
info	struct: mode, iterations, degree-variance model

Example

```
S = shLowLevel.buildSignalCov(Xres, N, idx, Mode = "isotropic");
```

shLowLevel.vceRescale

`s = shLowLevel.vceRescale(N, Xres, idx, Name=Value)`

Per-month variance factors for the noise covariance shape. s = shLowLevel.vceRescale(N, Xres, idx) estimates one variance factor per month such that N_t = s(t) * N. The factor is the robust (median-based, chi-square(1) corrected) ratio between the observed squared residuals and the formal variances, evaluated over noise-dominated coefficients (n >= MinDegree, default round(2/3 * idx.Lmax)). This is the single-component Helmert VCE specialization; it absorbs the month-to-month noise level changes (orbit geometry,...

input	size	type	default	kind
N	—	—	required	positional
Xres	—	double	required	positional
idx	1 x 1	struct	required	positional
MinDegree	1 x 1	double	round(2/3 * idx.Lmax)	name-value
Rows	—	logical	true(0, 1)	name-value

output	description
s	(T x 1) or (nBands x T) double monthly (band-wise) VCE noise variance factors

Example

```
s = shLowLevel.vceRescale(N, Xres, idx);          % monthly noise factors
```

shLowLevel.opApply

`Yout = shLowLevel.opApply(op, Z, t, mode)`

Apply the month-t filter operator W_t (or W_t') to columns of Z. Yout = shLowLevel.opApply(op, Z, t) computes W_t * Z Yout = shLowLevel.opApply(op, Z, t, 'transp') computes W_t' * Z

input	size	type	default	kind
op	1 x 1	struct	required	positional
Z	—	double	required	positional
t	1 x 1	double	required	positional
mode	—	— ∈ {'forward','transp'}	'forward'	positional

output	description
Yout	(P x T) double operator applied to every column of Y

Example

```
xf = shLowLevel.opApply(op, op.Xres(:, 1), 1);   % W_1 * x without the P x P matrix
```

shLowLevel.resolutionMap

`res = shLowLevel.resolutionMap(op, t, latDeg, lonDeg, Name=Value)`

Filter-kernel half-widths: per-point, per-azimuth resolution. RES = shLowLevel.resolutionMap(OP, T, LATDEG, LONDEG) evaluates, for each query point, the effective smoothing kernel of month T: xf(P) = y_P' * W_t * x => kernel coefficients k = W_t' * y_P and finds the great-circle distance psi_1/2 at which the kernel drops to half its peak, along NAz (8) azimuths. Replaces the single "equivalent Gaussian radius" with a spatially and azimuthally resolved resolution product (N-S vs E-W half-width ratio = stripin...

input	size	type	default	kind
op	1 x 1	struct	required	positional
t	1 x 1	double	required	positional
latDeg	: x 1	double	required	positional
lonDeg	: x 1	double	required	positional
NAz	1 x 1	double	8	name-value
PsiMax	1 x 1	double	20	name-value
NPsi	1 x 1	double	120	name-value

output	description
res	(nlat x nlon) double local resolution [km] of the filter

Example

```
res = shLowLevel.resolutionMap(op, 1, -60:30:60, 0:60:300);   % epoch 1
```

Basins, uncertainty & analysis tools

shLowLevel.basinKernel

`[b, info] = shLowLevel.basinKernel(idx, region, Name=Value)`

Band-limited basin kernel(s) from a region definition. [B, INFO] = shLowLevel.basinKernel(IDX, REGION) builds the SH coefficient vector of the band-limited indicator function of REGION by exact Gauss-Legendre quadrature, b = Y' * (w .* mask), ready for shSeries.basinAverage / shLowLevel.basinDeconvolve (P x 1, IDX ordering). Replaces the error-prone manual Y'*(w.*mask) construction.

input	size	type	default	kind
idx	1 x 1	struct	required	positional
region	—	—	required	positional
BufferKm	1 x 1	double	0	name-value
TaperKm	1 x 1	double	0	name-value
R	1 x 1	double	6378136.3	name-value
OverSample	1 x 1	double	2	name-value

output	description
b	(P,1) double kernel in IDX ordering
info	struct: areaFraction (quadrature area of the mask / 4pi), nGridPoints, buffered/tapered flags

Example

```
b = shLowLevel.basinKernel(idx, maskAmazon, TaperKm = 200);
```

shLowLevel.basinDeconvolve

`[avgHat, out] = shLowLevel.basinDeconvolve(B, op, Name=Value)`

Unbiased basin averages by exact operator deconvolution. [avgHat, out] = shLowLevel.basinDeconvolve(B, op) removes filter attenuation and inter-basin leakage from basin averages, exploiting that W_t is a KNOWN linear operator (no hydrology-model scale factors).

input	size	type	default	kind
B	—	double	required	positional
op	1 x 1	struct	required	positional
Ridge	1 x 1	double	0	name-value

output	description
avg	(K x T) double deconvolved basin averages
out	struct: c (K x T) deconvolved coefficients, sigma (K x T) 1-sigma incl. deterministic term (v2.5), avgNaive, attn, condA

Example

```
[avg, out] = shLowLevel.basinDeconvolve(B, op);   % op from ts.filter("tvANS")
errorbar(op.tYears, avg(1,:), out.sigma(1,:))
```

shLowLevel.basinScaling

`[k, info] = shLowLevel.basinScaling(op, b, tsModel, Name=Value)`

Leakage/attenuation scaling factors by forward modelling. [K, INFO] = shLowLevel.basinScaling(OP, B, TSMODEL) estimates the classic basin scaling (gain) factor: push a MODEL series (hydrology model, mascon-derived field, synthetic pattern) through the SAME filter operator OP that processed the data, and regress true against filtered basin averages: k = sum(aTrue .* aFilt) / sum(aFilt.^2) Scaled data series: k * (b' xFiltered)/(b'b). Complements shLowLevel.basinDeconvolve: deconvolution inverts the operator on a ...

input	size	type	default	kind
op	—	—	required	positional
b	: x 1	double	required	positional
tsModel	1 x 1	shSeries	required	positional
idx	—	struct	struct()	name-value
tYears	1 x :	double	[]	name-value
PerMonth	1 x 1	logical	false	name-value

output	description
k	(1,1) double regression scaling factor
info	struct: sigmaK (regression 1-sigma), kMonthly (T x 1, if PerMonth), matched (T x 1 logical), aTrue, aFilt

Example

```
k = shLowLevel.basinScaling(op, b, tsHydModel);   % scale factor vs a model series
```

shLowLevel.fitDeterministicModel

`[model, Xres, coef, A, coefSigma, resVar, ar1] = shLowLevel.fitDeterministicModel(X, tYears, Name=Value)`

Bias/trend/(semi-)annual (+extra periods) per SH coefficient. [MODEL, XRES, COEF, A, COEFSIGMA, RESVAR] = shLowLevel.fitDeterministicModel(X, TYEARS, ...) fits, for each coefficient time series (rows of X, P x T), the model x(t) = a + b*tc + c*cos(2*pi*tc) + d*sin(2*pi*tc) + e*cos(4*pi*tc) + f*sin(4*pi*tc) + sum_k [g_k*cos(2*pi*tc/p_k) + h_k*sin(2*pi*tc/p_k)] with tc = t - T0 (NaN) and optional extra periods p_k [years] - e.g. the GRACE tidal alias periods S2 = 161/365.25, K2 = 3.66, K1 = 7.48.

input	size	type	default	kind
X	—	double	required	positional
tYears	: x 1	double	required	positional
Robust	1 x 1	logical	false	name-value

input	size	type	default	kind
HuberK	1 x 1	double	1.5	name-value
MaxIter	1 x 1	double	10	name-value
T0	1 x 1	double	NaN	name-value
Periods	1 x :	double	double.empty(1,0)	name-value
Weights	: x 1	double	[]	name-value
ARCorrect	1 x 1	logical	false	name-value
Breaks	1 x :	double	double.empty(1,0)	name-value

output	description
model	P x T fitted deterministic part
Xres	P x T X - model
coef	ncol x P [bias; trend; cosA; sinA; cosSA; sinSA; cosP1; sinP1; ...], ncol = 6 + 2*numel(Periods)
A	T x ncol design matrix (unweighted)
coefSigma	ncol x P 1-sigma per coefficient: sqrt(resVar_p * diag(inv(A'WA))); OLS formula, validated by Monte Carlo (ratios 0.99-1.02). For Robust=true the final IRLS weights enter W (approximation, documented).
resVar	1 x P residual variance rss/(T - ncol)
ar1	1 x P bias-corrected lag-1 residual auto- correlation actually applied (clamped to [0, 0.99]; 0 when ARCorrect=false)

Example

```
[model, Xres, coef, A, sig] = shLowLevel.fitDeterministicModel(X, tYears, ARCorrect = true);
```

shLowLevel.mcPropagate

`out = shLowLevel.mcPropagate(fun, g, Name=Value)`

Monte-Carlo uncertainty propagation through arbitrary chains. OUT = shLowLevel.mcPropagate(FUN, G) draws coefficient samples consistent with the formal errors of the shCoefficients G, pushes each through FUN, and returns empirical statistics - uncertainty propagation for ANY functional (filtered grids, basin averages, amplitude maps, ...) without deriving analytic formulas, and the standard cross-check for every analytic sigma in this toolbox.

input	size	type	default	kind
fun	1 x 1	function_handle	required	positional
g	1 x 1	shCoefficients	required	positional
N	1 x 1	double	500	name-value
Cov	—	double	[]	name-value
Idx	—	struct	struct([])	name-value
Seed	—	double	[]	name-value
KeepSamples	1 x 1	logical	false	name-value

output	description
out	(1,1) struct fields: samples (K x nSamples double), mean, sigma (K x 1 double) Monte-Carlo moments of the functional

Example

```
out = shLowLevel.mcPropagate(@(gs) gs.C(3, 1), g, N = 200);   % any functional of g
```

shLowLevel.errorMap

`[sig, latDeg, lonDeg] = shLowLevel.errorMap(M, idx, latDeg, lonDeg, Name=Value)`

Analytic formal-error maps from a full coefficient covariance. [SIG, LAT, LON] = shLowLevel.errorMap(M, IDX, LAT, LON, quantity ("geoid")="ewh", kn ([])=kn) computes the pointwise 1-sigma of a synthesized field from a full P x P covariance (e.g. shLowLevel.readSINEX(...,

input	size	type	default	kind
M	—	double	required	positional
idx	1 x 1	struct	required	positional
latDeg	1 x :	double	required	positional
lonDeg	1 x :	double	required	positional
quantity	1 x 1	string	"geoid"	name-value
kn	—	double	[]	name-value
hn	—	double	[]	name-value
GM	1 x 1	double	3.986004415e14	name-value
R	1 x 1	double	6378136.3	name-value
Height	1 x 1	double	0	name-value

output	description
sig	(nlat x nlon) double 1-sigma of the synthesized quantity
latDeg	(1, nlat) double latitude grid used [deg]
lonDeg	(1, nlon) double longitude grid used [deg]

Example

```
sig = shLowLevel.errorMap(M, idx, -89:4:89, 0:4:356, quantity = "ewh", kn = kn);
```

shLowLevel.combineCenters

`[tsComb, info] = shLowLevel.combineCenters(tsCell, Name=Value)`

VCE-weighted combination of multi-center monthly series. [TSCOMB, INFO] = shLowLevel.combineCenters({tsITSG, tsCSR, tsGFZ}) combines monthly solutions from several processing centers into one series with one variance factor per (center, month), estimated by Foerstner variance-component iteration with PARTIAL REDUNDANCIES:

input	size	type	default	kind
tsCell	1 x :	cell	required	positional
Tolerance	1 x 1	double	0.05	name-value
AllowMissing	1 x 1	logical	true	name-value
Robust	1 x 1	logical	false	name-value
MaxIter	1 x 1	double	6	name-value
Tol	1 x 1	double	1e-3	name-value

output	description
tsComb	shSeries combined stacks on the reference epochs (first center), sigmaCs/Ss = posterior sqrt(diag(H)) per month, provenance in history
info	struct: s2 (C x T variance factors, NaN where missing), weights (C x T, normalized 1/s2), redundancy (C x T), interCenterCorr (C x C, empirical residual correlation - the honesty diagnostic: VCE assumes independence, GRACE centers share data and background models), nIter, epochs, centers, note

Example

```
tsC = shLowLevel.combineCenters({tsCSR, tsGFZ, tsJPL});   % VCE-weighted
```

shLowLevel.eofAnalysis

```
[modes, pcs, varExplained, info] = shLowLevel.eofAnalysis(ts, Name=Value)
```

Empirical orthogonal functions of a coefficient series. [MODES, PCS, VAREXPLAINED, INFO] = shLowLevel.eofAnalysis(TS) decomposes the (typically residual) series into spatial modes and principal components via SVD of the flattened, time-centered coefficient stack: X = U S V'; MODES{k} carries U(:,k)*S(k,k)/sqrt(T-1) as an shCoefficients (synthesize for the spatial pattern), PCS(:,k) = sqrt(T-1)*V(:,k) is the unit-variance temporal amplitude, so mode-k field * pc reconstructs the mode's contribution exactly.

input	size	type	default	kind
ts	1 x 1	shSeries	required	positional
NModes	1 x 1	double	6	name-value
Center	1 x 1	logical	true	name-value

output	description
modes	1 x K cell of shCoefficients (spatial patterns)
pcs	(T,K) double, unit variance columns
varExplained	(K,1) double, fraction in [0,1]
info	struct: singular values, total variance, T

Example

```
[modes, pcs, ve] = shLowLevel.eofAnalysis(resid);
plot(resid.epochs, pcs(:, 1))        % leading principal component
```

shLowLevel.slepianBasis

```
[G, lam, info] = shLowLevel.slepianBasis(idx, region, Name=Value)
```

Spatially concentrated (Slepian) basis for a region. [G, LAM, INFO] = shLowLevel.slepianBasis(IDX, REGION) solves the spherical concentration problem of Simons et al. (2006): find band-limited functions g = Y*x maximizing the energy fraction inside REGION, lambda = (x' K x) / (x' x), K = Y' diag(w .* mask) Y, by symmetric eigendecomposition of the localization kernel K (exact Gauss-Legendre quadrature). Columns of G are the Slepian taper coefficient vectors, orthonormal (G'G = I), sorted by concentration LAM (...

input	size	type	default	kind
idx	1 x 1	struct	required	positional
region	—	—	required	positional
NKeep	—	double	[]	name-value
BufferKm	1 x 1	double	0	name-value
R	1 x 1	double	6378136.3	name-value
OverSample	1 x 1	double	2	name-value

output	description
G	(P,NKeep) double taper coefficient vectors, G'G = I
lam	(NKeep,1) double concentrations, descending in [0,1]
info	struct: shannon (= areaFraction*P, exact by the addition theorem), areaFraction, lamAll (P x 1)

Example

```
[G, lam] = shLowLevel.slepianBasis(idx, capSpec);   % concentration basis
```

shLowLevel.seaLevelFingerprint

```
[S, grid, info] = shLowLevel.seaLevelFingerprint(loadRegion, ocean, idx, Name=Value)
```

Elastic sea-level fingerprint (sea-level equation). [S, GRID, INFO] = shLowLevel.seaLevelFingerprint(LOADREGION, OCEAN, IDX, kn (“”)=kn, hn (“”)=hn) solves the elastic sea-level equation for a land load: the ocean does NOT rise uniformly - it follows the perturbed geoid minus the deformed

crust under global mass conservation (Farrell & Clark 1976, elastic limit):

input	size	type	default	kind
loadRegion	—	—	required	positional
ocean	—	—	required	positional
idx	1 x 1	struct	required	positional
kn	—	double	required	name-value
hn	—	double	required	name-value
LoadValue	1 x 1	double	1	name-value
rho_water	1 x 1	double	1000	name-value
rho_ave	1 x 1	double	5517	name-value
R	1 x 1	double	6378136.3	name-value
OverSample	1 x 1	double	2	name-value
MaxIter	1 x 1	double	50	name-value
Tol	1 x 1	double	1e-8	name-value

output	description
S	(Ngrid,1) double relative sea-level change over the ocean [m per unit load], 0 on land
grid	struct latDeg/lonDeg rings of the solver grid
info	struct: iterations, dS, massResidual [kg/m^2 sphere mean], eustatic [m], kappa, areaFractionOcean, S2D (nlat x nlon)

Example

```
S = shLowLevel.seaLevelFingerprint(gLoad, oceanMask, idx, kn = LN.kn, hn = LN.hn);
```

shLowLevel.pctile

`v = shLowLevel.pctile(x, p)`

Percentile without the Statistics Toolbox (base MATLAB only). V = shLowLevel.pctile(X, P) returns the P-th percentile(s) (0..100) of the finite values of X(:), using the same linear interpolation between order statistics as the Statistics Toolbox prctile: sample k maps to percentile 100*(k-0.5)/n. P may be a vector (v2.5.1); V has the same size as P. NaN/Inf are ignored; empty input yields NaN(size(P)).

input	size	type	default	kind
x	—	double	required	positional
p	1 x :	double	required	positional

output	description
v	(size(p)) double percentile value per requested p

Example

```
q = shLowLevel.pctile(res, [16 50 84]);      % base-MATLAB percentiles
```

Comparison & validation metrics (v2.6.0)

shLowLevel.compareSolutions

`[rep, h] = shLowLevel.compareSolutions(g1, g2, Name=Value)`

Full spectral + spatial comparison of two solutions. REP = shLowLevel.compareSolutions(G1, G2) compares two shCoefficients on a COMMON BASIS - both are truncated to the smaller nmax and G2 is rescaled to G1's GM/R if they differ (otherwise the comparison measures processing chains, not solutions) - and returns the standard metric set: spectral: difference degree amplitude vs both signal spectra, per-degree correlation, crossover degree of agreement (shLowLevel.diffSpectrum) spatial: area-weighted bias/RMSD/cent...

input	size	type	default	kind
g1	1 x 1	shCoefficients	required	positional
g2	1 x 1	shCoefficients	required	positional
LatDeg	1 x :	double	-89:2:89	name-value
LonDeg	1 x :	double	0:2:358	name-value
Quantity	1 x 1	string	"geoid"	name-value
kn	—	double	[]	name-value
Mask	—	—	[]	name-value
Names	1 x 2	string	["solution 1", "solution 2"]	name-value
Plot	1 x 1	logical	false	name-value

output	description
rep	(1,1) struct fields: .spectral (1,1) struct shLowLevel.diffSpectrum output .spatial (1,1) struct shLowLevel.spatialStats output .chi2dof (1,1) double error-realism statistic (NaN when either solution lacks sigmas) .nmax (1,1) double common maximum degree used .quantity (1,1) string compared quantity .names (1,2) string solution labels .rescaled (1,1) logical true when GM/R were unified
h	(1,1) graphics handle figure handle (Plot = true only)

Example

```
rep = shLowLevel.compareSolutions(g, g.gaussian(350), Names = ["raw", "G350"]);
fprintf("crossover n = %d, spatial corr = %.3f\n", ...
    rep.spectral.ncross, rep.spatial.corr)
```

shLowLevel.compareSeries

`[rep, h] = shLowLevel.compareSeries(tsList, Name=Value)`

Temporal + component comparison of N solution series. REP = shLowLevel.compareSeries({TS1, TS2, ...}) compares N >= 2 shSeries against the FIRST (the reference) on a COMMON BASIS: epochs are matched to the reference within MatchTolerance (0.02) (unmatched epochs are dropped and reported - never interpolated across gaps), all series are truncated to the smallest nmax, and GM/R are unified. Per solution k >= 2 the standard temporal metric set is computed on a scalar comparison series y_k(t) - a basin average when ...

input	size	type	default	kind
tsList	1 x :	cell	required	positional
Names	—	string	strings(1, 0)	name-value
MatchTolerance	1 x 1	double	0.02	name-value
Basin	—	double	[]	name-value
Idx	—	—	[]	name-value
LatDeg	1 x :	double	-88:4:88	name-value
LonDeg	1 x :	double	0:6:354	name-value
Quantity	1 x 1	string	"geoid"	name-value
kn	—	double	[]	name-value
Mask	—	—	[]	name-value
Plot	1 x 1	logical	false	name-value

output	description
rep	(1,1) struct fields: .names (1,N) string .epochs (T,1) double common epochs used .nDropped (1,N) double reference epochs unmatched per series .y (T,N) double comparison series (col 1 = reference) .nse (1,N) double NSE vs reference (NaN for col 1) .corr (1,N) struct-array-free: Pearson r per series .pcorr (1,N) double AR(1)-corrected two-sided p .neff (1,N) double effective sample sizes .trendDiff (1,N) double trend - reference trend [units/yr] .trendSigma (1,N) double combined 1-sigma of trendDiff .trendZ (1,N) double significance z of trendDiff .ampRatio (1,N) double annual amplitude / reference .phaseLagDays (1,N) double annual phase lag vs reference .chi2dof (1,N) double error realism vs reference .trendDiffFields (1,N) cell shCoefficients (empty for col 1) .tch (1,1) struct shLowLevel.threeCorneredHat output (N >= 3) .nmax (1,1) double common maximum degree
h	(1,1) graphics handle figure handle (Plot = true only)

Example

```
    rep = shLowLevel.compareSeries({tsCSR, tsGFZ, tsJPL}, ...
        Names = ["CSR", "GFZ", "JPL"]);
    fprintf("TCH noise: %s\n", sprintf("%.3g ", rep.tch.sigma))
```

shLowLevel.diffSpectrum

`spec = shLowLevel.diffSpectrum(C1, S1, C2, S2)`

Spectral comparison of two coefficient sets. SPEC = shLowLevel.diffSpectrum(C1, S1, C2, S2) compares two solutions in the spectral domain: the difference degree amplitude localizes the disagreement in wavelength, the degree correlation separates "same pattern, different amplitude" from genuine disagreement, and the crossover of the difference against the signal spectrum gives the effective resolution of agreement as a single scalar. Inputs must share the same size; truncate beforehand for mixed degrees (the shLo...

input	size	type	default	kind
C1	—	double	required	positional
S1	—	double	required	positional
C2	—	double	required	positional
S2	—	double	required	positional

output	description
spec	(1,1) struct fields: .n (nmax+1 x 1) double degree axis 0..nmax .diffAmp (nmax+1 x 1) double degree amplitude of (1)-(2) .amp1 (nmax+1 x 1) double signal degree amplitude of (1) .amp2 (nmax+1 x 1) double signal degree amplitude of (2) .degCorr (nmax+1 x 1) double per-degree correlation (NaN where a degree is all-zero) .ncross (1,1) double first degree where diffAmp exceeds amp1 (NaN if never; agreement to nmax)

Example

```
    gF = g.gaussian(350);
    spec = shLowLevel.diffSpectrum(g.C, g.S, gF.C, gF.S);
    semilogy(spec.n, [spec.amp1, spec.diffAmp]); grid on
```

shLowLevel.spatialStats

`stats = shLowLevel.spatialStats(A, B, latDeg, lonDeg, Name=Value)`

Area-weighted spatial comparison of two grids. STATS = shLowLevel.spatialStats(A, B, LATDEG, LONDEG) compares two nlat x nlon grids with proper cos(latitude) area weighting - unweighted RMS on a lat/lon grid over-counts the poles and is simply wrong. Returns bias, RMSD, the CENTERED pattern statistics (centered RMSD, weighted correlation, standard deviations) that feed a Taylor diagram, and verifies the Taylor identity crmsd^2 = stdA^2 + stdB^2 - 2 stdA stdB corr internally.

input	size	type	default	kind
A	—	double	required	positional
B	—	double	required	positional
latDeg	1 x :	double	required	positional

input	size	type	default	kind
lonDeg	1 x :	double	required	positional
Mask	—	—	[]	name-value
Weights	—	double	[]	name-value

output	description
stats	(1,1) struct fields: .bias (1,1) double weighted mean of A - B .rmsd (1,1) double weighted RMS of A - B .crmsd (1,1) double centered (bias-free) RMSD .corr (1,1) double weighted pattern correlation .stdA (1,1) double weighted std of A .stdB (1,1) double weighted std of B .ratio (1,1) double stdB / stdA .nUsed (1,1) double number of grid points used

Example

```
A1 = g.synthesis(lat, lon); A2 = gF.synthesis(lat, lon);
st = shLowLevel.spatialStats(A1, A2, lat, lon);
fprintf("rmsd %.3g, corr %.3f\n", st.rmsd, st.corr)
```

shLowLevel.nashSutcliffe

`nse = shLowLevel.nashSutcliffe(ref, model)`

Nash-Sutcliffe efficiency of a model against a reference. NSE = shLowLevel.nashSutcliffe(REF, MODEL) computes the hydrology-standard skill score NSE = 1 - sum((model-ref).^2) / sum((ref-mean(ref)).^2) column-wise. NSE = 1 is perfect, 0 means "no better than the reference mean", negative is worse than the mean. More informative than correlation because bias and amplitude errors are punished. NaN pairs are removed per column.

input	size	type	default	kind
ref	—	double	required	positional
model	—	double	required	positional

output	description
nse	(1,K) double efficiency per column (K = 1 for vectors)

Example

```
nse = shLowLevel.nashSutcliffe(avg(1,:), avg(2,:));   % e.g. SH vs mascon basin
```

shLowLevel.effectiveCorr

`out = shLowLevel.effectiveCorr(x, y)`

Correlation with AR(1)-corrected significance. OUT = shLowLevel.effectiveCorr(X, Y) computes the Pearson correlation of two time series together with a significance test that accounts for serial correlation: monthly GRACE series are strongly autocorrelated, so the effective sample size Neff = T*(1-r1*r2)/(1+r1*r2) (r1, r2 = lag-1 autocorrelations) is well below T, and the naive t-test badly over-states significance (Monte-Carlo: ~30% false positives at phi = 0.7 vs ~6% corrected; validated in Python). The two-si...

input	size	type	default	kind
x	: x 1	double	required	positional
y	: x 1	double	required	positional

output	description
out	(1,1) struct fields: .r (1,1) double Pearson correlation .neff (1,1) double effective sample size (clamped [4, T]) .t (1,1) double t statistic on neff-2 dof .p (1,1) double two-sided p-value .n (1,1) double finite pairs used

Example

```
ec = shLowLevel.effectiveCorr(avg(1,:)', avg(2,:)');   % two basin series
fprintf("r = %.3f (p = %.3g, Neff = %.0f)\n", ec.r, ec.p, ec.neff)
```

shLowLevel.threeCorneredHat

`out = shLowLevel.threeCorneredHat(X)`

Per-solution noise levels from N >= 3 series. OUT = shLowLevel.threeCorneredHat(X) with X (T x N, N >= 3) estimates the individual noise standard deviation of each series by the classic (generalized) three-cornered hat: pairwise difference variances V(i,j) = var(x_i - x_j) cancel the common signal, and under the assumption of mutually independent noises V(i,j) = s_i^2 + s_j^2; the N unknowns are solved from the N*(N-1)/2 pairs by least squares (exact Grubbs solution for N = 3). This answers the question pairwise...

input	size	type	default	kind
X	: x :	double	required	positional

output	description
out	(1,1) struct fields: .sigma (1,N) double noise standard deviation per series .variance (1,N) double raw LS variance estimates (may be < 0) .pairVar (P,3) double [i j var(x_i - x_j)] per pair .clipped (1,N) logical true where variance was clipped to 0

Example

```
    out = shLowLevel.threeCorneredHat(Y3);   % T x 3 basin series (CSR/GFZ/JPL)
    fprintf("noise levels: %s\n", sprintf("%.3g ", out.sigma))
```

shLowLevel.taylorDiagram

`h = shLowLevel.taylorDiagram(refStd, stds, corrs, Name=Value)`

Taylor diagram (std / correlation / centered RMSD). H = shLowLevel.taylorDiagram(REFSTD, STDS, CORRS) draws the standard one-figure comparison of solutions against a reference: radius is the (weighted) standard deviation, the azimuth is acos(correlation), and by the law of cosines the distance to the reference point equals the centered RMSD - so pattern amplitude, pattern correlation and pattern error are read off a single plot. Feed it the .stdA/.stdB/ .corr fields of shLowLevel.spatialStats, or temporal statis...

input	size	type	default	kind
refStd	1 x 1	double	required	positional
stds	1 x :	double	required	positional
corrs	1 x :	double	required	positional
Labels	—	string	strings(1, 0)	name-value
Title	1 x 1	string	"Taylor diagram"	name-value
Normalize	1 x 1	logical	false	name-value

output	description
h	(1,1) graphics handle the axes handle

Example

```
    st = shLowLevel.spatialStats(grid, 0.8 * grid, lat, lon);
    shLowLevel.taylorDiagram(st.stdA, st.stdB, st.corr, Labels = "damped");
```

Plotting

shLowLevel.plotSHMap

`h = shLowLevel.plotSHMap(grid, latDeg, lonDeg, Name=Value)`

Global map plot with coastlines and a sane divergent scale. H = shLowLevel.plotSHMap(GRID, LAT, LON) renders a synthesized field the way a GRACE map should look by default: divergent blue-white-red colormap symmetric about zero, robust color limits (98th percentile of |value|), coastline overlay (base MATLAB 'coastlines'), correct aspect. The convenience method g.map(...) synthesizes and plots in one call.

input	size	type	default	kind
grid	—	double	required	positional

input	size	type	default	kind
latDeg	1 x :	double	required	positional
lonDeg	1 x :	double	required	positional
Projection	1 x 1	string ∈ ["plate","hammer"]	"plate"	name-value
Coast	1 x 1	logical	true	name-value
CLim	—	double	[]	name-value
Colormap	1 x 1	string	"divergent"	name-value
Title	1 x 1	string	""	name-value
Units	1 x 1	string	""	name-value
ax	—	—	[]	name-value

output	description
h	(1,1) graphics handle axes of the map plot

Example

```
shLowLevel.plotSHMap(grid, lat, lon, Title = "EWH 2008-04 [m]");
```

shLowLevel.plotSHSpectrum

`h = shLowLevel.plotSHSpectrum(specs, varargin)`

Log-log degree-amplitude spectrum plot. PLOTSHSPECTRUM(SPEC) plots SPEC.degree vs SPEC.degAmplitude (from SHDEGREERMS) on a log-log scale, in mm geoid-height-equivalent.

input	size	type	default	kind
specs	—	—	required	positional
varargin	—	—	required	positional

output	description
h	(1 x 1) graphics handle log-log spectrum axes

Example

```
spec = shLowLevel.shDegreeRMS(g.C, g.S);
shLowLevel.plotSHSpectrum(spec, Kaula = true, MarkCrossover = true);
```

shLowLevel.plotSHCoeffTriangle

`h = shLowLevel.plotSHCoeffTriangle(C, S, varargin)`

Coefficient-triangle diagnostic plot. PLOTSHCOEFFTRIANGLE(C, S) shows |Cnm| (left half) and |Snm| (right half) as a single degree/order "butterfly" triangle image on a log color scale -- the standard first diagnostic for spotting striping, leakage, or bad coefficients before spectral/spatial analysis. x-axis: order m (mirrored: Snm on the LEFT, Cnm on the RIGHT), y-axis: degree n with the apex (n=0) at the TOP - the triangle's peak points upwards. The data aspect ratio is fixed to 1:1, so the axes box is 2:1 (x ...

input	size	type	default	kind
C	—	—	required	positional
S	—	—	required	positional
varargin	—	—	required	positional

output	description
h	(1 x 1) graphics handle butterfly triangle image (abs log10 or signed diff with RefC/RefS)

Example

```
shLowLevel.plotSHCoeffTriangle(g.C, g.S);
shLowLevel.plotSHCoeffTriangle(gF.C, gF.S, RefC = g.C, RefS = g.S);   % signed diff
```

shLowLevel.plotBasinSeries

`h = shLowLevel.plotBasinSeries(tYears, c, sigma, Name=Value)`

Basin time series with 1-sigma band, gaps, trend. H = shLowLevel.plotBasinSeries(TYEARS, C, SIGMA) draws the standard basin- average figure: shaded 1-sigma band (fill), data line, the GRACE <-> GRACE-FO gap (or any gap > GapThreshold (0.3) years) hatched grey, and an annotated trend fitted with the toolbox climatology design (bias/ trend/annual/semi-annual, AR(1)-corrected sigma).

input	size	type	default	kind
tYears	: x 1	double	required	positional
c	: x 1	double	required	positional
sigma	—	double	[]	positional
Label	1 x 1	string	"basin average"	name-value
Units	1 x 1	string	""	name-value
GapThreshold	1 x 1	double	0.3	name-value
Trend	1 x 1	logical	true	name-value
ax	—	—	[]	name-value

output	description
h	(1,1) graphics handle axes of the basin series plot

Example

```
shLowLevel.plotBasinSeries(op.tYears, avg(1,:).', out.sigma(1,:).', Units = "m EWH");
```

shLowLevel.plotCovariance

`h = shLowLevel.plotCovariance(M, idx, Name=Value)`

Covariance/correlation image with order-block boundaries. H = shLowLevel.plotCovariance(M, IDX) shows a P x P covariance (e.g. from shLowLevel.readSINEX with Index=IDX) as an image in shIndex ordering, with the (order, C/S) block boundaries drawn - the one-look check of how good the block-diagonal approximation (tvANS block path, DDK structure) is for THIS matrix.

input	size	type	default	kind
M	—	double	required	positional
idx	1 x 1	struct	required	positional
Type	1 x 1	string ∈ ["correlation","covariance"]	"correlation"	name-value
BlockLines	1 x 1	logical	true	name-value
ax	—	—	[]	name-value

output	description
h	(1 x 1) graphics handle image of the covariance/correlation structure

Example

```
shLowLevel.plotCovariance(snx.M, idx);       % SINEX covariance structure
```

Utilities

shLowLevel.designFilter

`[W, info] = shLowLevel.designFilter(SigmaC, SigmaS, Name=Value)`

Build a DDK-class anisotropic filter from user covariances. W = shLowLevel.designFilter(SIGMAC, SIGMAS) constructs the Wiener- type decorrelation filter W = (N + Alpha * inv(S))^-1 * N per order/parity block - the same construction as the published DDK filters, but matched to YOUR solution's error information instead of a fixed release. N is the normal (inverse noise covariance) matrix built from the formal sigmas (diagonal) or from a full covariance (Noise=); S is the signal covariance from a Kaula-type degree ...

input	size	type	default	kind
SigmaC	—	double	required	positional
SigmaS	—	double	required	positional
Alpha	1 x 1	double	1	name-value
Kaula	1 x 1	double	1e-5	name-value
DegVar	—	double	[]	name-value
Noise	—	double	[]	name-value
Signal	—	double	[]	name-value
Idx	—	—	[]	name-value
MinDegree	1 x 1	double	2	name-value
Name	1 x 1	string	"designFilter"	name-value

output	description
W	(1,1) struct fields: nmax (1,1 double), name (char), blocks (1,K struct: m (1,1 double), cs ('c'|'s'), n (1,B double, degrees), M (B x B double)) - identical layout to shLowLevel.readDDK
info	(1,1) struct fields: alpha (1,1 double), mode (string: "diagonal" | "fullcov"), gainRange (1,2 double, min/max eigenvalue over all blocks)

Example

```
W = shLowLevel.designFilter(g.sigmaC, g.sigmaS, Alpha = 1);
gF = g.applyDDK(W);               % apply like any DDK
```

shLowLevel.fetchLoveNumbers

`[files, info] = shLowLevel.fetchLoveNumbers(names, Name=Value)`

Download load/deformation Love number files (GROOPS set). FILES = shLowLevel.fetchLoveNumbers() downloads the standard load Love number file (Gegout97, kn for degrees 0..1024) from the GROOPS data collection hosted by TU Graz into <dataFolder>/loveNumbers, with the usual safe-swap semantics. The toolbox deliberately never hardcodes Love numbers - this fetcher automates obtaining a PUBLISHED set, which you then pass explicitly (kn = ...) where required.

input	size	type	default	kind
names	—	string	"loadLoveNumbers_Gegout97.txt"	positional
Dest	1 x 1	string	""	name-value
BaseURL	1 x 1	string	"https://ftp.tugraz.at/pub/ITSG/groops/data/loading"	name-value
Timeout	1 x 1	double	30	name-value
Proxy	1 x 1	string	""	name-value
Update	1 x 1	logical	false	name-value
Quiet	1 x 1	logical	false	name-value

output	description
files	(1,K) string local paths
info	(1,1) struct fields: fetched/updated/skipped/failed (1,K string); parsed (1,K struct: name, n (N x 1), kn (N x 1); empty for deformation files)

Example

```
    [f, inf] = shLowLevel.fetchLoveNumbers();
    kn = inf.parsed(1).kn(1:61);        % degrees 0..60 for an n60 field
```

shLowLevel.gridScaling

`[k, info] = shLowLevel.gridScaling(model, latDeg, lonDeg, Name=Value)`

Per-pixel leakage/attenuation scaling factors from a model. [K, INFO] = shLowLevel.gridScaling(MODEL, LATDEG, LONDEG, Filter = ...) computes the classic gridded scaling factors: push a MODEL field series through the SAME processing chain the data saw and regress, per pixel, the true value on the filtered one,

input	size	type	default	kind
model	—	double	required	positional
latDeg	—	double	required	positional
lonDeg	—	double	required	positional
Filter	—	—	"gauss300"	name-value
Nmax	1 x 1	double	NaN	name-value
MinSignal	1 x 1	double	1e-3	name-value
Clip	—	double	[]	name-value
GM	1 x 1	double	3.986004415e14	name-value
R	1 x 1	double	6378136.3	name-value
Quiet	1 x 1	logical	false	name-value

output	description
k	(nLat x nLon) double scaling factors; NaN where the model carries too little signal to define one
info	(1,1) struct fields: covered (1,1 double, fraction of the grid with a finite k), kMedian / kMin / kMax (1,1 double, over the pixels the MODEL gives mass to - not over all finite pixels, where pure-leakage k = 0 would dominate), onSignal (1,1 double, how many those are), clipped (1,1 double, number of clamped pixels), nmax (1,1 double), filter (1,1 string), epochs (1,1 double)

Example

```
k = shLowLevel.gridScaling(modelEWH, -89:89, 0:359, Filter = "gauss300");
corrected = k .* filteredEWH;      % NaN where the model is blind
```

shLowLevel.leakageCorrect

`[mCorr, info] = shLowLevel.leakageCorrect(grid, latDeg, lonDeg, Name=Value)`

Iterative forward modelling of filter leakage/attenuation. [MCORR, INFO] = shLowLevel.leakageCorrect(GRID, LATDEG, LONDEG, Filter = ...) recovers the mass field that, pushed through the SAME processing chain the data saw, reproduces the observed filtered GRID. The chain F is analysis -> filter -> synthesis, and the correction is the fixed-point iteration

input	size	type	default	kind
grid	—	double	required	positional
latDeg	—	double	required	positional
lonDeg	—	double	required	positional
Filter	—	—	"gauss300"	name-value
Mask	—	—	[]	name-value
Nmax	1 x 1	double	NaN	name-value
Gain	1 x 1	double	1	name-value
NoiseLevel	1 x 1	double	0	name-value
Tau	1 x 1	double	1.2	name-value

input	size	type	default	kind
MaxIter	1 x 1	double	200	name-value
Tol	1 x 1	double	1e-4	name-value
GM	1 x 1	double	3.986004415e14	name-value
R	1 x 1	double	6378136.3	name-value
Quiet	1 x 1	logical	false	name-value

output	description
mCorr	(nLat x nLon) double leakage-corrected field, same units and grid as GRID
info	(1,1) struct fields: iterations (1,1 double), residual (1,1 double, final relative residual - with a mask this floors above Tol and that is correct), history (1,K double, the residual per iteration - plot it, a rising curve means Gain is too large), step (1,K double, the relative change of the solution per iteration; this is what Tol tests), stoppedBy (1,1 string: "discrepancy" | "step" | "maxIter" | "zeroField" - always check this, it says whether the result was regularised), residualRMS (1,1 double) and noiseLevel (1,1 double, as given), so the discrepancy ratio is inspectable, converged (1,1 logical), nmax (1,1 double), filter (1,1 string), masked (1,1 logical)

Example

```
ewh = g.synthesis(-89:89, 0:359, quantity = "ewh", kn = kn);
[m, info] = shLowLevel.leakageCorrect(ewh, -89:89, 0:359, ...
    Filter = "gauss300", Mask = basinMask, NoiseLevel = oceanRMS);
```

shLowLevel.listITSG

`[T, info] = shLowLevel.listITSG(Name=Value)`

Catalogue of ITSG gravity field series on the TU Graz server. T = shLowLevel.listITSG() enumerates the available ITSG releases and product folders (releases ITSG-Grace2014/2016/2018 and the continuously extended ITSG-Grace_operational; products monthly at n60/n96/n120 and daily Kalman solutions) by parsing the server directory indices - so mixing files from different releases, the trap of guessing folder names, becomes impossible. The catalogue row number is the selection handle for shLowLevel.fetchITSG(Catalog=...

input	size	type	default	kind
BaseURL	1 x 1	string	"https://ftp.tugraz.at/pub/ITSG/GRACE"	name-value
Timeout	1 x 1	double	30	name-value

output	description
T	(K x 5) table columns: idx (1,1) double selection number for fetchITSG release (1,1) string e.g. "ITSG-Grace2018" product (1,1) string "monthly" | "daily" | "static" nmax (1,1) double 60/96/120 (NaN for daily/static) url (1,1) string folder URL or mirror path
info	(1,1) struct fields: source (string), nReleases (double)

Example

```
    T = shLowLevel.listITSG();
    disp(T)     % pick rows, then fetch via Catalog= in fetchITSG
```

shLowLevel.poleTideConvert

`varargout = shLowLevel.poleTideConvert(varargin)`

Convert C21/S21 between mean-pole (pole tide) conventions. G2 = shLowLevel.poleTideConvert(G, From="IERS2010", To="IERS2018") adjusts the C21/S21 coefficients of an shCoefficients object (epoch taken from G) from one mean-pole convention to another. Processing centers remove the (solid + ocean) pole tide using a MEAN POLE MODEL, and the IERS 2010 conventional model (cubic until 2010.0, linear after) differs from the 2018 secular-pole update (linear, adopted for GRACE/GRACE-FO RL06) by up to ~1e-10 in C21/S21 wit...

input	size	type	default	kind
varargin	—	—	required	positional

output	description
G2	(1,1) shCoefficients converted object (object input; history appended)
C2, S2	(nmax+1 x nmax+1) double converted triangles (raw input)

Example

```
    g18 = shLowLevel.poleTideConvert(g, From = "IERS2010", To = "IERS2018");
    dC21 = g18.C(3, 2) - g.C(3, 2)     % ~1e-10 level, trend-like in t
```

shLowLevel.readSHM

`out = shLowLevel.readSHM(filename, Name=Value)`

Read a GRAVIS/GRACE SHM file (GRCOF2 fields, GRDOTA rates). OUT = shLowLevel.readSHM(FILENAME) reads the spherical-harmonic format used by the GRAVIS Level-2B products and by GRACE/GRACE-FO Level-2 files that are not in ICGEM layout:

input	size	type	default	kind
filename	—	—	required	positional
Nmax	1 x 1	double	NaN	name-value

output	description
out	(1,1) struct fields: kind (1,1) string "GRCOF2" (field) | "GRDOTA" (rate) C, S (nmax+1 x nmax+1) double coefficients in the C(n+1, m+1) house layout; for GRDOTA these are rates per year sigmaC, sigmaS (nmax+1 x nmax+1) double, or [] when the file carries none (GRDOTA never does) GM (1,1) double [m^3/s^2] from the header R (1,1) double [m] from the header nmax (1,1) double maximum degree actually read header (1,1) string the raw YAML header, kept so provenance is not lost on the way in

Example

```
gia = shLowLevel.readSHM("GRAVIS-2B_..._GIA_ICE-6G_D_VM5a.gz", Nmax = 96);
g   = shCoefficients(gia.C, gia.S, GM = gia.GM, R = gia.R);   % rates [1/yr]
```

shLowLevel.safeMove

`n = shLowLevel.safeMove(src, dst, Name=Value)`

Move a file into place, retrying while a scanner still holds it. SAFEMOVE(SRC, DST) renames SRC to DST, overwriting DST, and retries with exponential backoff while the move fails. Every shLowLevel.fetch* function downloads to "<target>.part", verifies the fresh file by parsing it, and only then swaps it into place - this function is that swap.

input	size	type	default	kind
src	1 x 1	string	required	positional
dst	1 x 1	string	required	positional
Retries	1 x 1	double	5	name-value
Pause	1 x 1	double	0.25	name-value

output	description
n	(1,1) double number of attempts used (1 = the move succeeded immediately, which is the normal case)

Example

```
shLowLevel.shReadGFC(tmpf);                % verify BEFORE the swap
shLowLevel.safeMove(tmpf, "ITSG-Grace2018_n60_2008-04.gfc");
```

shLowLevel.standardChain

`[ts, rep] = shLowLevel.standardChain(folder, Name=Value)`

Canonical GRACE/GRACE-FO post-processing in one call. [TS, REP] = shLowLevel.standardChain(FOLDER) encodes the standard chain as a single, correctly ordered entry point: 1. read all monthly solutions from FOLDER (shSeries.fromFolder) 2. replace C20/C30 with the SLR values (TN-14) 3. add the degree-1 (geocenter) coefficients (TN-13) 4. optionally subtract a GIA trend field 5. filter (Gaussian radius, a DDK, or a shLowLevel.designFilter W) The order matters (TN corrections BEFORE filtering, GIA on the corrected se...

input	size	type	default	kind
folder	1 x 1	string	required	positional
Pattern	1 x 1	string	"*.gfc*"	name-value
Truncate	1 x 1	double	NaN	name-value
TN14	1 x 1	logical	true	name-value
TN14File	1 x 1	string	""	name-value
Degree1	1 x 1	string ∈ ["GFZ", "CSR", "JPL", "none"]	"GFZ"	name-value
Degree1File	1 x 1	string	""	name-value
GIA	—	—	[]	name-value
GIAEpoch	1 x 1	double	NaN	name-value
Filter	—	—	"gauss300"	name-value
Tolerance	1 x 1	double	0.05	name-value
OnMissing	1 x 1	string ∈ ["drop", "error"]	"drop"	name-value
Quiet	1 x 1	logical	false	name-value

output	description
ts	(1,1) shSeries the processed series (full history)
rep	(1,1) struct fields: files (1,T string), steps (1,K string, applied stages with parameters), nmax (1,1 double), epochs (T,1 double), version (1,1 string, shLowLevel.version at run time), created (1,1 string)

Example

```
[ts, rep] = shLowLevel.standardChain("D:/grace/monthly", ...
    Filter = "DDK3", Degree1 = "GFZ");
disp(rep.steps')
```

shLowLevel.synthesisPoints

`val = shLowLevel.synthesisPoints(C, S, GM, R, latDeg, lonDeg, r, Name=Value)`

Gravity field quantities at arbitrary 3-D points. VAL = shLowLevel.synthesisPoints(C, S, GM, R, LATDEG, LONDEG, RGEO) evaluates the spherical harmonic expansion POINTWISE at positions given by equal-length vectors of geocentric latitude [deg], longitude [deg] and geocentric radius [m] - satellite altitudes, scattered stations, profiles - including the upward continuation (R/r)^n that grid synthesis on the sphere cannot provide: potential T = GM/r sum_n (R/r)^n sum_m Pnm (C cos + S sin) disturbance ...

input	size	type	default	kind
C	—	double	required	positional
S	—	double	required	positional
GM	1 x 1	double	required	positional
R	1 x 1	double	required	positional
latDeg	: x 1	double	required	positional
lonDeg	: x 1	double	required	positional
r	: x 1	double	required	positional

input	size	type	default	kind
Quantity	1 x 1	string ∈ ["potential", "disturbance", "anomaly"]	"potential"	name-value
MinDegree	1 x 1	double	2	name-value

output	description
val	(P,1) double quantity at each point

Example

```
dg = shLowLevel.synthesisPoints(g.C, g.S, g.GM, g.R, ...
    [10; 12], [100; 101], (g.R + 450e3) * [1; 1], ...
    Quantity = "disturbance") * 1e5;   % mGal at 450 km
```

shLowLevel.version

`v = shLowLevel.version()`

Toolbox metadata: name, version, date, root path, provenance. V = shLowLevel.version() returns the toolbox metadata as a struct. The version string and date are PARSED FROM Contents.m (the MATLAB toolbox convention line "% Version x.y.z ... dd-Mmm-yyyy"), so there is a single source of truth also honoured by MATLAB's own ver('shAnalysis'). Called without an output, prints a summary.

output	description
v	(1,1) struct fields: .Name (1,1) string "shAnalysis" .Version (1,1) string e.g. "2.5.1" .Date (1,1) string release date from Contents.m .Root (1,1) string toolbox root folder .Provenance (1,1) string authorship line

Example

```
v = shLowLevel.version();
fprintf("%s %s (%s)\n", v.Name, v.Version, v.Date)
```

Toolbox root

setup_shAnalysis

`summary = setup_shAnalysis(Name=Value)`

One-call setup: path + staged data downloads. setup_shAnalysis adds the toolbox to the path for THIS session only setup_shAnalysis(Permanent (false) = true) also persists it (savepath) setup_shAnalysis(Download ("none") = "core") + TN-14 and TN-13 files setup_shAnalysis(Download = "filters") + DDK (3) filter matrices setup_shAnalysis(Download = "starter") + ITSG starter months setup_shAnalysis(DryRun (false) = true, ...) print/return the plan, do NOTHING (no path change, no f...

input	size	type	default	kind
Permanent	1 x 1	logical	false	name-value
Download	1 x 1	string ∈ ["none", "core", "filters", "starter"]	"none"	name-value
DDK	1 x :	double	3	name-value
Providers	1 x :	string ∈ ["GFZ", "CSR", "JPL"]	["GFZ", "CSR", "JPL"]	name-value
Months	1 x :	string	["2008-04", "2025-12"]	name-value
Nmax	1 x 1	double ∈ [60, 96]	96	name-value
Docs	1 x 1	logical	false	name-value
DryRun	1 x 1	logical	false	name-value
DataFolder	1 x 1	string	""	name-value
FetchITSG	1 x 1	string	"none"	name-value
Proxy	1 x 1	string	""	name-value

input	size	type	default	kind
Update	1 x 1	logical	false	name-value
Quiet	1 x 1	logical	false	name-value

output	description
root	(1,1) string toolbox root folder
pathAction	(1,1) string "already-on-path" | "added-temporarily" | "added-permanently" | "planned"
dataFolder	(1,1) string download target root
plan	(1,:) string human-readable action list
fetched	(1,:) string files newly downloaded (verified)
skipped	(1,:) string files already present (re-verified)
failed	(1,:) string files that failed (download or parse)
ok	(1,1) logical isempty(failed)

Example

```
    setup_shAnalysis(Permanent = true, Download = "core")
    setup_shAnalysis(Download = "filters", DDK = [2 3 5])
    s = setup_shAnalysis(DryRun = true, Download = "starter")
```

runAllTests

`results = runAllTests(Name=Value)`

Run the complete shAnalysis v2 validation suite. RESULTS = RUNALLTESTS runs, in order: tests/testCorrectness.m numerics vs. golden/legacy/analytic values tests/testScience.m regression against PUBLISHED values (defining constants, load Love numbers, the closed-form EWH kernel, geocenter amplitudes) - so a consistently wrong formula has somewhere to show up tests/testContract.m error IDs, immutability, dimensions tests/testRobustness.m edge cases, degenerate inputs, file I/O documented patch: kern...

input	size	type	default	kind
SkipPerformance	1 x 1	logical	false	name-value
LogFile	—	—	""	name-value

output	description
results	(1,N) matlab.unittest.TestResult full result array (pass/fail/duration per test; log written to LogFile)

Example

```
runAllTests           % 169 tests; acceptance gate on R2026a
```

demo_shAnalysis

`reg = demo_shAnalysis(cases, Name=Value)`

Selectable demonstrations of every toolbox capability. demo_shAnalysis runs the "core" set (D01 D02 D04 D05 D15) demo_shAnalysis("all") runs every case demo_shAnalysis("list") prints the case table, returns registry demo_shAnalysis(["D04","D15"]) runs selected cases demo_shAnalysis(..., Visible (true)=false) hidden figures (CI/smoke tests) demo_shAnalysis(..., OutDir (string(tempdir))=tempdir) target for D16 file exports demo_shAnalysis(..., StopOnError (false)=true) rethrow ins...

input	size	type	default	kind
cases	1 x :	string	"core"	positional
Visible	1 x 1	logical	true	name-value
OutDir	1 x 1	string	string(tempdir)	name-value
StopOnError	1 x 1	logical	false	name-value

output	description
reg	(D x 3) table demo registry: id, title, exercised API

Example

```
demo_shAnalysis(["D01", "D06"])   % run selected demos, fail-and-continue
```

Appendix

A. Error-identifier conventions

Namespaces: `shLowLevel:<function>:<what>` for package functions, `shCoefficients:/shSeries:/shClimatology:` for class methods. Every guard has a unit test; every message names the offending quantity. Instrumentation asserts (nonFiniteVCE, nonFiniteEig, nonFiniteOperator, allZeroVariance) remain in the code as a permanent safety net after the v2.1 debugging campaign.

B. Known limitations (documented, tested where applicable)

- TN-13 verified against the real GFZ, CSR and JPL RL06.3 files (v2.5; 256 paired months each, cross-provider C10 correlation 0.995); SINEX against ITSG monthly files and synthetic fixtures — COST-G/other SINEX dialects may need parser tweaks (drop files into tests/test_data, the discovery tests pick them up).
- DDK: all eight released Wbd binaries parsed and gain-ordering validated (v2.5); a discovery test re-checks whatever subset is present locally. Other BIN dialects untested.
- readMascon validated against a synthetic JPL-style fixture; real product files were not openly reachable at build time (auth/503) — verify against yours (units, GIA convention, CRI flag).
- GIA models treated as exact; AR(1) correction is first-order (Kendall-corrected r_1 since v2.5, ~3% residual optimism).
- Multi-center combination: common-mode errors invisible → posterior sigmas a lower bound.
- Basin sigma: deterministic-part errors correlated across epochs (pointwise 1-sigma reported).
- Streaming gz SINEX needs the JVM (fallback: gunzip to temp).
- ICGEM catalogue parsers are fixture-tested against real page captures (2026-08-07); a future site redesign would need refreshed fixtures (shLowLevel:listICGEM:parseFailed errors loudly).

C. License & provenance

The toolbox is released under the MIT license (LICENSE in the package root; copyright Matthias Weigelt). The bundled DDK3 Wbd file carries its own MIT license from its upstream repository; ITSG/GFZ files in tests/test_data remain the property of their providers and are included as test fixtures only.

All code, tests, documentation and this guide: developed by Matthias Weigelt with the help of Claude (Fable 5), 2026-08-07 (v1 through v2.4.2 in working sessions the same week). Every numerical method was independently validated in Python before MATLAB implementation; MATLAB execution itself happens on the user's side with runAllTests as the acceptance gate (118/118 green at v2.1; 134 at v2.2.2; 167 at v2.4.1; 168 at v2.4.2; 179 at v2.5 — v2.5 adds the exact constrained posterior, the basin deterministic-sigma term, the Kendall AR(1) correction, shLowLevel.readLoveNumbers, setup_shAnalysis/shLowLevel.fetchTN with verified staged downloads, real CSR/JPL TN-13 + GSFC TN-14 fixtures and the DDK-pack validation).